

DRY ELECTRODE 4680

Complete Manufacturing Handbook — Process and Equipment

Solvent-Free Electrode Production for Automotive Cylindrical Cells

Process Physics · Machine Datasheets · Instrumentation and Control
Defect Engineering · Facility and Utilities · Patent Landscape
Procurement, Acceptance and Ramp-Up

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46-Series Cylindrical Format — 4680 · 4695 · 46105 · 46120

Dry route only — the wet slurry route is out of scope

Revision 1.0 · August 2026

Scope and How to Use This Handbook

What this handbook is

A single reference covering both the **process** and the **equipment** for building and running a dry electrode line for automotive 46-series cylindrical cells. It merges two things that are usually separate documents: the physics of why each operation works and fails, and the machine specifications you need to buy, accept and maintain the hardware.

SCOPE DECISION: DRY ROUTE ONLY

The conventional wet slurry route — NMP mixing, slot-die coating, multi-zone convection drying and solvent recovery — is **out of scope**. It is referenced only where a comparison genuinely clarifies a design decision, and never as a fallback option. Every process window, every machine datasheet, every acceptance test and every facility figure in this handbook assumes solvent-free electrode manufacture.

Why the merge matters

Process documents describe what happens to the material and leave the machine as a black box. Equipment documents list specifications with no explanation of which ones actually matter. On a dry electrode line neither is sufficient on its own, because the process is immature enough that *the machine specification is the process specification*. The residence time distribution of a mixer is not a mechanical detail — it is the variable that determines whether your binder network works.

The station template

Every production station is described in one place, with the same structure:

Block	What it answers
Purpose	The physical change this station makes and why it is needed
Process physics	The mechanism that governs the outcome — the reason the parameters matter
Machine datasheet	Throughput, footprint, utilities, environment, technology maturity
Subsystem breakdown	Every module inside, named as a supplier names it, with specification points
Instrumentation and control loops	Which sensors, where, and which measured variable drives which actuator
Recipe window	Numerical parameter ranges from published industrial references
Consumables and wear parts	What to stock and on what replacement basis
Acceptance criteria	What to prove at FAT and SAT before signing
Troubleshooting	Defect → probable cause → first parameter to move
Selection notes	The questions that separate a good machine from a bad one at this station

Equipment numbering

Series	Group	Notes
DE-1xx	Dry electrode manufacturing	Twelve machine classes, powder to slit conditioned coil
AS-3xx	Cell assembly	Dry room, −40 to −60 °C dew point
FN-4xx	Cell finishing	Formation, aging, test — a third of the plant capital
MT-xxx	Metrology and inspection	Including the dry-specific measurement gaps
FC-xxx	Facility and utilities	

Reader's map

If you are...	Read in this order
Building a capital estimate	Ch. 3 → Ch. 24 → datasheet headers only
Writing an RFQ	Ch. 5–9 → the relevant station chapters → Ch. 25
Commissioning a line	Ch. 27 → the station chapters, in process order
Running a line that is producing defects	Ch. 19 → Ch. 20 → the station chapter
Specifying metrology	Ch. 17–18
Planning the building	Ch. 21–23 → Appendix B
Doing IP groundwork	Ch. 2 → Ch. 10 → Ch. 28

Notation and conventions

Convention	Meaning
MD / CD	Machine direction (along web travel) / cross direction (across web width)
Areal loading	Dry active-material mass per unit area, mg/cm ² . The single most important electrode number
C-rate	Current normalised to nominal capacity. 1C discharges a cell in one hour
Ranges quoted	Published industrial reference windows, not a recipe. Every line needs its own DOE
“Reported”	From secondary sources; indicative, verify before relying on it
Currency	EUR, following the source study

RELIABILITY OF THE NUMBERS — READ BEFORE RELYING ON ANY OF THEM

Process windows come from public engineering literature. They give the order of magnitude and the direction of sensitivity. They are not a production recipe — a working recipe is chemistry-specific, machine-specific and established by designed experiment on your own equipment.

Footprints are indicative envelopes for layout blocking, excluding maintenance access, staging and operator space. Add 40-60 % for a realistic layout.

CAPEX figures are published station-group costs for a 10 GWh/year reference line. They do not decompose cleanly to individual machines and exclude building, utilities, installation and commissioning.

Patent numbers are for technical study only. Verify each against the official register. This is not legal advice and not a freedom-to-operate opinion.

TECHNOLOGY MATURITY — THE CONTEXT FOR EVERYTHING THAT FOLLOWS

Published assessment places dry coating at **TRL 6-7** — demonstration of a system prototype in the operating environment. One manufacturer has reached automotive scale. That is not the same as a mature, transferable, multi-supplier technology. The procurement approach in Part VIII is built around that reality, and it is why this handbook spends more space on acceptance testing and process guarantees than a conventional equipment reference would.

Sources and Public Availability

PROVENANCE DECLARATION

Every technical statement, process window, parameter range, cost figure and architectural description in this handbook derives from **publicly available sources**: peer-reviewed literature, institutional engineering references sold under ISBN, patent documents published by national patent offices, public corporate disclosures, published standards, and equipment supplier literature issued for general distribution.

This document contains **no confidential, proprietary or trade-secret information** from any manufacturer, supplier or institution. It contains no information obtained under non-disclosure agreement, no unpublished internal data, and no material derived from privileged access of any kind. Every source in the register at Appendix F can be located and read independently by any reader, and the verification route for each class of source is given below.

NAMING POLICY

Manufacturers, suppliers and patent assignees are named throughout this handbook wherever the underlying source names them. Names are not withheld, paraphrased or genericised anywhere in the body text, because a technical claim that cannot be traced to a named source cannot be checked. Where a statement originates with a company about its own process, it is attributed to that company and marked as a corporate disclosure (Tier D below) rather than presented as an independent finding.

S.1 Source classes and how to verify each

Class	What it supplies to this handbook	Independent verification route
Peer-reviewed literature	Cell architecture, teardown geometry, chemistry identification, measured energy density, thermal behaviour	DOI resolution via doi.org, or the publisher's own platform. Every entry carries a DOI or journal citation
Institutional engineering references	The large majority of process parameter windows, station sequences and machine descriptions	ISBN lookup; the PEM RWTH Aachen / VDMA series is distributed publicly and several volumes are free to download
Published patent documents	Process mechanisms, claimed parameter ranges, architectural principles, IP landscape	Espacenet (EPO), USPTO Patent Public Search, Google Patents, KIPRIS, CNIPA, J-PlatPat. Publication of a patent application <i>is</i> the act of making it public
Public corporate disclosure	Process economics claims, deployment status, capacity and timeline statements	Investor relations pages, earnings call transcripts, shareholder presentations, press releases — all issued for public consumption
Published standards	Safety, transport and abuse-test requirements	Purchasable from the issuing body (UN, IEC, ISO, SAE, UL, SAC). Scope and abstract are freely readable
Supplier technical literature	Machine subsystem descriptions, instrument capability, wear-part practice	Supplier public websites, downloadable brochures, conference material and trade-fair documentation
Open datasets	Metrology parameterisation, drying and calendering DOE data	DOI-indexed data repositories; the datasets used here are open-access
Analyst and trade reporting	Market context, programme status, competitive timeline	Named publication. Treated as the weakest tier and flagged as such wherever it is the sole support for a statement

S.2 Evidence tiers used in this handbook

Not all public information is equally reliable. Statements in this handbook rest on the following tiers, and the tier determines how a statement is worded.

Tier	Source type	How statements are worded	What it supports here
A	Peer-reviewed, DOI-indexed	Stated as measured findings	Teardown geometry, chemistry identification, measured energy density
B	Institutional reference, ISBN	Stated as reference windows requiring DOE validation	Nearly all process parameters in Parts III-IV
C	Published patent document	Stated as <i>claimed</i> in the patent, not as established fact	Fibrillation mechanism, binder fractions, pass counts
D	Public corporate disclosure	Attributed to the producer — “reports”, not “achieves”	CAPEX and energy reduction claims for the dry route
E	Supplier literature	Stated as capability, not as guaranteed performance	Machine subsystems, instrumentation
F	Analyst / trade press	Marked “ <i>reported</i> ”; indicative only	Market and programme status; a small number of secondary figures, each flagged in place

WHERE THE TIERS DISAGREE, THE HANDBOOK SAYS SO

Two documented cases: the electrode-step CAPEX reduction is reported at approximately 90 % by the producer (Tier D) and approximately 50 % by an independent source, and both figures are carried rather than averaged. One patent publication date is reported inconsistently between databases and both dates are shown. Public information is not automatically consistent information, and this handbook does not silently pick a winner.

S.3 What this handbook deliberately does not contain

The absence of certain content is itself evidence of the sourcing discipline. Appendix D lists in full what is **not** publicly available and is therefore not in this document. In summary:

Not included	Why
Formation current and voltage protocols	Described in the open literature itself as core proprietary knowledge of a cell manufacturer
Electrolyte additive formulations	Held as trade secret across the industry
Fibrillation setpoints for a specific chemistry	Patent claims give ranges; working setpoints are not published
Laser weld process windows	Machine- and joint-specific; not published
Grading algorithms and predictive quality models	Competitive advantage; not published
Station-level yield figures	Not published by any manufacturer

Where such information would have been needed, this handbook states that it is unavailable and identifies the legitimate route to obtain it — a research institute engagement or the reader's own designed experiment — rather than filling the gap with speculation.

S.4 Verifying any individual statement

1. Identify the chapter in which the statement appears.
2. Locate that chapter in the **Used in** column of the source register at Appendix F.
3. Retrieve the source by its DOI, ISBN or publication number.
4. For patent-derived statements, read the claims rather than the abstract — the claims define the disclosure.

A STANDING CAVEAT THAT THE PUBLIC SOURCING DOES NOT REMOVE

Public availability establishes that this information may be freely read and discussed. It does not establish that it is current, complete, correct for any particular chemistry, or free of third-party patent rights. Patent numbers require verification against the official register before any legal or commercial use. Nothing here is legal advice or a freedom-to-operate opinion.

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Part I — The Product

You cannot specify a dry electrode line without understanding what the cell demands of it. Part I establishes the format, the tabless architecture that the format requires, the dry route itself, and the material specifications that the equipment must handle.

1. The 46-Series Format: Rationale and Reality

1.1 Naming and the family

The number 4680 encodes geometry: 46 mm outer diameter, 80 mm height. It follows the same convention as 18650 (18 mm × 65 mm) and 21700 (21 mm × 70 mm). The final zero is a legacy digit indicating a cylindrical format.

The wider **46-series** family holds diameter constant and varies height: 4680, 4695, 46105 and 46120. Holding diameter fixed is deliberate. It lets a manufacturer reuse the same winding mandrels, the same current-collector disc tooling, the same can-forming die set and the same jelly-roll handling geometry across the whole family, changing only web width and stroke length. A supplier who has qualified a 4680 line can extend to 46120 with a fraction of the retooling that a diameter change would demand.

1.2 Why 46 mm and not more

Cell diameter is bounded above by heat transport, not by manufacturability. A cylindrical cell rejects heat through its outer surface. As diameter grows, volume rises with the square of the radius while lateral surface area rises only linearly. The surface-to-volume ratio therefore falls, and the temperature difference between the core and the skin rises for the same volumetric heat generation.

This matters most during fast charge, where volumetric heat generation is highest. Published thermal studies of 4680 cells confirm the point directly: because the surface-to-volume ratio of a 46 mm cell is smaller than that of smaller formats, thermal gradients within the cell are *exacerbated* under fast-charging conditions. The 46 mm figure represents an industry consensus about where the curve of increasing pack-level benefit crosses the curve of worsening thermal behaviour.

DESIGN CONSEQUENCE

The tabless architecture is not an optional refinement of the 46-series — it is what makes the diameter viable at all. Without a low-resistance, distributed current path and the thermal conduction that the collector discs provide, a 46 mm cell would overheat at automotive charge rates. Format and architecture are one decision, not two.

1.3 Timeline

Date	Event	Significance for manufacturing
2019	Tesla acquires Maxwell Technologies	The acquisition target was dry-electrode coating IP developed for supercapacitors, not the supercapacitor business itself
Sept 2020	4680 unveiled at Battery Day	Announced package: larger format, tabless design, silicon anode, dry electrode, structural pack
2021	Tesla divests the acquired ultracapacitor business, retains dry battery electrode technology	Confirms the strategic intent was process IP
2022	Limited production at Gigafactory Texas; first independent teardowns published	Teardowns reveal Gen 1 is a hybrid: dry anode, wet cathode, graphite anode with no meaningful silicon
Aug 2024	LG Energy Solution begins pilot 46-series production at Ochang, South Korea	Second credible large-format cylindrical programme
Q1 2025	Samsung SDI applies 46Φ cells in micromobility packs	First mass production of the family, but in a non-automotive application
Jan 2026	Tesla reports large-scale deployment of dry electrode for <i>both</i> anode and cathode	The five-year bottleneck — the dry cathode — is reported solved at production scale
Apr-Jul 2026	LG reports 46 mm order backlog above 440 GWh; Panasonic delays automotive 4680 mass production a second time	Demand signal strong on paper; execution still the constraint

1.4 What the format actually delivers

This section exists because the gap between the 2020 announcement and the 2026 measured reality is large enough to change engineering decisions.

1.4.1 Energy density did not improve as promised

Independent testing of Austin-built Tesla 4680 cells measured roughly **244 Wh/kg**, against **269 Wh/kg** for the Panasonic 21700 cells the format was intended to replace — about 13 % *lower*, not higher. Academic teardown work on first-generation cells determined 622.4 Wh/L and 232.5 Wh/kg at cell level.

Three design decisions account for the shortfall:

Decision	Cost	Why it was made anyway
Can wall thickness ~0.5 mm	Heavy inactive mass; reduces internal volume relative to 18650/21700 proportions	Structural pack integration — the can carries load
Pure graphite anode, no silicon	Foregoes the capacity uplift silicon offers	Silicon expands up to ~300 % on lithiation; in a tightly wound large jelly roll this destroys mechanical integrity and cycle life
Thicker electrodes (cathode ~20–25 % thicker than typical 21700)	Longer ion diffusion path; worse rate capability	Compensates for the inactive-mass penalty by raising the active fraction

1.4.2 What it does deliver

Benefit	Mechanism
Fewer cells per pack	Roughly 830 cells replace roughly 4,400 for an 80 kWh pack — fewer welds, simpler wiring harness, fewer BMS channels, fewer failure sites
Reported ~14 % cell cost reduction vs 21700	Lower ratio of casing mass to capacity; fewer structural parts and welds per unit of energy
Better current handling	Tabless design shortens the electron path, reducing internal resistance and heat generation
Enables cell-to-chassis	Structural efficiency of the EV platform improves when the module level is eliminated
Dry electrode process economics	Tesla reports roughly 90 % reduction in equipment capital expenditure and energy consumption for the electrode step, and roughly 70 % reduction in cell production emissions

THE HONEST SUMMARY

The 46-series advantage is **structural and economic, not gravimetric**. If a business case rests on the assumption that switching to 4680 automatically raises pack energy density, the teardown data does not support it. Build the case on cell count, assembly labour, pack architecture and electrode-step CAPEX instead — those are real and measurable.

1.4.3 Where each manufacturer actually stands (mid-2026)

Company	Status	Notes
Tesla	Only automotive-scale producer	Dry electrode on both electrodes reported at scale from Jan 2026; cells in Model Y
LG Energy Solution	Pilot since 2024, Arizona expansion planned	46 mm backlog reported above 440 GWh; customers undisclosed. Plans span 4680 to 46120
Samsung SDI	Mass production of 4695 at Cheonan	Modules assembled in Vietnam for a US customer; initial run targets micromobility, not passenger EV. In discussion with automakers
Panasonic	Lines installed at Wakayama, mass production delayed twice	Waiting on a firm purchase order from its primary customer; Kansas plant renovated for large 4680
CATL, EVE	Testing 46-series for structural compatibility	No announced automotive-scale 46-series production
BYD	Developing a similarly sized large-format cell	LFP chemistry rather than nickel-based

WHAT THIS MEANS FOR A NEW ENTRANT

Four of the world's most capable cell manufacturers have taken four to six years each to reach limited production of this format. That is not a comment on their competence — it is a measure of how many simultaneous novelties the format bundles together: new geometry, new current-collection architecture, new welding regime, new electrode process.

The practical lesson for anyone planning a line: **do not introduce more than one novelty per programme**. A conventional-electrode 4680 with proven wet coating is a hard but tractable project. A dry-electrode 4680 with silicon anode is four research programmes wearing a trench coat. Tesla's own leadership has publicly acknowledged that pursuing the dry electrode process for the 4680 from the outset was a mistake that delayed the programme.

1.5 Comparison to 21700

Property	21700	4680 (Gen 1)
External volume	~0.024 L	~0.133 L
Nominal capacity	~5 Ah	~24–26 Ah
Energy	~17–19 Wh	~96–99 Wh
Cell mass	~68 g	~355 g
Gravimetric energy density (measured)	~269 Wh/kg (Panasonic reference)	~232–244 Wh/kg
Volumetric energy density	—	~622 Wh/L
Current collection	1–2 welded tabs	Tabless / full-tab
Can wall thickness	~0.2–0.3 mm	~0.5 mm
Cathode coating thickness	~60–70 μm	~80–90 μm
Cells for an 80 kWh pack	~4,400	~830
Assembly welds per pack	~8,800	~1,660

The last row is the one that pays for the programme. Halving pack assembly cycle time and eliminating 80 % of the interconnect welds removes an entire category of pack-level defect and an entire category of pack-level capital equipment.

2. Cell Architecture: The Tabless Design Decoded

2.1 The problem tabless solves

In a conventional cylindrical cell, current generated anywhere on the electrode must travel *along the foil* to reach a discrete welded tab, then out through that tab. Three penalties follow.

Penalty	Mechanism	Scaling with cell size
Ohmic resistance	Resistance rises with the distance current must travel along the thin foil	Worsens linearly with electrode length — and electrode length in a 4680 can exceed 4 m
Current concentration	All current funnels through a small tab cross-section	Worsens with total current, creating a localised hot spot
Manufacturing burden	Tabs are extra components, add thickness, and must themselves be wound into the jelly roll	Adds cost and creates a mechanical discontinuity in every wrap

These penalties are what bound conventional cylindrical cells to small diameters. Removing them is what unlocks the format.

2.2 How the tabless architecture works

The mechanism, as described in Tesla's published patent family:

1. **Leave an uncoated margin.** The electrode foil is coated so that a strip of bare foil remains along one edge — the top edge for the anode, the bottom edge for the cathode.
2. **Cut flags.** A laser cuts a series of tongues, called *flags*, into this bare margin. The patent describes providing an electrode layer with active material disposed over a foil, then forming a series of flags in the foil to create a flagged electrode.
3. **Wind.** The flagged electrode is wound into a jelly roll, producing a roll with a series of rolled flags projecting from each end.
4. **Fold.** The flags at each end are folded over, interleaving into what the patent describes as a flower-like shape — a dense, continuous conductive plane at each end of the roll.
5. **Weld.** A current collector disc is laser-welded onto this folded plane.

The result: the current path changes from *metres along the foil* to *millimetres across the foil width*. Internal resistance drops, heat generation falls, and the collector plane itself becomes a thermal conduction path out of the cell.

2.2.1 The manufacturing consequence nobody mentions in the marketing

Because the jelly roll is a spiral, the circumference of each wrap increases with radius. The flags on the outer wraps must therefore be **longer** than those on the inner wraps if they are all to meet at the same folded plane. The notching pattern is not a repeating stamp — it is a *pattern that varies continuously along the web*, synchronised to consumed web length.

WHY THIS DRIVES MACHINE SPECIFICATION

Variable-pitch notching requires closed-loop synchronisation between the laser scanner and a high-resolution measurement of actual consumed web length — not commanded length. Any slip, stretch or tension-induced elongation between the length encoder and the cut point accumulates as a position error that appears downstream as a gap in the folded plane. This single requirement is why 4680 notching machines are substantially more expensive than 21700 notching machines.

2.3 What teardowns actually reveal

Independent academic disassembly and characterisation of first-generation Tesla 4680 cells has produced the most reliable public description of the architecture. Key findings:

2.3.1 It is not fully tabless

Peer-reviewed teardown work reports that the Tesla 4680 architecture is **not fully tabless** — only the jelly-roll design is engineered in a tabless manner. Approximately 25 % of the cathode and anode edge is engaged in current collection. This is an important distinction when reading patent claims and when modelling internal resistance.

2.3.2 Measured internal geometry

Feature	Measured value
Current collector discs	Hexagonal symmetry
Cathode disc connection point	Joined at the outer ring
Anode disc connection point	Joined toward the centre
Anode outer ring	Connected to the cell housing
Cathode disc centre	Ultrasonically welded to the positive terminal
Weld geometry per connector	Five laser-welded zig-zag seams, approximately 0.5 mm wide
Insulating plastic disc	0.65 mm thick
Jelly roll height (tabs removed)	71 mm
Jelly roll diameter	44.5 mm
Mandrel	None — the winding needle is withdrawn
Can wall thickness	0.5 mm
Electrode winding	Double-sided, homogeneously coated cathode and anode, two separators
Cathode chemistry (confirmed by EDX, dV/dQ, dQ/dV, three-electrode)	NMC811
Anode chemistry	Pure graphite, no meaningful silicon
Anode fabrication	A solvent-free fabrication and coating process can be derived from the material evidence

2.3.3 The two discs act as springs

Because the cathode connector attaches at the outer ring and the anode connector attaches toward the centre, the two discs can act as **compensating elements — effectively springs — when the jelly roll is attached to the housing**. This absorbs the mechanical stress of insertion and the dimensional change of the roll during cycling.

This is a subtle and excellent piece of design. It is also a manufacturing constraint: the disc geometry must be produced to tight tolerance because its compliance is a functional property, not just a shape.

2.3.4 Coating thickness varies along the electrode

Measurement along the unwound electrodes shows a characteristic profile: electrode thickness is **highest at the core** where winding begins, then decreases significantly, and rises again beyond roughly one metre of electrode length. Areal loading, measured gravimetrically from electrode samples, remains *almost constant* along the entire length.

INTERPRETATION — THIS IS A COMPACTION GRADIENT, NOT A COATING DEFECT

Constant areal mass with varying thickness means constant coating and varying **density**. The inner wraps are less compacted because they experienced lower radial pressure during winding and because winding tension is highest at small radius. This tells you two things: your calendering was uniform (good), and your winding tension profile is imprinting a porosity gradient into the finished roll. That gradient is a real electrochemical inhomogeneity — the core and the outer wraps will not age identically.

2.4 Generation 1 versus Generation 2

Feature	Generation 1	Generation 2
Bottom lid	Crimped with a gasket	Welded to the cell can
Anode current collector connection	Via an intermediate tab	Welded directly to the bottom lid using 12 welding seams
Anode process	Dry	Dry
Cathode process	Wet slurry	Dry (reported at scale from January 2026)
Thermal path to base	Through gasket — poor	Metallic through weld — substantially better
Sealing method at base	Mechanical (crimp)	Metallurgical (laser weld)

The change from crimped to welded base is not cosmetic. It converts the bottom of the cell from a thermal insulator into a thermal conductor, which is what makes bottom cooling viable at all — and it removes an elastomeric seal whose long-term permeability was a lifetime limiter.

2.5 Thermal behaviour and cooling strategy

Experimental work on a state-of-the-art tabless 4680 with steel housing, validated against a 3D CFD model, produced two conclusions that directly drive pack design:

- **Side cooling outperforms bottom cooling** with respect to surface temperatures.
- **Tab design and housing material** both have measurable effect on thermal performance — the parameter study covers both.

Comparative analysis also indicates the 4680 generates roughly **2.3× more heat per unit volume at 1C** than a large flat prismatic LFP cell, which is why its cooling architecture is more demanding.

2.5.1 Housing material trade-off

Property	Nickel-plated steel (Hilumin)	Aluminium
Corrosion resistance	From nickel plating	Native oxide; generally good
Structural strength	High — handles internal pressure and mechanical stress	Lower; needs thicker wall for equivalent burst
Mass	Heavy — direct energy-density penalty	Roughly one third the density
Thermal conductivity	~50 W/m·K	~200 W/m·K — far better heat spreading
Forming route	Deep drawing	Impact extrusion
Module interconnect welding	Well understood	Aluminium-to-steel joining across a module is a genuine process challenge due to differing thermal properties and material behaviour during laser joining

Some 46 mm cells are built from different materials — aluminium and Hilumin — to meet specific design and performance requirements. The choice is not settled industry-wide.

2.6 Design trade-off map

A compact summary of the coupled decisions. Moving any one of these moves at least two others.

Decision	Push it this way...	...and you gain	...and you lose
Electrode thickness	Thicker	Higher active fraction, higher energy density, fewer metres of foil	Longer ion diffusion path, worse fast charge, higher risk of lithium plating
Calendered porosity	Lower	Higher volumetric energy density, better electronic contact	Slower electrolyte wetting, worse rate capability, higher cracking risk
Can wall thickness	Thicker	Structural pack capability, burst safety margin	Direct gravimetric energy density penalty
Silicon content in anode	Higher	Capacity uplift	Up to ~300 % volumetric expansion; mechanical failure of a tightly wound large roll; cycle-life loss
Tab-plane compaction	Flatter	Easier, more reliable welding	Closes capillary paths; electrolyte wetting time rises sharply
Formation rate	Faster	Less rack capital, less floor space, shorter cash cycle	Less uniform SEI, higher first-cycle loss, worse long-term cycle life
Winding tension	Higher	Tighter roll, better inter-layer contact, better fit in can	Compaction gradient core-to-edge; risk of foil yield or tear
Dry vs wet electrode	Dry	~90 % lower electrode-step CAPEX and energy; no NMP; ~40 % smaller footprint	Lower process maturity; harder inline inspection; fibrillation consistency is the new critical variable

3. The Dry Route: Principle, Economics and Line Architecture

3.6.1 The principle

Dry coating is a family of methods based on dry processing of materials to manufacture an active material layer. The absence of solvents is the significant difference from conventional mixing. Conventional wet coating requires large ovens and NMP solvent recovery, both energy-intensive. The dry process mixes PTFE binder with active material and then calenders the powder directly onto foil.

The sequence, as described in industrial literature:

1. The active material is **homogenised and preconditioned** in a mixing process.
2. The active material is fed to a **pair of rolls** as powder or granules.
3. The rolls rotate in opposite directions with **optional differential circumferential speed**.
4. A dry film consisting of active material **forms on the faster-rotating roll**.
5. Depending on system type, the dry film is either generated first and then applied to the substrate foil, or **calendered directly onto it**.
6. Before the dry film is applied, it is usually cut to the desired width.

3.6.2 The commercial case

Claimed benefit	Reported figure	Equipment consequence
Electrode-step capital expenditure	~90 % reduction (Tesla figures); ~50 % reduction (other source)	The 100 m dryer and the solvent recovery plant — the two largest items in the electrode block — are deleted
Energy consumption	~90 % reduction for the electrode step	Drying is the single largest thermal load on a wet electrode line
Factory footprint	~40 % reduction	Layout is dominated by the drying oven length in the wet route
Cell production emissions	~70 % reduction	NMP is toxic and expensive to recover; removing it removes the recovery and abatement train
Binder content	Below 2 wt%, reported as low as 1.25 wt%	Higher active fraction — up to 98 % active material content is claimed
Cycle life	~90 % capacity retention after 2,000 cycles (patent claim)	Attributed to improved particle bonding reducing cracking over many cycles

3.6.3 Where the technology actually stands

Dimension	Status
Technology readiness level	TRL 6-7 — demonstration of the system prototype in the operating environment
Automotive-scale production	One manufacturer. Tesla reported large-scale deployment of dry electrode for <i>both</i> anode and cathode from January 2026; prior to that the 4680 was a hybrid — dry anode, wet-slurry cathode
Equipment availability	Several equipment suppliers now offer production-scale dry coating lines. Named in public sources: Wuxi Lead, Manz
Named remaining challenges	Increasing web speed; adhesion of the layers; adjustment of porosity. Binder fibrillation consistency and coating uniformity at high line speed (above ~50 m/min) remain the main challenge
Programme risk precedent	Tesla's leadership has publicly acknowledged that pursuing the dry battery electrode process for the 4680 from the outset was a mistake that delayed the programme

3.6.4 Qualitative comparison against the wet route

Criterion	Dry process	Equipment implication
Space requirement	Better	Drying oven and solvent plant deleted
Operating cost	Better	No solvent purchase, no recovery, far lower thermal energy
Investment cost	Better	Fewer, smaller machines in the electrode block
Power demand	Better	Drying is the dominant electrode-step load
Environment	Better	No NMP; no solvent abatement train
Performance	Comparable / unclear at scale	Claims of thicker, denser electrodes; limited independent confirmation
Quality and uniformity	Currently worse	This is the honest state of the art and it drives the metrology and acceptance sections of this handbook

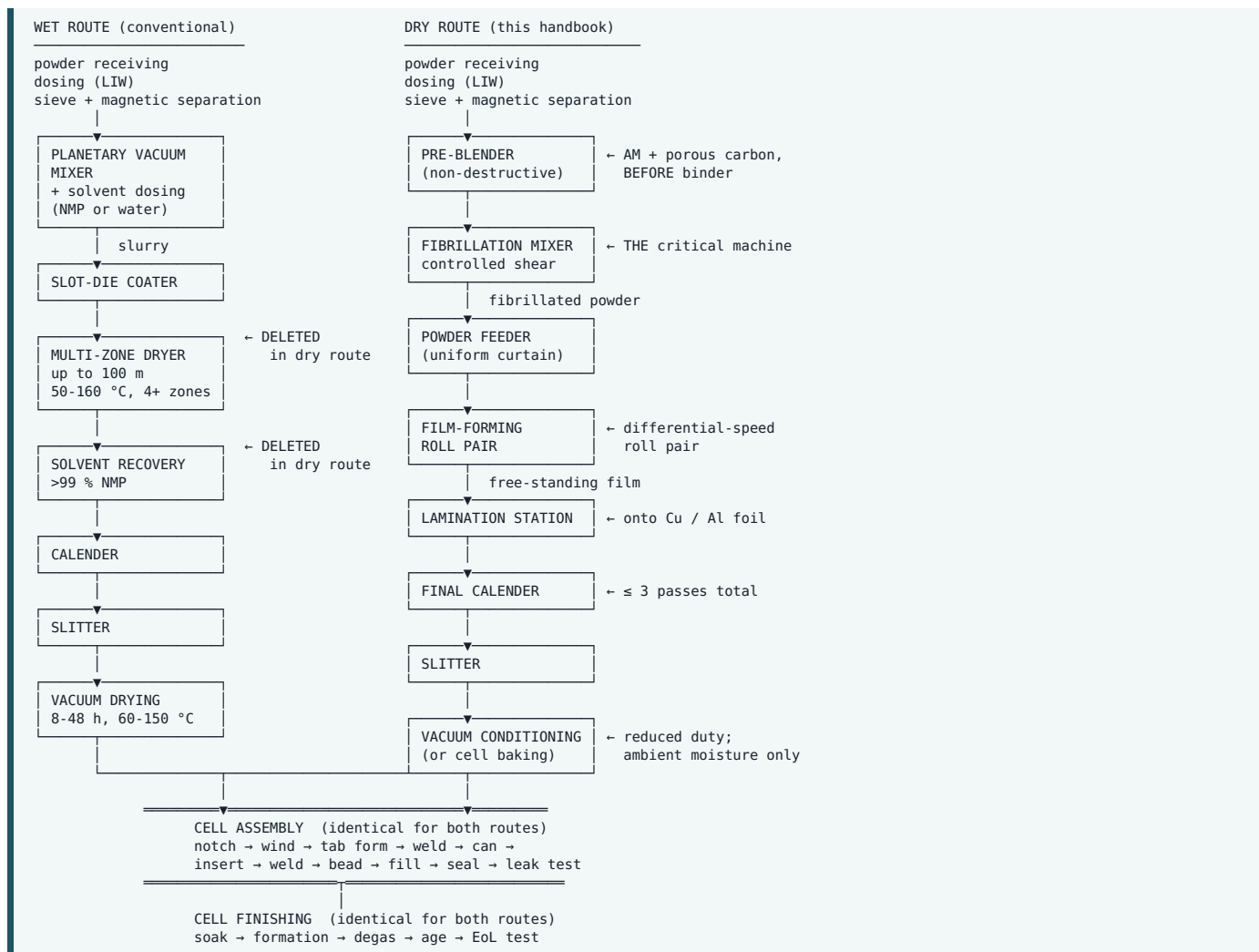
THE PROCUREMENT CONSEQUENCE OF TRL 6-7

A mature machine class has multiple suppliers, published performance data, transferable process windows and a service network. Dry electrode film-forming equipment has none of these to the degree that a slot-die coater does.

Practically, this means: **a longer and more technical qualification phase, process guarantees written into the contract rather than assumed, a pilot or demonstration line before the production order, and far more of your own engineering effort spent on the fibrillation mixer than the price would suggest.** Part 7 builds the procurement approach around this.

3.6.5 Line architecture

3.5.1 Side-by-side block diagram



3.5.2 What the diagram shows

THE DRY ROUTE IS NOT A MODIFICATION OF THE WET ROUTE — IT IS A DIFFERENT BLOCK

Two large machines are deleted outright. Four new machines appear. The mixer is replaced by a different machine doing a different job with different physics. Only calendaring, slitting and everything downstream carry over. Anyone treating a dry-electrode project as “the same line with a different coater” will mis-plan the layout, the utilities and the engineering effort.

3.6.6 The machine park in detail

3.6.1 Eliminated

Machine	Wet-route CAPEX band	What its removal means for the plant
Multi-zone convection dryer (up to 100 m, 4+ temperature zones, 50-160 °C, air 5-15 m/s)	Part of EUR 30-57 m (coating and drying) for a 10 GWh/y line	Removes the single longest machine in the building. Layout is no longer dictated by a 100 m straight run. Removes the largest thermal load and the largest exhaust volume. Removes the entire class of drying defects — skin formation, cracking, binder migration
Solvent recovery plant (>99 % NMP recovery, condensation or adsorption)		Removes a chemical process plant from a manufacturing site, with its permits, its inventory, its maintenance discipline and its emissions monitoring
Slot-die coater (pump, filter, manifold, shims, lips, vacuum box, backing roll)	Part of the same band	Removes the highest-precision fluid-handling machine on the line and its associated rheology control burden
Slurry infrastructure (buffer tanks, transfer pumps, filtration skids, CIP)	Included above	Removes wet cleaning, solvent-compatible piping and the cross-contamination risk that comes with liquid transfer

A SECOND-ORDER EFFECT WORTH PLANNING FOR

Deleting the dryer removes the process step that *also* served as a de facto residual-moisture control. In the wet route, the coating leaves the oven bone-dry. In the dry route, the electrode has never been heated, so it carries whatever moisture it picked up from the powder and the room. This is why DE-220 (vacuum conditioning) survives in reduced form and why powder storage humidity control becomes more important, not less.

3.6.2 Added

Machine	Maturity	Why it is new and what makes it hard
DE-130 Pre-blender	Mature hardware, novel duty	Performs the non-destructive mixing of active material with porous carbon <i>before</i> the binder is added. Standard blending hardware, but the duty is defined by a patent claim about sequence
DE-140 Fibrillation mixer	Low — the critical machine	Must deliver a controlled, uniform specific shear energy at controlled temperature to fibrillate PTFE. Under-fibrillation gives a film that tears; over-fibrillation blocks lithium ion paths. Neither is visible in a mass or thickness measurement
DE-160 Powder feeder	Moderate	Must deliver a uniform powder curtain across the full web width. Cross-direction feed uniformity becomes cross-direction film uniformity directly — there is no fluid to average it out
DE-170 Film-forming roll pair	Moderate to low	Differential circumferential speed, temperature-controlled rolls, precise line pressure. The film forms on the faster roll; the speed ratio determines whether it does so cleanly
DE-190 Lamination station	Moderate	Bonds the free film to foil with <i>purely mechanical</i> adhesion — no solvent, no chemical anchoring. Foil surface topography and lamination pressure/temperature carry the whole bond

3.6.3 Modified in duty

Machine	How the duty changes in the dry route
Calender (DE-200)	Now part of a constrained pass budget — self-supporting films are claimed in three or fewer calender passes including the film-forming pass. Porosity must be <i>engineered</i> by roll pressure rather than inherited from solvent evaporation
Vacuum dryer (DE-220)	Duty reduced from solvent-and-moisture removal to ambient-moisture removal only. This is the strongest candidate for conversion to the <i>cell baking</i> approach — shifting the operation to immediately before electrolyte filling to reduce chamber size
Slitter (DE-210)	Unchanged in principle, but the dry electrode edge behaves differently: no binder film to hold the edge together, so dust generation and edge friability need re-qualification
Dust extraction (FC-series)	Substantially increased duty. The entire electrode process now handles dry powder rather than wet slurry, raising both contamination and combustible-dust risk

3.6.4 Unchanged

Everything from laser notching onward is common to both routes: notching, winding, tab forming, collector welding, can forming, insertion, can and terminal welding, beading, electrolyte filling, sealing, leak test, soaking, formation, degassing, aging, end-of-line test and grading. These are catalogued in Part 4 because a dry-electrode plant still needs all of them.

3.6.5 Net equipment count

Block	Wet route	Dry route
Material preparation	4 machine classes	4 machine classes
Mixing	1 (planetary vacuum mixer)	2 (pre-blender + fibrillation mixer)
Film / coating formation	3 (coater + dryer + solvent recovery)	3 (feeder + roll pair + lamination)
Compaction and separation	2 (calender + slitter)	2 (final calender + slitter)
Conditioning	1 (vacuum dryer, full duty)	1 (vacuum conditioning, reduced duty)
Machine classes in electrode block	11	12
Floor area of electrode block	Baseline	~60 % of baseline
Capital of electrode block	Baseline	~10-50 % of baseline depending on source

READ THIS TABLE CAREFULLY

The dry route has *slightly more* machine classes but far less floor area and far less capital. The savings do not come from having fewer machines — they come from the fact that a 100 m thermal oven and a solvent recovery plant are enormously expensive objects, and a roll pair is not. Do not expect a simpler line. Expect a smaller, cheaper, and in some respects *harder to control* line.

4. Materials and Incoming Quality Control

Everything downstream inherits the properties of what arrives on the loading dock. A defect introduced at incoming inspection cannot be removed by any subsequent process step — a point the industry states plainly about particle contamination in coating, which no later cleaning method can remove.

4.1 Cathode active material

4.1.1 Chemistries in the 46-series

Chemistry	Used by	Characteristics
NMC811	Tesla 4680 Gen 1 — confirmed by EDX, differential voltage, incremental capacity and three-electrode analysis	~80 % Ni, 10 % Mn, 10 % Co of the transition-metal fraction. High capacity, moderate cost, moisture-sensitive
NCMA (quaternary, high-Ni)	LG Energy Solution 46-series	Aluminium doping improves structural and thermal stability at high nickel content
Mid-nickel “Pure NCM”	LG (award-recognised technology)	Lower cost, better stability, lower capacity
Manganese-rich, doped	Tesla patent direction (WO 2024/147993)	Manganese is abundant and affordable but suffers manganese dissolution; specific dopants lock the manganese structure in place
LFP	BYD large-format development	Thermal stability, long cycle life, lower cost, lower energy density

4.1.2 Incoming specification and test methods

Parameter	Typical window	Instrument / method	Why it matters
Particle size D50	5–15 µm	Laser diffraction (wet dispersion)	Sets packing density and rate capability
Distribution width D90/D10	Ratio < 5	Laser diffraction	Broad distributions cause inhomogeneous compaction in calendaring
BET specific surface area	0.2–0.6 m ² /g	Nitrogen physisorption	High BET means high side-reaction area and high binder demand
Moisture, as received	< 500 ppm	Karl Fischer coulometric titration with oven sampler	Water reacts with LiPF ₆ to form HF
Residual lithium (LiOH, Li ₂ CO ₃)	< 0.5 wt%	Acid–base titration (Warder method)	Residual base gels PVDF slurry and causes gassing
Magnetic impurity (Fe, Ni, Cr, Zn)	< 50 ppb	Magnetic extraction followed by ICP-MS	The single largest root cause of internal short circuit
Tap density	2.0–2.6 g/cm ³	Tap density tester (fixed tap count)	Predicts achievable calendered density
pH of aqueous slurry	Chemistry-dependent	pH meter, fixed solid-to-water ratio	Correlates with residual lithium
Crystal structure / phase purity	No secondary phases	XRD	Detects incomplete calcination at the supplier
Morphology	Intact secondary particles	SEM	Cracked secondary particles indicate over-milling; they crack further under calendaring

MAGNETIC IMPURITY IS THE ONE TO BE UNCOMPROMISING ABOUT

A metallic particle in the cathode slurry will eventually penetrate the separator and create an internal short. It may do so during formation, or during aging, or eighteen months into field service. Two-stage magnetic separation with field strength above 10,000 gauss on the slurry line is standard practice and is not optional. Verify supplier certificates by independent ICP-MS on incoming lots — do not accept them on trust.

4.2 Anode active material

4.2.1 The silicon question

Battery Day promised a silicon anode. Teardown analysis of production cells found **pure graphite with no meaningful silicon content**. Differential capacity analysis of Model Y 4680 cells confirms an NMC–graphite chemistry with no meaningful silicon in the negative electrode — in contrast to cells such as the Samsung 50G, which shows some silicon, or the Molicel P45B, which uses significantly more.

The absence of silicon means the 4680 **trades some peak energy density for better cycle stability**, as silicon-containing anodes are more prone to degradation mechanisms driven by volume change. LG, by contrast, has announced high-efficiency silicon in combination with graphite for its 46-series.

4.2.2 Incoming specification

Parameter	Typical window	Method / consequence
Particle size D50	10–20 µm	Laser diffraction. Larger particles reduce first-cycle loss but hurt rate capability
BET surface area	1–4 m ² /g	Nitrogen physisorption. Directly proportional to SEI-forming lithium consumption
Specific capacity	340–360 mAh/g	Half-cell test at C/20
First-cycle coulombic efficiency	> 92 %	Half-cell test. Every lost percent is permanently lost cell capacity
Moisture, final	< 20 ppm	Karl Fischer. Tighter than cathode because graphite is more hygroscopic per unit area
Ash content	< 0.05 %	Combustion
Tap density	0.9–1.1 g/cm ³	Tap density tester
Magnetic impurity	< 50 ppb	Same as cathode — same consequence

4.3 Binder systems — where wet and dry diverge

Route	Cathode binder	Anode binder
Wet (conventional)	PVDF dissolved in NMP, ~2–4 wt%	CMC + SBR in water, roughly 2 wt% + 2 wt%
Dry (Tesla / Maxwell)	PTFE, below 2 wt%, reported as low as 1.25 wt%	PTFE

Some published analysis attributes a PAA + PEO binder system to the Tesla 4680 specifically, versus CMC + SBR in competing designs. Binder identification from teardown is inherently indirect, so treat specific attributions with caution.

4.3.1 PTFE fibrillation — the mechanism that makes dry electrodes possible

Under controlled shear, PTFE particles are drawn into nanoscale fibrils — a web-like network that mechanically stitches active material particles together without any solvent. This is the entire basis of the dry process. It is treated in depth in Chapter 8; what matters here is the material implication:

THE BINDER WINDOW IS NARROW IN BOTH DIRECTIONS

Too little binder and the film lacks mechanical integrity — it cannot survive cell assembly or cycling. Too much binder and it blocks the paths lithium ions need to travel, reducing performance and raising internal resistance. There is no safe direction to err. This is why fibrillation control, not binder quantity alone, is the critical variable.

4.4 Electrolyte

Component	Typical	Function
Salt	LiPF ₆ , 1.0–1.2 M	Ion transport. Hydrolyses in the presence of water
Cyclic carbonate	EC (ethylene carbonate)	High dielectric constant; dissolves the salt; participates in SEI
Linear carbonates	EMC / DMC / DEC blends	Lower viscosity; improves low-temperature conductivity
SEI former	VC (vinylene carbonate)	Forms a more stable, more elastic SEI
Silicon-compatible additive	FEC (fluoroethylene carbonate)	Essential where silicon is present; largely unnecessary for pure graphite
Salt stabiliser	LiPO ₂ F ₂	Reduces impedance growth
Life-extension additive	Dioxazalone-type compounds	Tesla has filed on a method for more convenient preparation of 3-substituted-1,4,2-dioxazol-5-one compounds, described as an electrolyte additive that significantly extends lithium-ion battery life during long cycling

4.4.1 Electrolyte incoming control

Parameter	Limit	Method and consequence
Water content	< 20 ppm	Karl Fischer. Above this, HF generation accelerates
Free HF	< 50 ppm	Titration. HF attacks the cathode and dissolves transition metals
Ionic conductivity at 25 °C	8–12 mS/cm	Conductivity cell
Colour	Clear and colourless	Visual. Yellowing indicates decomposition — reject the lot
Density	Per specification ±0.5 %	Oscillating U-tube densitometer. Cross-check on formulation
Metallic impurities	< 1 ppm each	ICP-MS

WHY THE ENTIRE DRY ROOM EXISTS

Water inside the cell leads to severe quality loss over lifetime *and* a safety risk through the formation of hydrofluoric acid. HF corrodes the cathode, dissolves transition metals, and poisons the SEI. Every square metre of dry room, every kilowatt of desiccant regeneration and every dew-point sensor in the building exists to prevent that one reaction. When someone proposes relaxing the dew point to save energy, this is the trade being made.

4.5 Separator

Property	Typical	Consequence
Base material	PE, or PP/PE/PP trilayer	Trilayer gives a two-stage shutdown behaviour
Thickness	9–16 µm	Thinner raises energy density and puncture risk simultaneously
Ceramic coating	Al ₂ O ₃ or boehmite, one or both sides	Improves puncture resistance and thermal stability. Directly mitigates burr-induced shorts
Porosity	40–45 %	Ion transport vs mechanical strength
Gurley number	150–350 s/100 cc	Air permeability proxy for ionic resistance
Shutdown temperature	~130 °C	Pores close, ion transport stops
Meltdown temperature	> 160 °C with ceramic	Mechanical integrity retained above shutdown
Thermal shrinkage, 120 °C / 1 h	< 5 %	Shrinkage exposes electrode edges — direct short path
Puncture strength	> 300 gf	The direct defence against burrs and particles
MD/TD tensile strength	> 100 MPa MD	Survives winding tension without necking

Separator technology is significant enough commercially that it appears in patent litigation: of the five US patents asserted by LG Energy Solution against EVE Energy in 2026, four cover cylindrical cell construction including tabless designs, and **one covers separator technology**.

4.6 Current collector foils

Property	Cathode (aluminium)	Anode (copper)
Alloy / type	1060 / 1235 series	Electrolytic (ED) or rolled-annealed
Thickness	10–15 µm	6–8 µm
Tensile strength	> 150 MPa	> 250 MPa
Elongation at break	> 2 %	> 4 %
Surface roughness Ra	0.1–0.5 µm	0.2–0.6 µm
Camber over 10 m	< 2 mm	< 2 mm
Pinhole count	Zero at specified detection limit	Zero
Surface cleanliness	No rolling oil residue — kills adhesion	No oxide bloom — kills weldability

THE FOIL THICKNESS TRADE-OFF SHOWS UP AS A TENSION SETTING

Thinner foil means higher energy density and higher risk of web break at line speed. Surface roughness is equally consequential in both directions: too smooth and coating adhesion fails; too rough and thin-foil pinholes appear. Camber — lateral curvature of the coil — is the parameter most often overlooked at incoming inspection and the one that most reliably defeats an otherwise well-tuned edge position control system.

4.7 Can and cap components

Component	Specification	Notes
Can material	Nickel-plated steel (Hilumin), or aluminium	Nickel gives corrosion resistance, steel gives strength to handle internal pressure and mechanical stress
Can wall thickness	~0.5 mm, tolerance ±20 µm	Measured ultrasonically or by micrometer on sectioned samples
Roundness	< 50 µm	Out-of-round cans jam at jelly-roll insertion
Forming route	Deep drawing (steel) or impact extrusion (aluminium)	Multi-stage; 3–6 draws for an 80/46 aspect ratio
Internal cleanliness	No drawing lubricant residue	Residue contaminates electrolyte and defeats welding
Cap assembly	Includes an etched vent score	The explosion-proof line is etched on the cover plate; the cover plate connects to a notch in the housing
Vent score depth	Tolerance in the tens of microns	Too shallow: no vent at design pressure. Too deep: premature leak. Verified by profilometry and sampled burst test
Gasket (Gen 1 crimped design)	PP or PFA	Compression set is the long-term failure mode

4.8 Building an incoming inspection plan

A practical structure that scales from pilot to production:

Tier	Applies to	Content
Tier 1 — every lot	All materials	Certificate of analysis review, identity confirmation, moisture (Karl Fischer), visual inspection, quantity and packaging integrity
Tier 2 — every lot, safety-critical	Cathode, anode, electrolyte	Magnetic impurity by ICP-MS, particle size by laser diffraction, free HF for electrolyte
Tier 3 — skip-lot	Active materials	BET, tap density, XRD, SEM morphology — on a reduced frequency once a supplier is qualified
Tier 4 — qualification only	New supplier or new grade	Full electrochemical half-cell characterisation, rate capability, first-cycle efficiency, and a full-cell build
Retain samples	All lots	Sealed, dated retains for every lot, stored to shelf life plus one year. When a field failure appears eighteen months later, this is the only evidence that exists

PRACTICAL NOTE ON RETAINS

Retain samples cost almost nothing and are the difference between a targeted recall and a total one. Store them under the same environmental conditions as production material, with the lot number, receipt date, supplier certificate and the identity of every production coil that drew from that lot. Link the retain identifier into the MES genealogy described in Chapter 13.

Part II — Subsystem Building Blocks

These components recur across many machines. Specifying them once, properly, prevents the same argument at every station. Chapter 5 gives the model used to read every machine in Parts III and IV.

5. How to Read Any Machine: The Six-Subsystem Model

Every machine in this catalogue decomposes into the same six subsystems. Learning to identify them turns an unfamiliar machine into a familiar one within an hour, and it structures both the RFQ and the fault-finding.

Subsystem	What it does	Where it fails
1. Material transport	Moves product through the machine: web path, powder conveying, conveyor, indexing table, robot, carrier	Tension excursions, tracking drift, powder bridging, jams, carrier wear
2. Process actuator	Does the physical work: roll pair, mixer tool, laser head, sonotrode, press, pump, heater	Wear, thermal drift, contamination, alignment loss
3. Metrology	Measures the result: gauges, cameras, load cells, thermocouples	Calibration drift, fouling, cabling noise, wrong installation position
4. Control	Closes the loop: PLC, motion controller, PID, recipe manager	Loop tuning, sample rate too slow, deadband too wide
5. Environment and utility	Supplies and conditions: vacuum, dry air, inert gas, chilled water, extraction, filtration	Usually the actual root cause when a “process problem” is intermittent
6. Safety and interlock	Guards, light curtains, laser enclosure, e-stop chain, dust explosion protection, gas detection	Nuisance trips that operators then defeat — the classic path to an incident

USE THIS AS A DIAGNOSTIC ORDER, NOT JUST A DESCRIPTION

When a station starts producing defects, walk the six subsystems in order before touching a recipe parameter. Subsystem 5 — utilities and environment — is the root cause of a disproportionate share of intermittent problems and is the one nobody checks first. A chilled-water temperature swing, a compressed-air dew-point excursion or a partially blocked extraction duct looks exactly like a process drift.

5.1 Use it to structure the RFQ too

A specification organised by these six subsystems is harder for a supplier to answer vaguely. For each subsystem, ask: what is supplied, what is the performance, what is the interface, what is excluded. Most disputes at acceptance trace to subsystem 3 (metrology — whose gauge, calibrated how) and subsystem 5 (utilities — who supplies what to which specification at which boundary).

6. Powder Handling and Dosing Hardware

In the wet route, powder handling ends at the mixer inlet and everything downstream is a liquid. In the dry route, **the product is a powder all the way to the roll pair**. Powder handling is therefore not a peripheral utility; it is a core part of the process train, and its shortcomings appear directly in the finished electrode.

6.1 Component catalogue

Component	Function	Specification points
Big-bag (FIBC) discharge station	Contained emptying of bulk bags	Integral dust extraction; bag massagers for bridging; glovebox or split-butterfly valve where containment class demands it; load cells for inventory
Drum tipper / sack tip station	Smaller-quantity additions (binder, conductive carbon)	Downflow booth or extracted hood; operator exposure control
Loss-in-weight (LIW) feeder	Continuously weighs the hopper and controls screw speed to hold a mass flow rate	Accuracy typically better than gain-in-weight for small additions. Specify accuracy as a percentage of <i>setpoint</i> , not of full scale
Twin-screw volumetric feeder	Handles cohesive, poor-flowing powders	Screw geometry must match the powder. Get the supplier to trial <i>your</i> powder, not a surrogate
Vibratory / sieve feeder	Gentle handling of friable material	Low attrition — important for brittle cathode powder
Sieve / screen	Removes lumps and oversize	Mesh in the 100–200 µm band typical. Ultrasonic debinding for fine cohesive powders
Magnetic separator	Removes ferrous and paramagnetic contamination	At least two stages, field strength above 10,000 gauss. Rare-earth grid or rotary drum. Non-negotiable — see the warning below
Dense-phase pneumatic conveying	Transfers powder between stations	Preferred over dilute phase: lower velocity means less particle attrition , which matters because the cathode powder is brittle
Mechanical conveying (screw, aero-mechanical)	Short transfers	Lower attrition still; harder to clean between chemistries
Intermediate bulk containers (IBC)	Batch transfer between blending and fibrillation	Dedicated per chemistry. Tracked by ID for genealogy
Nitrogen inerting system	Blankets vessels and conveying lines	Both a moisture control and a dust-explosion control. Oxygen concentration monitored, not assumed

MAGNETIC SEPARATION IS A SAFETY SYSTEM, NOT A QUALITY SYSTEM

Metallic particle contamination — iron, nickel, copper — is one of the three primary causes of internal short circuit, because the particle penetrates the separator. In the dry route there is no slurry filtration step to act as a second line of defence, so the magnetic separators *are* the defence. Specify at least two stages above 10,000 gauss, schedule cleaning by interval rather than on demand, and verify supplier certificates by independent ICP-MS on incoming lots rather than accepting them on trust.

6.2 Powder handling problems specific to the dry route

Problem	Mechanism	Hardware answer
Segregation	Three components of different density and particle size separate during conveying, storage and discharge	Minimise free-fall height; use mass-flow hopper geometry; blend as close to use as possible; avoid intermediate storage of blended powder
Attrition of brittle cathode	The cathode powder is brittle and abrasive; conveying fractures secondary particles and changes the size distribution	Dense-phase or mechanical conveying at low velocity; minimise transfer count; monitor D50 before and after transfer during qualification
Abrasive wear	The same abrasiveness wears screws, bends and valve seats — and the wear debris becomes metallic contamination	Hardened or ceramic-lined wear parts at every impact point; wear inspection on a schedule; magnetic separation <i>after</i> conveying, not only before
Moisture pickup	No drying oven downstream to remove it	Humidity-controlled storage below 20 % RH; nitrogen blanketing; enforce a maximum queue time between blending and film forming
Electrostatic charging	Dry powder in motion charges; charged powder adheres to surfaces and clumps	Full bonding and earthing of all metallic components; conductive or dissipative flexible connections; ionisation where permitted
Bridging and rat-holing	Cohesive fine powders arch over outlets	Mass-flow hopper design from measured flow properties, not from a standard drawing. Get a shear-cell characterisation of your powder before the hopper is designed

7. Shear and Fibrillation Equipment

This is the technical heart of the dry route and the section to read most carefully.

7.1 What fibrillation is, in equipment terms

Under controlled shear, PTFE particles draw out into nanoscale fibrils forming a three-dimensional web that mechanically entangles the active material particles. There is no solvent, no dissolution and no drying — the film's integrity is purely mechanical. The machine's job is to deliver a **specific shear energy**, at a **controlled temperature**, with a **narrow residence time distribution**.

WHY THIS IS THE HARDEST MACHINE SPECIFICATION ON THE LINE

Fibrillation is a function of accumulated shear energy, temperature and residence time. All three vary spatially inside any real mixer. The consequence: under-fibrillated and over-fibrillated material can coexist in the same batch.

Under-fibrillated — the film lacks mechanical integrity, tears at the roll and cannot survive assembly.

Over-fibrillated — the binder network becomes dense enough to block the paths lithium ions need to travel, raising internal resistance and reducing performance.

Neither condition is visible in a mass measurement or a thickness measurement. **Binder fibrillation consistency is the named principal challenge of the dry process**, alongside coating uniformity at high line speed. A mixer that produces a narrow residence time distribution is therefore worth a substantial premium over one that does not.

7.2 Candidate machine architectures

Architecture	Shear mechanism	Residence time distribution	Assessment for fibrillation duty
High-shear blade mixer (plough, ribbon with chopper)	Impeller tip speed against the bulk	Broad (batch)	Widely available. Shear is highly non-uniform — material near the chopper sees far more energy than material at the wall. Needs long cycles to homogenise, which risks over-working part of the batch
Intensive / cyclone mixer	High-speed rotor in a conditioned chamber, often with wall scraper and jacket	Moderate	Better temperature control and better bulk turnover than a simple blade mixer. Commonly cited as an intensive mixing option
Twin-screw extruder / compounder	Configurable screw elements deliver defined shear along a defined path	Narrow (continuous)	The strongest candidate on residence-time grounds. Every particle follows a similar path and receives a similar energy. Continuous mixing by twin-screw extruder is a recognised industry direction. Screw configuration becomes a process parameter
Jet mill / fluidised bed opposed jet	Particle-particle collision in high-velocity gas streams	Narrow	Very uniform energy input and no contact wear parts, so no metallic contamination from tooling. High gas consumption; needs careful separation downstream
Mechanofusion / dry particle coating	Compression and shear in a narrow gap between rotor and wall	Moderate	Well suited to producing composite particles; used in dry particle-coating applications

THE SPECIFICATION QUESTION THAT MATTERS MORE THAN ANY OTHER

Ask every supplier: “**What is the residence time distribution of my material in your machine, and how did you measure it?**” A supplier who answers with a tracer study and a distribution curve understands the problem. A supplier who answers with a mixing time understands a different problem. This single question separates the candidates faster than any other.

7.3 Specification points for a fibrillation mixer

Parameter	Why it matters	How to specify it
Specific shear energy (kWh/kg or J/kg)	The primary process variable	Demand it as a <i>measured and recorded</i> quantity per batch, computed from net motor power integrated over time and divided by mass. Not as a proxy like “mixing time”
Residence time distribution	Determines whether the batch is uniformly fibrillated	Require a tracer study on your material at your throughput. Specify a maximum acceptable spread
Temperature control	PTFE fibrillates poorly cold and degrades hot; shear input generates heat continuously	Jacketed with active chilled-water control; specify the maximum permissible temperature rise across the process, not just a setpoint
Product temperature measurement	Jacket temperature is not product temperature	Require in-product temperature measurement, not only wall RTDs
Net power measurement	Enables shear energy calculation and batch fingerprinting	Torque or net motor power logged at high sample rate, with a no-load tare
Contact material and wear	Abrasive cathode powder wears tooling; wear debris is contamination	Hardened or ceramic contact surfaces; wear-part inspection interval; magnetic separation downstream of the mixer
Atmosphere	Moisture and oxygen control; dust explosion control	Nitrogen inerting with oxygen monitoring
Cleanability and changeover	Cross-contamination between anode and cathode chemistries	Separate machines for anode and cathode is standard practice for wet mixers and applies with equal force here. If a shared machine is unavoidable, demand a validated cleaning procedure with a residue acceptance limit
Scale-up basis	Pilot results must predict	Require the supplier to state the scale-up rule they use (constant tip speed,

7.4 The pre-blender and why it is a separate machine

A SEQUENCE CLAIM, NOT A COMPOSITION CLAIM

Published patent material describes **non-destructive mixing of the active material with porous carbon *before* adding a single PTFE binder**. The engineering insight: once PTFE is present and shear is applied, fibrillation begins. Any mixing needed to disperse the conductive carbon must therefore be completed *first*, or the carbon dispersion step will simultaneously over-fibrillate the binder.

Separating the two operations decouples two variables that otherwise fight each other. In equipment terms this means **two machines, not one**: a low-shear, non-destructive blender (DE-130) followed by a controlled high-shear fibrillation machine (DE-140). A single-machine solution that does both is combining two incompatible duties.

8. Roll Technology, Fragile-Web Handling and Lamination

8.1 The differential-speed roll pair

The defining machine of the dry route. Powder is fed to a pair of counter-rotating rolls with **optional differential circumferential speed**. A dry film forms on the **faster-rotating roll**.



8.2 Why the speed ratio exists

THE MECHANISM TO UNDERSTAND BEFORE SPECIFYING

Two identical counter-rotating rolls compact the powder symmetrically, and the resulting film has no preference for either roll — it splits, sticks to both, and tears. Introducing a speed differential breaks the symmetry: the faster surface imparts more shear and more adhesion, so the film consolidates onto it and can be taken off cleanly as a coherent web. **The speed ratio is therefore not a fine-tuning parameter; it is what makes film take-off possible at all.**

8.3 Roll pair specification points

Element	Function	Specification points
Roll bodies	Compact the powder into a coherent film	Hardened forged steel, ground and polished. Surface finish transfers directly to the film. Specify surface roughness explicitly — it affects both film release and film surface quality
Independent drives	Enable the speed ratio	Each roll individually servo-driven with its own encoder. A geared pair with a fixed ratio removes your most important process variable
Speed ratio control	Sets which roll the film forms on and how strongly	Continuously adjustable, recorded per batch. Typical band 1.2:1 to 2:1
Line pressure / loading	Sets film density and cohesion	Hydraulic or servo-electric. Reference range for battery calendering extends to 2,500 N/mm . Measure with load cells on the bearing housings — specify measured, not commanded
Gap axis	Alternative control mode to load	Micron-resolution servo with linear scale feedback
Roll crown / deflection compensation	Compensates roll bending under load	A straight roll deflects under high line load, giving a thicker centre and thinner edges. Fixed crown works at one load only. Adjustable crown compensation is what separates a production machine from a laboratory one
Roll temperature control	PTFE behaviour is temperature-dependent	Rotary union feeding internal channels; reference range for tempered rolls approximately 50–250 °C . Independent control per roll is preferable
Doctor blade / take-off	Removes the film from the fast roll	Blade material, angle and load. A wear item. Poor take-off is a common cause of film defects
Roll cleaning	Prevents build-up printing a repeating defect	Continuous compact roll cleaners are a distinct product category. Essential with tacky fibrillated material
Edge containment	Prevents powder escaping at the roll ends	Side dams or profiled roll ends. Determines edge quality of the free film

DIAGNOSTIC WORTH BUILDING INTO THE OPERATING CULTURE

If a defect recurs at a fixed interval along the film, measure the interval and compare it to the circumference of every roll in the path. A match identifies the offending roll immediately. This one technique resolves a large share of film and calendering defect investigations in minutes.

8.4 The three-pass constraint

Published patent material claims self-supporting films achieved in **three or fewer calender passes**. This is a real design constraint on the machine train, not a marketing figure:

Pass	Machine	Purpose
Pass 1	DE-170 film-forming roll pair	Consolidate powder into a coherent self-supporting film
Pass 2	DE-190 lamination station	Bond film to foil; some densification occurs
Pass 3	DE-200 final calender	Set final thickness and porosity

Every additional pass adds cost, adds a defect opportunity and works the binder network further. If a supplier's proposal requires five passes to reach specification, that is a signal about the maturity of their film-forming step.

8.5 Handling a free-standing film: fragile-web hardware

Between DE-170 and DE-190 the product is an unsupported film held together only by a fibril network. This is the most fragile web in the entire plant. Standard web-handling hardware is not adequate without modification.

8.5.1 What is different

Property	Coated foil web (wet route)	Free-standing dry film
Tensile strength	Carried by the metal foil — 150-250 MPa	Carried by the fibril network only — orders of magnitude lower
Tension window	50-200 N/m	Very low; must be established experimentally and held tightly
Failure mode	Stretch, then tear	Immediate tear with little warning; no plastic region to warn you
Handling damage	Coating flake-off at rollers	Film disintegration at any stress concentration
Recoverability after a break	Splice and continue	Often requires clearing the whole path

8.5.2 Hardware requirements

Component	Requirement for free film
Web path length	As short as possible. The strongest argument for a machine where film forming, trimming and lamination are one integrated unit rather than three coupled machines
Tension control	Load-cell closed loop with a very low setpoint and high resolution. Standard load cells sized for foil webs will be operating at the bottom of their range — specify a load cell sized for the actual tension
Idler rollers	Minimum inertia, minimum count, minimum wrap angle. Every roller is a tension riser and a contact point
Support	Consider air-support (floatation) or vacuum-belt conveyance instead of free spans
Spreader rollers	Bowed rollers apply lateral stress — may be unusable on a fragile film. Verify with the supplier
Edge guiding	Non-contact edge sensing; steering guides with gentle correction rates. Aggressive EPC correction can tear the film
Trimming	The film is usually cut to the desired width before application. Non-contact (laser) or very sharp rotary knives; edge quality of a free film is more fragile than a supported one
Static control	Ionising bars. A charged fragile film clings to surfaces and tears
Break detection and safe stop	Immediate detection with a controlled stop sequence that does not pile film into the nip
Access for clearing	Design for it. Film breaks will happen during ramp-up; a machine that takes four hours to clear will destroy your learning rate

THE ARCHITECTURE DECISION THIS DRIVES

The fragility of the free film is the strongest argument for the **direct calendering** route (powder compacted directly onto the foil, no free film ever exists). The counter-argument is that a free film can be measured and rejected *before* it is committed to expensive foil. Both routes are described as the most promising technologies at present. Decide this consciously — it determines the entire machine train from DE-170 onward.

8.6 Lamination and bonding hardware

8.6.1 The adhesion problem

THERE IS NO CHEMISTRY HOLDING THE FILM TO THE FOIL

In the wet route, the binder is dissolved and re-precipitates in intimate contact with the foil surface, giving a chemically anchored bond. In the dry route the bond is **purely mechanical interlocking**. This is why *adhesion of the layers* is one of the three named challenges of the process, and why foil surface topography matters far more than it does in the wet route.

8.6.2 Lamination station elements

Element	Function	Specification points
Foil unwind with tension control	Feeds the current collector	Cu 6-8 μm (anode) or Al 10-15 μm (cathode). Standard web handling applies here — the foil is the strong web
Foil surface preparation	Creates mechanical anchoring sites	Options: as-supplied roughened foil, corona or plasma treatment, primer coating, or texturing. Specify measured surface roughness Ra as an incoming parameter
Foil pre-heat	Improves bond formation	Heated roller or IR. Temperature is a process parameter
Heated nip	Bonds film to foil under pressure and temperature	Independent temperature control per roll; line pressure measured on bearing housings; crown compensation as for DE-170
Registration / alignment	Positions the film on the foil, leaving the uncoated margin	Critical for the tabless format — the bare

		margin is where the flags will be cut. Margin width and position tolerance must be specified explicitly
Second-side station	Applies film to the reverse face	Sequential (two passes or tandem station) or simultaneous. Simultaneous is mechanically harder but halves the handling of a part-finished electrode
Cooling section	Sets the bond before winding	Chilled rollers. Winding a warm laminate causes blocking in the coil

8.6.3 Adhesion verification hardware

Test	What it tells you
90° peel test	The direct measure of film-to-foil bond strength. Should be an inline-sampled routine test on a dry line, not an occasional laboratory curiosity
Tape test (cross-hatch or plain)	Quick pass/fail for operators. Crude but useful at shift start
Bend / mandrel test	Detects brittleness and delamination under the curvature the electrode will see at winding
Scratch or stylus test	Cohesive strength of the film itself, as distinct from its adhesion to the foil

BUILD PEEL TESTING INTO THE LINE, NOT THE LAB

Adhesion is the failure mode most specific to the dry route and it is invisible to every inline gauge. A thickness gauge, a mass gauge and a camera will all report a perfect electrode that has almost no bond to its foil. Specify an automated sampling peel tester adjacent to the line with results feeding the MES, and set a control limit on it like any other critical parameter.

9. Drives, Vacuum, Lasers, Welding and Dust Protection

These subsystems are common to both routes and appear across many machines. Specified here once.

9.1 Drives and motion

Element	Function	Specification notes
Servo motor and drive	Torque and velocity control	Size on peak torque during acceleration, not average. Under-sized drives produce following error under transients
Rotary encoder	Position and velocity feedback	Resolution sets the smallest achievable registration step. 20–24-bit absolute is typical for notching and winding
Linear encoder / scale	Direct position feedback on a linear axis	Removes screw and coupling error from the loop. Specify on calender and roll-pair gap axes
Electronic camming	Slave axis follows a master position via a stored profile	The mechanism that lets a notching laser fire at variable pitch while the web runs continuously (see AS-300)
Master measuring wheel	Measures <i>actual</i> web length	Must measure the web, not the motor. Motor position and web length diverge as soon as any slip or stretch occurs. This is a frequent and expensive specification omission
Registration / mark control	Corrects cam phase from a sensed mark	Compensates cumulative web stretch
Fieldbus	Determinism for motion	EtherCAT (<1 ms, sub- μ s jitter) or PROFINET IRT. Distributed clocks are what make camming work

9.2 Vacuum equipment

Pump type	Range	Application
Liquid-ring	30–1000 mbar	Rough vacuum tolerant of vapour
Rotary vane (oil-sealed)	0.1–1000 mbar	General rough vacuum. Oil back-migration is a contamination risk near cells — use with a trap or avoid
Dry screw	0.01–1000 mbar	Preferred near the product: no oil, handles vapour, low maintenance
Roots blower (booster)	Boosts a backing pump	Stacked with a screw pump to reach vacuum-conditioning pressures at practical speed
Turbomolecular	<10 ^{−3} mbar	Only inside the helium leak detector

Vacuum instrument	Range	Notes
Capacitance manometer	1–1000 mbar	Gas-species independent. The correct choice for controlling electrolyte filling at ~150 mbar
Pirani	10 ^{−3} –10 mbar	Gas-species dependent; reads incorrectly in vapour. Trending only
Combined Pirani/capacitance	Wide	Standard on vacuum dryers, whose working band spans 0.07 to 1000 mbar

9.3 Laser sources and beam delivery

Source	Use	Characteristics
Pulsed fibre (ns, MOPA)	Notching, film trimming	Tens to hundreds of kHz; short pulses limit heat input and heat-affected zone. ~1064 nm
Ultrashort pulse (ps/fs)	Highest-quality edges	Near-athermal, minimal HAZ and burr. Limited by average power and cost at production speed
CW / QCW fibre	Welding collector discs, can, terminal, sealing	Hundreds of watts to several kW; deep-penetration keyhole welding
Green (515/532 nm)	Copper welding	Copper absorbs far better at green than IR, greatly stabilising the weld pool
Blue diode (~450 nm)	Copper welding	Same rationale; different power and beam-quality trade-off

Beam delivery element	Function	Process consequence
Galvanometer scanner	Steers the beam by two mirrors	High speed, small field
F-theta lens	Keeps focus flat across the scan field	Without it, penetration varies across the seam
Dynamic focus (z-shifter)	Moves focus along the beam axis in software	Enables 3D seams and large fields
Beam shaping (ring / multi-spot)	Core plus surrounding ring	The most effective spatter-reduction tool in battery welding. Preheats and stabilises the keyhole
Wobble / oscillation	Small superimposed circular motion	Widens the seam, bridges fit-up gaps, improves tolerance
Protective cover slide	Sacrificial optic	Contamination here is the most common cause of gradual power loss at the workpiece — see the warning below
Crossjet and fume extraction	Deflects spatter, removes plume	Plume absorbs and defocuses the beam. Extraction is a process parameter, not housekeeping

THE COVER-SLIDE TRAP — SPECIFY AGAINST IT AT PURCHASE

A dirty cover slide reduces delivered power gradually and asymmetrically. The process drifts toward under-penetration over hours or days while every commanded parameter reads correct. Teams without a power meter at the workpiece or a cover-slide contamination monitor will chase this as a “material problem” for weeks. **Specify an integrated cover-slide monitor and a means of measuring delivered power at the process point in every laser machine RFQ.**

9.4 Welding equipment

Technology	Use on a 46-series cell	Key control parameters
Laser welding	Collector disc to folded tab plane; collector to can wall or base; can to cap; sealing nail. The most important welding process in 4680 production	Power, scan speed, focus position, beam shape, wobble amplitude and frequency, shielding gas, clamp force
Ultrasonic welding	Cathode collector centre to positive terminal; tab-to-tab joints	Frequency, amplitude, pressure, time — or preferably energy/displacement control
Resistance welding	Anode tab to can base in conventional designs	Current, time, electrode force, electrode condition

9.4.1 Ultrasonic welder stack

GENERATOR (20/40 kHz drive) → CONVERTER (piezo stack) → BOOSTER (amplitude scaling) → SONOTRODE (vibration + force) → WORKPIECE (foils) → ANVIL (knurled)

Parameter	Typical	Effect
Frequency	20 or 40 kHz	20 kHz for thicker multi-layer stacks; 40 kHz for delicate thin foil
Amplitude	10–60 μm	Primary energy control. Too high shears the foil; too low gives cold joints
Control mode	Energy (J) or displacement	Substantially better repeatability than fixed-time control because it compensates stack-height variation. Specify this explicitly
Sonotrode and anvil knurl	Consumable	Knurl wear changes coupling efficiency. Change on counted cycles, not on defect

9.5 Dust extraction, filtration and explosion protection

THIS CHAPTER CARRIES MORE WEIGHT IN A DRY PLANT THAN IN A WET PLANT

In the wet route, the powder is wetted out at the mixer and everything downstream is a liquid or a bonded coating. In the dry route, **dry combustible powder is present throughout the electrode block** — graphite, conductive carbon and PTFE. The dust hazard assessment for a dry electrode plant is a different document from that of a wet plant, and it is not a formality.

9.5.1 Why the dust matters twice

Concern	Consequence
Product contamination	Cross-contamination between anode and cathode powder ruins cells. Airborne electrode dust settling on a foil surface becomes a defect that no later cleaning can remove
Operator exposure	Respirable fine powders; occupational exposure limits apply per material
Combustible dust explosion	Metallic and carbon dust is both a contamination and an explosion hazard. Requires appropriately rated collectors and a formal hazardous-area assessment
Equipment reliability	Abrasive dust in bearings, guides and optics

9.5.2 Equipment

Equipment	Function	Specification points
Local exhaust ventilation (LEV) hoods	Capture at source	Capture velocity specified at the source, not at the fan. Verified by measurement at commissioning and periodically
Negative-pressure de-dusting on machines	Removes dust generated in-process	Standard on notching and slitting machines: brush, negative-pressure de-dusting, FFU and dust collector
Cartridge or bag dust collector	Central collection	Must be rated for combustible dust. Explosion venting, suppression or isolation as determined by the hazard assessment
Explosion isolation valve	Prevents flame propagation back along the duct	On the duct between the process and the collector. Frequently omitted in first designs
Explosion relief panel or suppression	Protects the collector vessel	Relief must vent to a safe place — which constrains where the collector can be sited, which constrains the building layout
Spark detection and extinguishing	Detects an ignition source in the duct	Particularly relevant downstream of laser processes and mechanical cutting
Earthing and bonding	Removes electrostatic ignition sources	Every metallic component bonded and earthed; continuity tested and recorded, not assumed

Nitrogen inerting	Removes the oxygen leg of the fire triangle	On enclosed vessels; oxygen concentration monitored continuously
HEPA final filtration	Protects the cleanroom	Extraction that discharges into the room defeats the cleanroom. Filter differential pressure trended
Cleaning equipment	Prevents dust accumulation on surfaces	Certified vacuum equipment for combustible dust. No compressed-air blow-down — it creates a dust cloud, which is the precondition for an explosion

9.5.3 Design rules

1. **Commission the hazard assessment before the layout is frozen.** Explosion relief venting requirements determine where collectors can go, which determines duct routing, which determines machine positions.
2. **Anode and cathode extraction must be separate systems.** A shared duct is a cross-contamination path and mixes two different dust hazards in one collector.
3. **Design for cleaning.** Horizontal ledges accumulate dust. Specify sloped or eliminated ledges on machine frames in the RFQ.
4. **Measure capture velocity at commissioning** and record it as a baseline. A slowly blocking duct looks exactly like a process drift.
5. **Never permit compressed-air blow-down for cleaning.** Write it into the operating procedures and enforce it.

Part III — The Dry Electrode Line (DE-1xx)

Twelve machine classes covering the solvent-free electrode block, from powder receiving to a slit, conditioned electrode coil ready for the dry room.

10. Material Preparation — DE-100 to DE-130

DE-100 Powder Receiving and Containment Station

Function:

contained discharge of bulk active material, conductive carbon and PTFE binder |

Route:

shared, higher duty in dry

Throughput:

batch, sized to blender capacity |

Footprint:

~4 × 4 m per station incl. hoist |

Environment:

ISO 7-8, <20 % RH

Utilities:

extraction, compressed air, N₂, power, earthing |

Maturity:

mature, multiple suppliers

What it does

Empties FIBC bulk bags, drums and sacks into the process without releasing dust to the room or admitting moisture to the powder. Weighs the delivered quantity for inventory and genealogy.

Subsystem breakdown

Module	Specification points
Bag lifting frame and hoist	Rated for full bag mass with margin; controlled descent to avoid shock loading the load cells
Discharge spout and clamp	Dust-tight iris or split-butterfly valve. Containment class selected from the material's occupational exposure limit
Bag massagers / agitators	Pneumatic paddles that break bridging without over-compacting the powder
Receiving hopper	Mass-flow geometry derived from measured powder flow properties — not from a standard drawing
Integral extraction	Capture velocity specified at the spout; connected to the correct (anode or cathode) collector
Load cells	Inventory and receipt confirmation; class C3 or better
Nitrogen purge	Displaces humid air during and after discharge
Earthing and bonding	Continuity-tested; conductive FIBC liners specified in the material purchase order

Key specifications

Parameter	Target
Storage / operating relative humidity	< 20 %
Storage temperature	20-25 °C
Containment performance	Specified against the material OEL, verified by surrogate monitoring at SAT
Anode / cathode segregation	Physically separate stations. No shared spouts, hoppers or extraction

Acceptance criteria

- Surrogate dust monitoring during a full discharge cycle, at the operator breathing zone
- Earthing continuity measured and recorded at every bonded point
- Weighing accuracy verified against certified check weights across the working range
- Extraction capture velocity measured at the spout and recorded as the commissioning baseline

DE-110 Loss-in-Weight Dosing System

Function:

gravimetric metering of active material, conductive carbon and PTFE to the blender |

Route:

shared

Throughput:

matched to blender / continuous line rate |

Footprint:

~2 × 2 m per feeder group |

Environment:

ISO 7-8

Utilities:

power, compressed air, N₂, extraction |

Maturity:

mature

What it does

Holds each constituent in a weighed hopper and modulates a screw or vibratory feeder so that mass flow follows a setpoint. Because the hopper mass is measured continuously, the system self-corrects for changes in bulk density — which is why it outperforms volumetric dosing on powders whose density varies with fill level and compaction.

Subsystem breakdown

Module	Specification points
Weigh hopper on load cells	Must be mechanically decoupled from the frame — flexible connections, no rigid pipework bridging the weighing element
Twin-screw or single-screw feeder	Screw geometry matched to the powder. Insist on a trial with your actual powder , not a surrogate
Agitation in hopper	Prevents bridging without compacting or attriting the powder
Refill system	Refill causes a weighing blind period; the controller runs volumetrically through it. Refill frequency therefore affects average accuracy
Vibration isolation	Load cells at this resolution are sensitive to floor and machine vibration. Specify isolation and verify at SAT with the neighbouring machines running
Controller	Logs actual delivered mass per batch to MES

Key specifications

Parameter	Target
Dosing accuracy	±0.1 % of batch mass. Specify as percentage of setpoint, not of full scale
Load cell class	C3 or better
Turndown ratio	Must cover the smallest addition (binder at <2 wt%) and the largest (active material at up to 98 %) — usually requires differently sized feeders, not one large one
Refill blind time	Minimised; disclosed by supplier; accounted for in the accuracy statement

WHY DOSING ACCURACY MATTERS MORE IN THE DRY ROUTE

In a wet slurry, a small dosing error is diluted through a large liquid volume and partially averaged out by long mixing. In the dry route there is no liquid phase to average anything. A binder dosing error of a few tenths of a percent, on a formulation where binder is below 2 wt% in total, is a *large* proportional error in the variable that governs film integrity. Specify the binder feeder to a tighter accuracy than the active material feeder.

Acceptance criteria

- Catch-and-weigh test at 25 / 50 / 100 % of rated flow, minimum ten samples per point
- Accuracy verified across a refill cycle, not only between refills
- Repeat with the adjacent machines running to confirm vibration isolation

DE-120 Sieving and Magnetic Separation Unit

Function: removal of oversize agglomerates and ferrous/paramagnetic contamination | **Route:** shared, **safety-critical in dry**
Throughput: matched to line rate | **Footprint:** ~1.5 × 1.5 m | **Environment:** ISO 7-8
Utilities: power, extraction, N₂ | **Maturity:** mature

What it does

Screens out lumps that would appear as defects in the film, and extracts metallic particles that would penetrate the separator and cause an internal short circuit. In the dry route this is the *only* particle removal step — there is no slurry filtration downstream.

Subsystem breakdown

Module	Specification points
Vibratory or centrifugal sieve	Mesh typically 100–200 µm. Ultrasonic deblinding for cohesive fine powders
Oversize collection	Weighed and logged — a rising oversize rate is an early warning of incoming material change
Stage 1 magnetic separator	Rare-earth grid or rotary drum, field strength above 10,000 gauss
Stage 2 magnetic separator	Independent second stage. Positioned <i>after</i> any conveying that could generate wear debris
Automatic cleaning mechanism	Self-cleaning grids allow scheduled cleaning without opening the system to the room
Captured-material collection	Collected and weighed. This is your contamination trend data — do not discard it unmeasured

Key specifications

Parameter	Target
Magnetic stages	Minimum two, independent
Field strength at the working surface	> 10,000 gauss
Sieve mesh	100–200 µm band
Incoming material magnetic impurity limit	< 50 ppb (Fe, Ni, Cr, Zn), verified by ICP-MS on incoming lots
Cleaning basis	Scheduled interval, not on demand. A saturated separator passes contamination silently

Acceptance criteria

- Field strength measured with a gauss meter at the working surface, at multiple points, and recorded
- Seeded-particle challenge test: introduce a known count of ferrous particles of specified size and confirm capture
- Sieve integrity check — a torn mesh is invisible and passes everything

DE-130 Pre-Blender (Non-Destructive Mixing)

Function: homogenise active material with porous conductive carbon *before* binder addition | **Route:** **dry-specific**
Throughput: batch or continuous, matched to DE-140 | **Footprint:** ~3 × 3 m | **Environment:** ISO 7-8, N₂ blanketed
Utilities: power, chilled water, N₂, extraction | **Maturity:** mature hardware, novel duty

What it does

Distributes the conductive carbon uniformly through the active material **without** applying the shear that would begin fibrillating binder — because the binder is not present yet. Published patent material describes non-destructive mixing of active material (particles larger than 10 µm) with porous carbon before adding a single PTFE binder.

WHY THIS IS A SEPARATE MACHINE AND NOT A FIRST PHASE OF DE-140

Once PTFE is present and shear is applied, fibrillation begins. Any mixing needed to disperse the conductive carbon must therefore be finished before the binder enters, or the carbon dispersion step will simultaneously over-fibrillate the binder. Combining both duties in one machine forces a compromise on the one variable you cannot afford to compromise. **Two machines. This is a sequence decision with equipment consequences.**

Process physics

Two distinct mixing regimes exist and they are not interchangeable. **Distributive** mixing moves material around so composition is uniform on a coarse scale, breaking up heterogeneous phases; it is achieved by low-shear, large-displacement action. **Dispersive** mixing applies shear stress high enough to break agglomerates into primary particles; it needs high shear in a small gap.

This station must deliver distributive mixing *only*. The conductive carbon must be spread through the active material, but no significant shear may be applied — because the next station will apply a precisely metered amount of shear, and any shear delivered here is an uncontrolled addition to that budget. In a wet process the two regimes are separated by adding solvent between them. Here they are separated by adding the binder between them.

Subsystem breakdown

Module	Specification points
Low-shear mixing element	Ribbon, plough or conical screw. Chopper or high-speed disperser must be omissible or independently controllable — you do not want incidental high shear here
Jacket	Chilled water; keeps product temperature stable and low
Wall scraper	Prevents a stagnant, under-mixed boundary layer
Nitrogen blanket	Moisture and explosion control; oxygen monitored
Net power / torque logging	Batch fingerprint. Deviation from the reference signature is an early indicator of off-spec incoming material
Discharge	Gentle; minimise segregation and free-fall

Key specifications

Parameter	Target
Shear regime	Distributive (low shear), explicitly not dispersive
Product temperature rise	Minimal; specify a maximum, jacket-controlled
Blend uniformity	Verified by sampling at multiple vessel locations; carbon content by loss-on-ignition or image analysis
Anode / cathode segregation	Separate machines. Cross-contamination is unrecoverable
Batch-to-batch repeatability	Torque signature within a defined envelope of the reference

Acceptance criteria

- Blend uniformity from a minimum of ten sample points across the vessel, on three consecutive batches
- Three reference torque signatures recorded and accepted as the baseline for production monitoring
- Verification that particle size distribution is not degraded — D50 measured before and after, confirming the mixing really is non-destructive

11. Fibrillation — DE-140 and DE-150

DE-140 Fibrillation Mixer — CRITICAL MACHINE

Function:

apply controlled shear to fibrillate PTFE into a binding fibril network |

Route:

dry-specific, defining machine

Throughput:

batch or continuous |

Footprint:

~3 × 5 m incl. feed and discharge |

Environment:

ISO 7-8, N₂

Utilities:

power (high), chilled water (high), N₂, extraction |

Maturity:

low — TRL 6-7, few suppliers

What it does

Draws PTFE particles into nanoscale fibrils under controlled shear, producing a web that mechanically entangles and holds the active material particles. This single operation replaces the entire function of solvent dissolution, coating and drying in the wet route. Everything downstream depends on it being done uniformly.

THE SPECIFICATION CHALLENGE, STATED PLAINLY

Fibrillation depends on accumulated shear energy, temperature and residence time. All three vary spatially inside any real mixer, so under- and over-fibrillated material can coexist in one batch.

Under-fibrillated: the film lacks mechanical integrity, tears at the roll pair, cannot survive assembly. **Over-fibrillated:** the binder network blocks the paths lithium ions need to travel, raising internal resistance and reducing performance.

Neither condition is detectable by any inline mass or thickness gauge. Binder fibrillation consistency is the named principal challenge of the process. Specify this machine on residence time distribution and specific energy control, not on installed power or batch size.

Process physics

PTFE under controlled shear does something unusual: its particles draw out into **nanoscale fibrils**, forming a three-dimensional web that mechanically entangles and holds the active material particles. There is no solvent, no dissolution and no drying. The film's integrity is purely mechanical, produced by this fibril network and nothing else.

The governing variable is **specific shear energy** — energy input per unit mass — modulated by temperature and residence time. Fibrillation is a kinetic process: fibrils nucleate, extend and eventually break or over-densify. The process therefore has an optimum, not a monotonic response, which is why both too little and too much are failure modes.

THE BINDER WINDOW IS NARROW IN BOTH DIRECTIONS

Too little binder and the film lacks mechanical integrity — it cannot survive cell assembly or cycling. Too much binder, or too much fibrillation, and the network blocks the paths lithium ions need to travel, reducing performance and raising internal resistance. There is no safe direction to err. This is why fibrillation *control*, not binder quantity alone, is the critical variable.

A second-order effect matters at production scale: all shear energy input becomes heat. Without effective removal, the batch temperature rises through the process and the fibrillation state drifts with it — so a machine that fibrillates correctly for the first ten minutes may over-work the material by the thirtieth. Temperature control is not a comfort setting; it is part of the shear specification.

Architecture options — select before RFQ

Architecture	Residence time distribution	Assessment
High-shear blade mixer	Broad (batch)	Widely available; shear highly non-uniform. Long cycles needed to homogenise, risking over-working part of the batch
Intensive / cyclone mixer	Moderate	Better temperature control and bulk turnover. Intensive mixing is a recognised industry direction
Twin-screw extruder	Narrow (continuous)	Strongest candidate on residence-time grounds; every particle follows a similar path. Continuous mixing by twin-screw extruder is a recognised direction. Screw configuration becomes a process parameter you own
Jet mill	Narrow	Very uniform energy input; no contact wear parts so no tooling-derived metallic contamination. High gas consumption

Subsystem breakdown

Module	Specification points
Shear element	Rotor, screw set or nozzle array. In an extruder, the screw configuration is a documented, version-controlled process parameter
Net power measurement	Torque or net motor power at high sample rate with no-load tare. This is how specific shear energy is computed — without it you cannot control the process
Jacket / barrel temperature control	Multi-zone where the geometry allows. Shear input becomes heat continuously; without effective removal the process temperature rises through the batch and the fibrillation state drifts with it
In-product temperature measurement	Not only wall RTDs. Jacket temperature is not product temperature and the difference is the whole point
Feed system	From DE-130 plus separate binder addition. Binder addition point is a process parameter in a continuous machine
Contact surfaces	Hardened or ceramic. The cathode powder is abrasive; wear debris becomes contamination
Nitrogen inerting	Oxygen concentration monitored continuously
Discharge and transfer	Minimum free-fall and minimum residence before DE-160 — fibrillated powder can consolidate on standing
Cleaning access	Fibrillated PTFE is tacky and adheres to surfaces. Design for disassembly and cleaning; ask to see it done at FAT

Key specifications for the RFQ

Parameter	How to specify

Specific shear energy	Measured and recorded per batch in kWh/kg or J/kg, computed from integrated net power over mass. Not as a proxy such as mixing time
Residence time distribution	Tracer study on your material at your throughput, with a stated maximum acceptable spread. Demand the measurement method
Temperature control band	Setpoint and maximum permissible rise across the process, measured in-product
Throughput at specification	Guaranteed kg/h at a stated fibrillation quality, verified by a defined test (see acceptance below)
Scale-up basis	Supplier must state the scale-up rule used (constant tip speed / specific energy / shear rate) and the supporting evidence
Wear part life	Stated in operating hours or tonnes processed for your abrasive material, not a generic figure
Cleaning and changeover time	Demonstrated at FAT, timed

Integrated instrumentation

Instrument	Purpose
Net torque / power, high sample rate	Specific energy calculation; batch fingerprinting
In-product temperature	The actual controlled variable
Jacket inlet and outlet temperature and flow	Heat removal rate — an independent check on energy input
Oxygen analyser	Inerting verification
Mass flow (continuous machines)	Denominator of the specific energy calculation
Sampling port	For the offline fibrillation quality tests below — specify a port that can be used without breaking inerting

How to actually verify fibrillation quality

There is no inline sensor for fibrillation state. The practical qualification set:

Test	What it reveals
SEM imaging of the powder	Direct visual evidence of fibril formation, length and distribution. The reference method
Green-film tensile strength	Functional consequence of fibrillation. Under-fibrillated material gives a weak film
Film formability trial on a pilot roll pair	Whether a self-supporting film forms at all, and in how many passes
Electrode resistivity / cell impedance	Detects over-fibrillation, which blocks ion paths
Powder flow / bulk density change	A quick proxy that correlates once calibrated against SEM on your material

Consumables and wear parts

Item	Replacement basis
Screw elements / rotor blades	Counted hours or tonnage; inspect dimensionally, do not wait for a defect
Barrel liners	Wall thickness measurement at interval
Seals	Scheduled; a failed seal releases combustible dust
Filter elements on inerting and extraction	Differential pressure

Acceptance criteria

- **Residence time distribution study** on your material at guaranteed throughput, with the measured spread inside the specified limit
- Three consecutive batches at guaranteed throughput; SEM fibrillation assessment and green-film tensile on samples from the start, middle and end of each
- Specific energy recorded and reproducible across the three batches within a stated band
- In-product temperature held within the specified band throughout, at full throughput
- Particle size distribution measured before and after — confirm the machine is not over-attriting the brittle cathode
- Full disassembly and cleaning demonstrated and timed
- Wear parts inspected after the FAT run and dimensional change recorded as the wear baseline

Selection notes

FIVE QUESTIONS THAT SEPARATE THE CANDIDATES

1. What is the residence time distribution of my material in your machine, and how did you measure it?
2. How do you measure and control specific shear energy, and can I see the per-batch record?
3. What is your scale-up rule from pilot to production, and what evidence supports it?
4. What is the wear-part life on an abrasive high-nickel cathode powder specifically?
5. How many production installations of this machine on this duty exist, and may I speak to one?

DE-150 De-agglomeration and Conditioning Unit

Function: break soft agglomerates formed during fibrillation and condition the powder for feeding | **Route:** dry-specific
Throughput: matched to DE-160 | **Footprint:** ~1.5 × 2 m | **Environment:** ISO 7-8, N₂
Utilities: power, N₂, extraction | **Maturity:** moderate

What it does

Fibrillated powder is tacky and tends to form soft lumps. Those lumps feed unevenly and appear in the film as thickness variation or as visible inclusions. This unit gently breaks them without applying further shear that would continue fibrillation.

THE NARROW WINDOW THIS MACHINE OPERATES IN

It must apply enough mechanical action to break agglomerates and not enough to fibrillate further. That is a genuinely difficult specification, and it is the reason some suppliers integrate this function into DE-140's discharge or into DE-160's feeding rather than offering a separate machine. Either integration is acceptable; what is not acceptable is ignoring the function and hoping the roll pair copes.

Subsystem breakdown and specification

Module	Specification points
Conditioning element	Low-intensity: rotating screen, gentle pin mill at low tip speed, or vibratory deagglomerator. Tip speed is the controlled variable
Screen	Sized to the maximum acceptable agglomerate for the target film thickness
Residence control	Short and defined; no accumulation
Buffer hopper	Minimal. Fibrillated powder consolidates on standing — specify a maximum permissible hold time and design so it cannot be exceeded

Acceptance criteria

- Agglomerate size distribution before and after, on your material
- Confirmation by SEM or green-film tensile that fibrillation state is *unchanged* through the unit
- Demonstration that no accumulation occurs over a sustained run

12. Film Forming — DE-160 and DE-170

DE-160 Powder Feeder to Roll Pair

Function:

 deliver a uniform powder curtain across the full working width into the roll nip |

Route:

 dry-specific

Throughput:

 matched to line speed and areal loading |

Footprint:

 integrated with DE-170 |

Environment:

 ISO 7–8, N₂

Utilities:

 power, N₂, extraction |

Maturity:

 moderate

What it does

Meters fibrillated powder into the nip of the film-forming roll pair as an even curtain across the width. In the wet route, the slot-die manifold spreads a fluid and self-levels to a degree. Here there is no fluid.

CROSS-DIRECTION FEED UNIFORMITY IS CROSS-DIRECTION FILM UNIFORMITY

There is no liquid phase and no levelling mechanism. Whatever variation exists in the powder curtain across the width is transferred directly into the areal loading of the film, and it cannot be corrected downstream by any means except scrapping. This machine deserves far more attention than its cost implies.

Process physics

In a wet process, the slot-die manifold distributes a fluid across the width and surface tension provides a degree of self-levelling. Here there is neither. The powder arrives as discrete particles and is compacted where it lands.

Two consequences follow. First, cross-direction variation in the powder curtain transfers *directly* into areal loading, with no averaging mechanism anywhere downstream. Second, because fibrillated powder is cohesive and tacky, it does not flow like a free-flowing granular solid — it bridges, it arches over outlets, and it consolidates when held. The feeder must therefore both meter accurately and keep the material moving.

Subsystem breakdown

Module	Specification points
Metering element	Grooved roll, multiple screws across the width, or vibratory tray. Multiple independently driven screws allow CD profile trimming; a single element does not
CD profile adjustment	The high-value option. Segmented gates, individually driven screw sections, or an adjustable blade. Ask specifically whether CD profile is adjustable and how
Curtain forming and guidance	Chutes or plates directing powder into the nip without segregation or re-agglomeration
Edge dams	Define the film width and contain the powder at the roll ends
Anti-bridging	Vibration or agitation in the feed hopper
Static control	Earthing and ionisation; charged powder adheres to chute surfaces and builds up until it releases as a lump
Level control	Maintains a constant head over the metering element — a varying head is a varying flow

Key specifications

Parameter	Target
CD feed uniformity	Specify as an areal loading tolerance on the resulting film. Industry reference for electrode loading uniformity is ±1.5 % or better
MD feed stability	No periodic variation attributable to the metering element geometry
Working width	Matched to roll face; usable width to be stated separately from mechanical width
CD adjustment resolution	Number of independently adjustable zones across the width
Maximum hold time in feeder	Specified and designed against, to prevent consolidation

Acceptance criteria

- **CD profile of areal loading** measured across the full width on the resulting film, at three line speeds. This is the single most important FAT test for this machine
- MD stability over a sustained run, with the gauge trace examined for periodicity matching the metering element
- CD adjustment demonstrated: deliberately introduce a profile error and correct it using the machine's own adjustment
- Verification that no bridging or accumulation occurs over a full shift-length run

DE-170 Film-Forming Roll Pair — CRITICAL MACHINE

Function:

 compact fibrillated powder into a coherent self-supporting film |

Route:

 dry-specific, defining machine

Throughput:

 speed-limited — uniformity degrades above ~50 m/min |

Footprint:

 ~4 × 6 m |

Environment:

 ISO 7–8, N₂

Utilities:

 power (high), hot/chilled water, N₂, extraction |

Maturity:

 moderate to low

What it does

Two counter-rotating rolls with a controlled speed differential compact the powder curtain into a film. The film consolidates on the faster-rotating roll and is taken off as a free-standing web, or is calendered directly onto the substrate foil.

Process physics

Powder entering a nip between two counter-rotating rolls is compacted by the converging geometry. Particles rearrange, the fibril network is drawn taut, and inter-particle contact area grows until the mass becomes a coherent solid film.

WHY THE SPEED DIFFERENTIAL EXISTS

Two identical counter-rotating rolls compact the powder symmetrically, and the resulting film has no preference for either roll — it splits, adheres

to both and tears. Introducing a speed differential breaks the symmetry: the faster surface imparts more shear and more adhesion, so the film consolidates onto it and can be taken off cleanly as a coherent web.

THE SPEED RATIO IS NOT A FINE-TUNING PARAMETER

It is what makes film take-off possible at all. A machine with geared, fixed-ratio rolls has removed the variable that governs whether a film forms. Reject that design.

POROSITY MUST BE ENGINEERED, NOT INHERITED

In a solvent process, the space occupied by evaporating solvent becomes porosity almost automatically, and calendering trims it. Here there is no solvent, so every void in the finished electrode is a void that the roll pressure deliberately left behind. Line pressure, roll temperature and speed ratio together set the porosity, and *adjustment of porosity* is one of the three named challenges of the dry process for exactly this reason.

THE SPEED LIMIT

Uniformity degrades above roughly 50 m/min. The mechanism is residence time in the nip: at higher web speed the powder has less time to rearrange and consolidate before it exits, so the film forms less completely and less uniformly. This is a physical constraint, not a machine deficiency, and it is why line speed must be a contractual performance figure rather than a datasheet aspiration.

Subsystem breakdown

Module	Specification points
Roll bodies	Hardened forged steel, ground and polished. Surface finish transfers directly to the film — specify Ra explicitly, both for film quality and for release behaviour
Independent servo drives	One per roll, each with its own encoder. A geared fixed-ratio pair removes your most important process variable — reject such a design
Speed ratio control	Continuously adjustable, recorded per batch. Typical band 1.2:1 to 2:1
Loading system	Hydraulic or servo-electric. Reference range for battery calendering line pressure extends to 2,500 N/mm
Load cells on bearing housings	Measure <i>actual</i> line pressure. Specify measured, not commanded — this is a frequent omission
Gap axis with linear scale	Micron-resolution servo; alternative control mode to load
Crown compensation	Adjustable bending or internal hydraulic zones. Fixed crown works at one load only, which bounds the usable process window to a single operating point
Roll temperature control	Rotary union to internal channels. Reference range for tempered rolls approximately 50–250 °C, independently controlled per roll
Doctor blade / take-off	Blade material, angle and load. A wear item and a common source of film defects
Continuous roll cleaning	Essential with tacky fibrillated material. Build-up prints a defect repeating at roll circumference
Edge dams	Define film width; determine edge quality of the free film
Static discharge	Before and after the nip

Key specifications

Parameter	Target / note
Line speed at specified uniformity	The number to make contractual. Uniformity degrades above roughly 50 m/min; challenge every supplier on this and define exactly how uniformity will be verified
Line pressure, measured	Up to 2,500 N/mm reference; stability band specified
Speed ratio range and resolution	1.2:1 to 2:1 typical, continuously adjustable
Roll temperature range and uniformity	50–250 °C reference; CD temperature uniformity across the roll face specified
Film areal loading tolerance	±1.5 % or better
Roll runout (TIR)	Specify; runout prints an MD variation at roll frequency
Working width	Usable width stated separately from mechanical width
Passes to self-supporting film	Patent claim references three or fewer calender passes total including downstream. A proposal needing five is a maturity signal

Integrated instrumentation

Instrument	Purpose
Load cells, bearing housings	Actual line pressure — the primary control variable
Encoders on both rolls	Speed and ratio; detects slip
Roll surface RTD / rotary union temperature	Temperature control
Paired laser triangulation at film exit	Film thickness. Requires a thermally stable C-frame — frame growth appears directly as thickness error
Beta or XRF gauge	Areal mass. With thickness, gives density — the variable you actually want
Multispectral / polarisation camera	See §18 — conventional RGB vision does not work on dry electrode
Web tension load cell (free-film route)	Very low setpoint; specify a load cell sized for the actual tension

Options and variants

Option	Worth paying for?
Adjustable crown compensation	Yes. Without it the machine has one usable operating point
Independent roll temperature control	Yes. Asymmetric temperature is another lever on which roll the film releases from
Direct-to-foil configuration	Architecture decision — see §9.2. Decide before RFQ, not during
Integrated film trimming	Yes on the free-film route — shortens the fragile web path
Quick roll change	Yes if running multiple products; roll change is otherwise a multi-shift operation

Consumables and wear parts

Item	Replacement basis
Doctor blades	Counted hours; film surface defects are the symptom of a worn blade
Roll cleaner media	Continuous consumable
Edge dam seals	Counted hours; leakage is the symptom
Roll regrind	Long interval, long lead time. Hold a spare roll set — a damaged roll otherwise stops the line for months
Rotary union seals	Scheduled; leakage of tempering fluid onto the product is a serious contamination event

Acceptance criteria

- **Areal loading CD profile and MD stability** at the guaranteed line speed, sustained for a defined run length
- Line speed ramp with uniformity measured at each step, to establish and document the actual speed limit
- Measured line pressure compared with commanded across the working range
- Crown compensation demonstrated: run at two very different line loads and show CD thickness profile held
- Roll surface temperature uniformity measured across the face by thermography
- Film formability: self-supporting film produced within the specified number of passes
- Green-film tensile strength measured, meeting the specification for downstream handling
- Repeating-defect check: run a length and confirm no defect matches any roll circumference

Selection notes

QUESTIONS THAT MATTER AT THIS STATION

1. What line speed do you guarantee at $\pm 1.5\%$ areal loading uniformity, and how will that be verified?
2. Is crown compensation adjustable, and over what load range does it hold the CD profile?
3. Are the rolls independently driven, and what is the speed ratio resolution?
4. Is line pressure measured on the bearing housings or inferred from hydraulic pressure?
5. What is the roll regrind interval and lead time, and what does a spare roll set cost?
6. How is the film taken off, and what is the doctor blade life on this material?

13. Film Handling and Lamination — DE-180 and DE-190

DE-180

Free-Film Handling and Trimming Unit

Function:

transport and trim the self-supporting film between forming and lamination

|

Route:

dry-specific

(free-film architecture only)

Throughput:

matched to DE-170

|

Footprint:

integrated, minimise span

|

Environment:

ISO 7-8

Utilities:

power, extraction, N₂

|

Maturity:

moderate — the hardest handling problem on the line

What it does

Carries the unsupported film from the take-off point to the lamination nip and cuts it to the required width. Before the dry film is applied it is usually cut to the desired width. This is the most fragile web in the plant — its tensile strength is carried entirely by the fibril network.

Subsystem breakdown

Module	Specification points
Web path	As short as possible. Every metre of free span is a failure opportunity. This argues strongly for an integrated forming-trimming-lamination machine
Support conveyance	Air-support (floatation) or vacuum belt in preference to free spans and idler rollers
Tension control	Load-cell closed loop at a very low setpoint. Specify a load cell sized for the actual tension , not one sized for a foil web operating at the bottom of its range
Idler rollers	Minimum count, minimum inertia, minimum wrap angle. Each roller is a tension riser
Edge guiding	Non-contact sensing with gentle correction rate. Aggressive EPC correction tears the film
Trimming	Laser (non-contact, preferred) or very sharp rotary knives. Edge quality of a free film is more fragile than a supported one
Trim waste removal	Suction; recycled or scrapped depending on your material policy
Static control	Ionising bars — a charged fragile film clings to surfaces and tears
Break detection and safe stop	Immediate detection with a controlled stop that does not pile film into a nip
Clearing access	Design for it. Breaks will happen throughout ramp-up; a machine that takes four hours to clear destroys your learning rate

Key specifications

Parameter	Target / note
Free span length	Minimised; state the total unsupported path length in the quotation
Web tension setpoint and resolution	Established experimentally on your film; load cell range matched
Trim width tolerance	Specified; feeds directly into lamination registration
Mean time between film breaks	Ask for it. A supplier who has run this duty will have a number
Clearing time after a break	Demonstrated and timed at FAT

Acceptance criteria

- Sustained run at guaranteed speed with break count recorded
- Deliberate break induced; clearing procedure demonstrated and timed
- Trim width and edge quality measured; edge friability assessed
- Tension stability recorded through a start, ramp and stop cycle

DE-190

Lamination Station — CRITICAL MACHINE

Function:

bond the dry film to the current collector foil

|

Route:

dry-specific

Throughput:

matched to DE-170

|

Footprint:

~4 × 8 m incl. foil unwinds

|

Environment:

ISO 7-8

Utilities:

power, hot/chilled water, compressed air, extraction

|

Maturity:

moderate

What it does

Presses the dry film onto aluminium (cathode) or copper (anode) foil under controlled temperature and line pressure, forming a purely mechanical bond, and positions it so that the required uncoated margin remains for the tabless flags.

TWO THINGS MAKE THIS STATION HARDER THAN IT LOOKS

1. There is no chemistry holding the film to the foil.

In the wet route the binder dissolves and re-precipitates in intimate contact with the foil, giving a chemically anchored bond. Here the bond is mechanical interlocking only. *Adhesion of the layers* is one of the three named challenges of the dry process.

2. The margin position is a tabless-format requirement.

The uncoated strip left at the foil edge is where the notching laser will cut the flags. Its width and position tolerance propagate all the way to the collector weld. Specify them here, explicitly.

Process physics

In a solvent process the binder dissolves and re-precipitates in intimate contact with the foil surface, giving a chemically anchored bond that penetrates surface irregularities at the molecular scale. Here the bond is **purely mechanical interlocking**: fibrils and particles are pressed into the foil's surface topography and held by friction and deformation.

Three variables therefore carry the entire bond: foil surface roughness (the anchoring sites), lamination pressure (how far the film is pressed into them), and temperature (how readily the film deforms to fill them). Change any one and adhesion changes. This is why *adhesion of the layers* is one of the three named challenges of the process, and why incoming foil roughness must be specified and inspected rather than assumed.

ADHESION IS INVISIBLE TO EVERY INLINE GAUGE

A thickness gauge, a mass gauge and a camera will all report a perfect electrode that has almost no bond to its foil. The failure surfaces later — as delamination at the calender, as flaking at the winder, or as capacity loss in service. Peel testing is not a laboratory nicety on this route; it is the primary quality control for this station.

Subsystem breakdown

Module	Specification points
Foil unwind with tension control	Cu 6–8 µm or Al 10–15 µm. Standard web handling; tension band 50–200 N/m depending on foil
Foil edge position control	Sets the datum for the margin. Prefer a vision guide tracking the actual foil edge with sub-0.1 mm resolution
Foil cleaning	Contact or air-knife plus extraction. Particles trapped at the interface are both an adhesion defect and a potential short
Foil surface preparation	Corona, plasma, primer or texturing. Specify measured surface roughness Ra as an incoming foil parameter — it carries the whole bond
Foil pre-heat	Heated roller or IR; temperature is a process parameter
Heated lamination nip	Independent temperature control per roll; line pressure measured on bearing housings; crown compensation as for DE-170
Film-to-foil registration	Sets margin width and position. Closed-loop from a vision sensor, not open-loop mechanical setting
Second-side station	Sequential (tandem) or simultaneous. Simultaneous is mechanically harder but halves handling of a part-finished electrode
Cooling section	Chilled rollers. Winding a warm laminate causes blocking in the coil
Rewind with taper tension	Constant tension crushes the inner wraps; taper to typically 50–70 % of starting tension at full diameter

Key specifications

Parameter	Target / note
Peel adhesion	The defining specification. Set a minimum 90° peel value derived from downstream handling, calendaring and winding stresses. Make it a contractual performance guarantee
Lamination temperature	Per roll, independently controlled; range and CD uniformity specified
Line pressure, measured	Bearing-housing load cells
Uncoated margin width tolerance	Specify explicitly — this is a tabless-format requirement and it propagates to the notcher
Margin position tolerance relative to foil edge	Specify explicitly
Foil surface roughness Ra (incoming)	Cathode Al 0.1–0.5 µm; anode Cu 0.2–0.6 µm reference. Too smooth kills adhesion; too rough risks pinholes in thin foil
Line speed	Matched to DE-170; the whole train runs at the slowest station

Integrated instrumentation

Instrument	Purpose
Beta or XRF areal mass gauge	Post-lamination loading, both sides
Paired laser triangulation	Total laminate thickness; with mass gives density
Vision, margin measurement	Margin width and position, continuously — this is a tabless-critical dimension
Multispectral / polarisation camera	Surface and interface defects on a low-contrast dry surface
Automated sampling peel tester	Specify this. Adhesion is the dry route's characteristic failure mode and it is invisible to every other inline gauge. Results should feed the MES with a control limit like any critical parameter
Roll temperature RTDs and thermography	CD temperature uniformity
Load cells on nip bearings	Actual line pressure

Consumables and wear parts

Item	Replacement basis
Nip roll covers (if elastomer-covered)	Counted hours; hardness measured at interval
Corona / plasma electrodes	Counted hours; treatment level verified by dyne test
Web cleaner media	Continuous
Rotary union seals	Scheduled

Acceptance criteria

- **90° peel adhesion** measured at multiple CD positions and multiple MD positions, meeting the specified minimum with a stated capability index
- Bend / mandrel test at the curvature the electrode will experience at winding — no delamination
- Margin width and position measured continuously over a sustained run, meeting tolerance
- Areal loading and thickness CD profiles, both faces
- Adhesion verified after a deliberate thermal excursion, to establish process sensitivity
- Peel tester correlated against laboratory reference method

Selection notes

QUESTIONS THAT MATTER AT THIS STATION

1. What peel adhesion do you guarantee, on my film and my foil, and how is it measured?
2. What foil surface preparation do you recommend and what Ra do you require incoming?
3. Is margin registration closed-loop from a vision sensor, or set mechanically?
4. Is the second side laminated simultaneously or sequentially, and why?
5. Is an automated sampling peel tester included, or is it my scope?

14. Compaction, Separation and Conditioning — DE-200 to DE-220

DE-200 Final Calender

Function:

 set final electrode thickness, density and porosity |

Route:

 shared, **different duty in dry**

Throughput:

 60–100 m/min reference |

Footprint:

 ~4 × 8 m |

Environment:

 ISO 7–8

Utilities:

 power, hot/chilled water, extraction |

Maturity:

 mature hardware

What it does

Compacts the laminated electrode between a roll pair under precisely defined line pressure to reach the target porosity. In the wet route, porosity is largely inherited from solvent evaporation and calendering trims it. In the dry route, **porosity must be engineered here from first principles** — *adjustment of porosity* is one of the three named challenges of the process.

Process physics

Under line pressure the porous film deforms plastically: particles rearrange, contact area between particles grows, and porosity falls. Two effects compete directly.

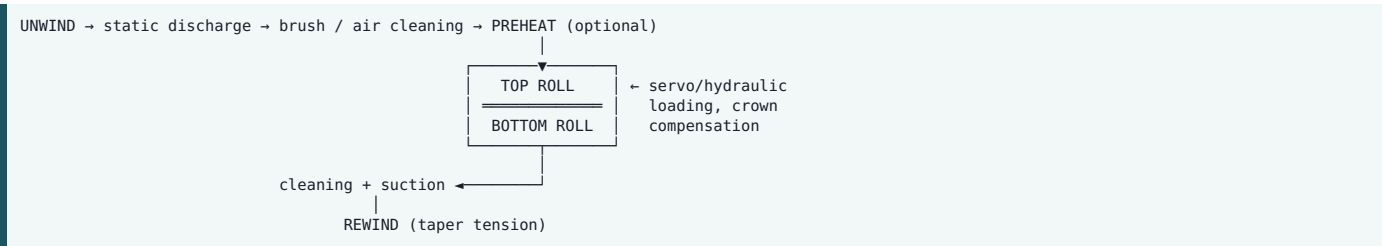
Lower porosity gives...	Higher porosity gives...
Higher volumetric energy density Better electronic conductivity within the electrode More active material per unit volume	Faster electrolyte infiltration Better rate capability and fast charge Lower risk of particle cracking

There is no universally correct target. The porosity setpoint is a function of chemistry, electrode thickness and the intended duty cycle, and must be established by experiment for each product. On the dry route the setting is more consequential than on the wet route, because the whole porosity budget is created mechanically here and at DE-170 rather than partly inherited from drying.

THE PASS BUDGET

Published patent material claims self-supporting films in **three or fewer calender passes** total. Counting DE-170 (film forming), DE-190 (lamination) and DE-200 (final calender), the budget is already spent. Every additional pass adds cost, adds a defect opportunity and works the binder network further. A supplier proposal that needs five passes is telling you something about the maturity of their film-forming step.

Subsystem breakdown



Module	Specification points
Roll bodies	Hardened forged steel, ground and polished; surface finish transfers to the electrode
Crown compensation	Adjustable bending or internal hydraulic zones. A straight roll deflects under high load giving a thick centre and thin edges. Fixed crown works at one load only
Loading system	Line pressure up to 2,500 N/mm reference. Measure with bearing-housing load cells
Gap axis	Micron-resolution servo with linear scale feedback
Roll temperature control	Preheat sections and roll tempering optionally available, approximately 50–250 °C. Heated rolls reach the same density at lower load , which reduces cracking risk
Static discharge	Before the nip — the electrode is statically discharged and cleaned by brushes or air streams before calendering
Continuous roll cleaning	A distinct product category. Debris prints a defect repeating at roll circumference
Rewind with taper tension	Prevents crushing inner wraps

Key specifications

Parameter	Reference range
Line pressure, held constant	up to 2,500 N/mm
Calendering speed	60–100 m/min
Roll temperature (optional tempering)	approx. 50–250 °C
Target cathode porosity	25–35 %
Target anode porosity	30–40 %
Cathode density after calendering	3.2–3.6 g/cm ³
Anode density after calendering	1.5–1.7 g/cm ³
Total pass budget	Three or fewer calender passes claimed, counting DE-170 and DE-190

Integrated instrumentation

Instrument	Purpose
Bearing-housing load cells	Actual line pressure — primary control loop
Paired laser triangulation at exit	Post-calender thickness
Beta / XRF gauge	Areal mass; combined with thickness gives density
Ultrasonic pulse-echo	Specify this on a dry line. Sensitive simultaneously to thickness, density and porosity — and porosity is the variable you must engineer rather than inherit. See §19
Rotary encoders both rolls	Speed and ratio; detects slip
Roll surface RTD	Tempering control
Multispectral / polarisation camera	Wavy edge, cracking, foreign matter on a low-contrast surface

Acceptance criteria

- Density and porosity achieved to target, verified against pycnometry on sampled material
- Measured line pressure versus commanded across the working range
- Crown compensation demonstrated at two very different line loads with CD thickness profile held
- No coating cracking or delamination at target density — peel test after calendaring, not only after lamination
- Repeating-defect check against every roll circumference in the path

DE-210 Electrode Slitter

Function: divide the wide mother coil into daughter coils at the width required by the cell format | **Route:** shared
Throughput: 80–150 m/min mechanical | **Footprint:** ~3 × 8 m | **Environment:** ISO 7–8
Utilities: power, extraction (high duty), compressed air | **Maturity:** mature

What it does

Cuts the wide electrode web into narrower webs using rolling knives, with edge-strip suction, and rewinds the daughter coils.

WHY THE EDGE IS A SAFETY-CRITICAL DIMENSION

A 20 µm burr on the edge of a 6 µm copper foil is more than three times the foil thickness and comparable to half the thickness of a 12 µm separator. Under winding tension it will penetrate. **Electrode edge burrs from poor notching and slitting are one of the three primary causes of internal short circuit.** The resulting short may appear during formation, during aging, or eighteen months into field service.

Process physics

Rolling knives cut by localised shear between an upper and lower knife with controlled overlap and side load. The cut quality is governed by the overlap, the side load, the knife edge condition and the support the web receives at the cut point.

WHY A BURR IS A SAFETY DEFECT, NOT A COSMETIC ONE

A 20 µm burr on the edge of a 6 µm copper foil is more than three times the foil thickness and comparable to half the thickness of a 12 µm separator. Under winding tension and the radial pressure of assembly it will penetrate. **Electrode edge burrs from poor notching and slitting are one of the three primary causes of internal short circuit.** The resulting short may appear during formation, during aging, or eighteen months into field service.

A dry-route-specific complication: there is no continuous polymer binder film holding the coating edge together, so the cut edge of a dry electrode is more friable than a solvent-cast one. Dust generation and edge integrity must be re-qualified rather than assumed from wet-route experience.

Subsystem breakdown

Module	Specification points
Unwind with tension control	Feeds the mother coil
Rotary knife assembly	Upper and lower knives with controlled overlap and side load. Micron-level setting
Motorised knife positioning	Substantially shortens width changeover; verify positions against encoders
Edge-trim removal	Suction for the cut edge strips — mandatory
Dust extraction	Higher duty on a dry electrode: no binder film holds the edge together, so more friable dust is generated. Re-qualify against wet-route experience
Web cleaner	Daughter coils cleaned after slitting
Multi-shaft rewind	Alternating shafts so adjacent slits wind without interference

Key specifications

Parameter	Reference
Mechanical cutting speed	80–150 m/min
Cutting width tolerance	±150 to ±250 µm
Burr height	Specify a maximum and the measurement method. Zero-tolerance in practice
Edge-strip suction	Mandatory
Knife life	Stated in metres of web for your material. Change on count, not on defect

Integrated instrumentation

Instrument	Purpose
Line-scan camera with dark-field illumination	Burr detection. Grazing illumination is what makes a burr visible; bright field will not see it
Laser width measurement	Slit width verification
Inline sampling microscope	Periodic quantitative burr height
Particle counter in extraction duct	Rising dust is the early warning of knife wear, before burrs appear
Knife position encoders	Commanded versus actual slit positions
Tension load cells per daughter web	Each web needs its own tension

Options and variants

Option	Assessment
Laser slitting	No tool wear, no mechanical burr. Introduces a heat-affected zone and vaporised material to manage. A recognised innovation direction
Integration into upstream process	Integrating the slitter into upstream steps eliminates the manual coil transport that otherwise exists between calendaring and slitting — a known contamination and damage point

Acceptance criteria

- Burr height measured by microscopy at multiple CD positions — not one
- Slit width tolerance verified over a sustained run
- Dust generation baseline recorded at the extraction duct for future trending
- Knife life established in metres, then converted to a scheduled change interval
- Daughter coil quality: no telescoping, correct taper tension

DE-220 Vacuum Conditioning Oven / Cell Baking Unit
Function: remove ambient moisture from the electrode before it enters the dry room Route: shared, reduced duty in dry Throughput: batch, 8–48 h reference Footprint: ~3 × 5 m per chamber Environment: encapsulated Utilities: power, vacuum, N ₂ , chilled water Maturity: mature

What it does

Removes residual moisture by evaporation at low temperature under low total pressure. Frequently also serves as the **air lock into the dry room** — one machine doing two jobs, which is worth designing for deliberately.

WHY THIS MACHINE SURVIVES IN THE DRY ROUTE — WITH A SMALLER DUTY
Deleting the drying oven also deletes the step that incidentally guaranteed a bone-dry electrode. In the dry route the electrode has never been heated, so it carries whatever moisture the powder and the room gave it. The duty here changes from <i>solvent-and-moisture removal</i> to <i>ambient-moisture removal only</i> — which usually means shorter cycles and lower temperatures, and makes this the strongest candidate for conversion to the cell baking approach: shifting the operation to immediately before electrolyte filling to reduce chamber size and optimise the process.

Process physics

Lowering total pressure lowers the boiling point of water, so moisture can be removed at a temperature that would not otherwise drive it off — and without the thermal damage an equivalent atmospheric bake would cause. Because the coils are wound, moisture must also diffuse laterally out of the roll, which is why the process takes hours rather than minutes and why coil tightness affects cycle time.

WHY THIS STATION SURVIVES ON A ROUTE WITH NO SOLVENT
Deleting the drying oven also deletes the step that incidentally guaranteed a bone-dry electrode. On the dry route the electrode has never been heated, so it carries whatever moisture the powder and the room gave it. Water inside a finished cell reacts with LiPF ₆ to form hydrofluoric acid , which corrodes the cathode and dissolves transition metals. That single reaction is why the dry room, the desiccant plant and this oven all exist.

Subsystem breakdown

Module	Specification points
Vacuum chamber	Sized for a rack of coils on a goods carrier
Heated shelves or radiant panels	Supply the latent heat of evaporation. <i>Constant heat supply</i> is a stated process requirement
Pump set	Dry screw plus Roots booster to cover the wide band from 0.07 to 1000 mbar
Cold trap / condenser	Protects the pump from condensing vapour
Inert gas supply	Vacuum dryers can be operated with inert gas to prevent corrosion — specifically copper corrosion on the anode foil
Air-lock function	Double-door configuration into the dry room

Key specifications

Parameter	Reference range
Working pressure	0.07 mbar < p < 1,000 mbar
Drying time	8-48 h per batch (expect the lower end in the dry route)
Drying temperature	60-150 °C
Target residual moisture, cathode	< 50 ppm
Target residual moisture, anode	< 20 ppm
Temperature uniformity across the load	Specify; a cold corner of the rack is an under-dried coil

Integrated instrumentation

Instrument	Purpose
Combined Pirani + capacitance vacuum gauge	Covers the full 0.07-1000 mbar band
Dew-point hygrometer in the exhaust line	Real-time endpoint indication — drying is finished when exhaust dew point stops falling. Far better than a fixed timer
RTDs on heater plates and in the load	Temperature control and uniformity verification
Karl Fischer titrator (offline)	Confirmatory residual moisture on sampled material

Acceptance criteria

- Residual moisture target achieved on a fully loaded chamber, verified by Karl Fischer on coils from the centre and the extremities of the load
- Temperature uniformity mapped across the loaded chamber
- Exhaust dew-point curve recorded to establish the endpoint criterion and the achievable cycle time
- Copper foil inspected for discolouration after a full cycle — confirms inerting adequacy
- Air-lock function verified: dry-room dew point held during a transfer cycle

14.4 Dry electrode block — summary register

Tag	Machine	Route status	Maturity	Governing specification
DE-100	Powder receiving	Shared	Mature	Containment; <20 % RH; segregation
DE-110	LIW dosing	Shared	Mature	±0.1 % of setpoint; binder feeder tighter
DE-120	Sieve + magnetic separation	Shared	Mature	2 stages >10,000 G; <50 ppb incoming
DE-130	Pre-blender	Dry-specific	Mature hw	Low shear; no attrition; blend uniformity
DE-140	Fibrillation mixer	Dry-specific	Low	Residence time distribution; specific shear energy
DE-150	De-agglomeration	Dry-specific	Moderate	Break lumps without further fibrillation
DE-160	Powder feeder	Dry-specific	Moderate	CD feed uniformity → ±1.5 % areal loading
DE-170	Film-forming roll pair	Dry-specific	Moderate-low	Line speed at uniformity; crown; speed ratio
DE-180	Free-film handling	Dry-specific	Moderate	Minimum span; low tension; clearing time
DE-190	Lamination station	Dry-specific	Moderate	Peel adhesion; margin tolerance
DE-200	Final calender	Shared	Mature	Porosity engineered; crown; ≤3 total passes
DE-210	Slitter	Shared	Mature	Burr-free edge; ±150-250 µm width
DE-220	Vacuum conditioning	Shared, reduced	Mature	<20/50 ppm moisture; air-lock function

WHERE TO CONCENTRATE ENGINEERING EFFORT

Four machines carry the technical risk of this block: **DE-140 (fibrillation)**, **DE-160 (feed uniformity)**, **DE-170 (film forming)** and **DE-190 (adhesion)**. Together they are a minority of the block's cost and the overwhelming majority of its risk. Allocate qualification time, pilot trials and contractual performance guarantees accordingly — not in proportion to purchase price.

Part IV — Assembly and Finishing

Everything from here on is common to wet and dry routes. A dry electrode plant still needs all of it. These datasheets are condensed relative to Part 3, because the equipment is mature and multi-sourced — but three stations (AS-300, AS-330, FN-410) carry the bulk of the assembly-side technical risk on a tabless 46-series cell and are treated at full depth.

15. Cell Assembly — AS-300 to AS-390

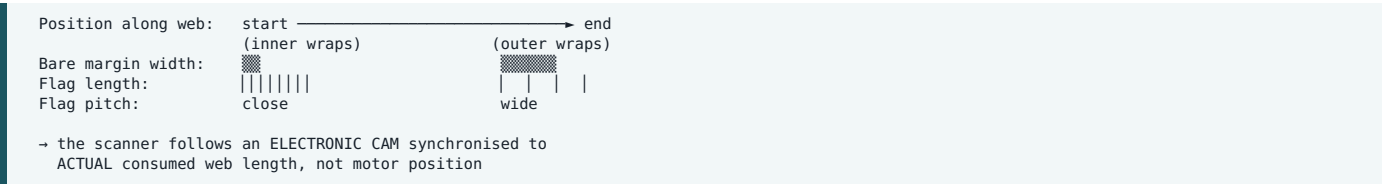
From here the process runs inside the **dry room at –40 to –60 °C dew point**. Cell assembly must be carried out under dry conditions, as water inside the cell leads to severe quality losses over lifetime and a safety risk through the formation of hydrofluoric acid.

THE THREE TECHNOLOGIES THAT GATE 46-SERIES MASS PRODUCTION
Published industry analysis names exactly three: **full-tab forming** (AS-320), **current collector plate welding** (AS-330), and **laser sealing** (AS-390). All three are in this chapter. If a 4680 programme is behind schedule, the cause is almost always one of them.

AS-300 Laser Notching Machine — CRITICAL
Function: cut the variable-pitch tabless flags into the uncoated foil margin | **Route:** shared
Throughput: ~0.2 s per sheet reference | **Footprint:** ~4 × 10 m | **Environment:** min ISO 7, dry room –40 to –60 °C
Utilities: power, N₂, chilled water, extraction, laser safety | **Maturity:** moderate — tabless variant is specialised

What it does
Pre-forms the electrode contour in the web state, cutting the series of flags that will later be folded into the current-collecting plane. Available as a standalone machine or integrated with the winder as a combined notch-and-wind unit.

The variable-pitch requirement



Because the jelly roll is a spiral, each wrap's circumference grows with radius, so outer flags must be longer. The arc-shaped edge of the full-tab coating requires higher equipment precision — the outer ring shows progressively more bare foil than the inner ring, and tab length grows toward the outer ring. **This single requirement is why 4680 notching machines cost substantially more than 21700 notching machines.**

Subsystem breakdown

Module	Specification points
Active unwind with tension control	Automatic tension detection and control
EPC deviation correction	In-process correction and clamping correction
Pulsed fibre laser + galvo scanner	F-theta lens for flat field; dynamic focus for large fields
Master measuring wheel	Low-slip, independent of the drive. The single most important specification item on this machine — motor position and web length diverge as soon as any slip or stretch occurs
Registration mark sensor	Corrects cumulative cam phase error from web stretch
CCD die-cutting size inspection	100 % verification of each flag's size and position, immediately post-cut
Dust removal	Brush, negative-pressure de-dusting, FFU and dust collector
Fume extraction at the cut point	Suction extraction of vaporised material is a stated quality requirement
Nitrogen assist shroud	Prevents edge oxidation, which degrades weldability at AS-330

Key specifications

Parameter	Reference
Separation time	~0.2 s per sheet
Dimensional tolerance	approx. ±200 µm
Focus position	±50 µm — critical
Pulse repetition rate	Tens to hundreds of kHz
Laser power	Set by foil: Al 10–15 µm and Cu 6–8 µm require different settings
Cam profile	Variable pitch, synchronised to measured web length
Format compatibility	Machines are available compatible with 4680, 32140, 40135 and other all-tab and multi-tab cylindrical cells

Quality parameters

- Edge geometry — minimisation of burr formation and defect prevention by suitable radii in the tab area
- Minimisation of thermal and mechanical influences during separation
- Minimisation of the heat-affected zone and extraction of vaporised material

THE DEFECT THAT PROPAGATES AND MULTIPLIES
If flag edges are not aligned, there will be gaps in the tab bonding when they are folded at AS-320. Multi-tab position errors are easily magnified. A small position error here becomes a large gap two stations later, which becomes a high-resistance joint, which becomes a hot spot in service. **Specify 100 % CCD inspection immediately after the cut, not sampling.**

Consumables and acceptance

Consumable	Basis
Laser cover slides	Interval established from measured power decay at the workpiece
Extraction filters	Differential pressure
Brushes / de-dusting media	Counted hours

- Cumulative flag position error over a full coil length — must not accumulate with registration correction enabled
- Burr and HAZ measured by microscopy on both foil types
- Focus repeatability verified across the *full* scan field
- Delivered power measured at the workpiece; cover-slide interval established

AS-310 Jelly Roll Winder

Function: wind anode, cathode and two separators into the jelly roll | **Route:** shared
Throughput: up to 30 cells/min (round cell) | **Footprint:** ~5 × 8 m | **Environment:** min ISO 7, dry room
Utilities: power, compressed air, extraction, ionisation | **Maturity:** mature

Subsystem breakdown



Module	Specification points
Four independent unwinds	Each needs its own tension loop. Separator setpoint is roughly one fifth the electrode setpoint — a single shared setpoint across all four is a design error causing separator necking
Multi-needle turret	Three-needle, three-station machines are standard; this is what makes the cycle time achievable
Winding needle	The Tesla 4680 has no mandrel — the needle is withdrawn after winding
Adaptive tension taper	Radial pressure from each wrap is proportional to tension divided by radius, so constant tension crushes the inner wraps. Closed loop from a roll-diameter laser
End-tape applicator	Ultra-long tape rolls allow 24-hour operation without a tape change, reducing waste and raising utilisation
Tab protection and flattening	Protects projecting flags during handling; some machines include tab folding inspection
Ionisers	Avoidance of electrostatic charging is a stated quality parameter
Dust control	Brush, negative-pressure de-dusting, FFU, dust collector

Key specifications

Parameter	Reference
Machine throughput	up to 30 cells/min (round cell)
Alignment accuracy	< 200-300 µm
Electrode web tension	30-100 N/m, tapering with radius
Separator web tension	10-40 N/m
Finished jelly roll diameter	44.5 mm (4680)
Finished jelly roll height	~71 mm

Acceptance criteria

- Alignment verified by X-ray on finished rolls from start, middle and end of a coil
- Tension taper profile recorded against measured diameter
- Separator width measured before and after — confirms no necking
- Ioniser effectiveness verified with an electrostatic field meter
- Roll dimensional consistency over a sustained run

AS-320 Tab-Forming Press

Function: convert projecting flags into a dense, flat, weldable conductive plane | **Route:** shared
Throughput: matched to winder | **Footprint:** ~2 × 3 m | **Environment:** dry room
Utilities: power, compressed air, extraction | **Maturity:** moderate

Two methods — a genuine architecture choice

Method	Description	Advantage	Disadvantage
Cutting and folding	Flags trimmed and folded inward	Shorter full-tab electrolyte injection time — folding preserves capillary paths	Demands high flag position accuracy; folding must be neat
Flattening	The end face of the wound core is mechanically flattened into a dense plane	Mechanically simpler; more tolerant of flag position error	Closes capillary paths; electrolyte penetration is slower

Tesla adopted the cutting-and-folding full-tab technology, with the stated aim of improving production efficiency.

THIS IS A DIRECT TRADE BETWEEN TWO LATER STATIONS

Flattening makes welding easier and filling harder. Folding makes welding harder and filling easier. Because electrolyte filling and wetting sits on the critical path of a three-week finishing phase, while welding sits on a station that can be duplicated, the economics generally favour folding. Model it rather than assuming it.

Subsystem breakdown and instrumentation

Module	Specification points
Trim station	Cuts flags to a defined length
Folding tool	Progressive die or rotary folding head; uniform fold around the circumference
Compaction press	Load cell recording the force-displacement curve for every part
3D laser profilometer	Flatness and height of the finished plane — the direct predictor of AS-330 weld quality
Tab folding inspection camera	Detects unfolded, mis-folded or torn flags before the welder

Key specifications and acceptance

Parameter	Requirement
Compaction force	Sufficient for metal-to-metal contact; below foil tearing threshold
Final tab-plane height	Tolerance in the tens of microns — it sets the weld focus position at AS-330
Plane flatness	Directly determines weld penetration consistency
Fold angle uniformity	Uniform around the full circumference

- Tab-plane height and flatness measured on a statistically meaningful sample and correlated to AS-330 OCT depth
- Force-displacement curves recorded and a reference envelope established
- No torn flags at the specified force

AS-330 Collector Disc Laser Welder — CRITICAL

Function: join the current collector disc to the folded tab plane | **Route:** shared
Throughput: matched to line; multiple heads typical | **Footprint:** ~3 × 4 m per station | **Environment:** dry room, Class 1 laser enclosure
Utilities: power, chilled water, Ar/N₂, extraction | **Maturity:** moderate — the lowest-yielding station on the line

THE HARDEST PROCESS ON THE LINE

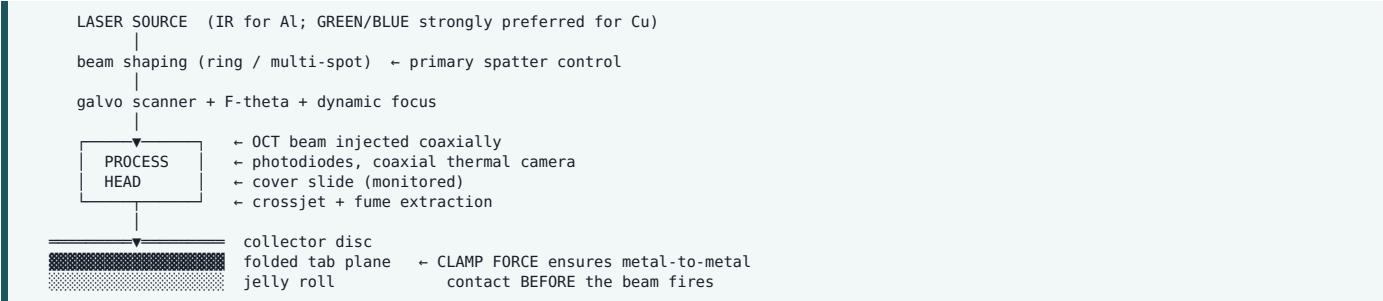
Laser welding is the most important welding process in 4680 production, and the collector disc welding process is very complicated. The transition is from *spot welding* (two conventional tabs) to *surface welding* (full tabs). Both weld process and weld volume increase substantially, the number of joints increases, **laser intensity and focal length are not easy to control**, and it is easy to cause **false welding or over welding**. Reported cell yield during ramp has been around 80 %, traceable substantially to this station.

Over-welding is the dangerous one. Excess penetration reaches through the tab plane into the jelly roll and can perforate the separator. The cell may pass every end-of-line test and fail catastrophically in service.

Production geometry (reference)

Feature	Configuration
Weld pattern, Gen 1	Five zig-zag seams per connector, approximately 0.5 mm wide
Cathode side	Connector joined at the outer ring; centre ultrasonically welded to the positive terminal
Anode side	Connector joined toward the centre; outer ring connected to the housing
Gen 2 anode	Collectors welded directly to the bottom lid using 12 welding seams

Subsystem breakdown



Key specifications

Parameter	Setting / note
Laser power	Several hundred watts to several kW, set for controlled penetration
Wavelength	IR acceptable for aluminium. Green (515/532 nm) or blue (~450 nm) strongly preferred for copper — cold copper absorbs several times more, so the process starts smoothly and stays stable. If anode-side penetration is inconsistent on an IR system, wavelength is the parameter to reconsider, not power
Focus position	±50 µm — deviation causes false welds

Beam shaping	Ring or multi-spot. Flexible beam guidance and shaping is a stated requirement
Wobble amplitude / frequency	Bridges fit-up gaps; widens the tolerance window
Shielding gas	Argon or nitrogen
Clamp force	Must guarantee metal-to-metal contact before welding. A gap of tens of microns causes a false weld
Target penetration depth	Must be less than the tab-plane thickness. The hard safety constraint

Integrated instrumentation — this station needs the most

Instrument	Role
Inline OCT (optical coherence tomography)	Measures keyhole depth in real time at the process point. The only method that measures penetration <i>directly</i> rather than inferring it. Enables closed-loop penetration control. The single highest-value instrument on a 46-series assembly line
Multispectral photodiodes	Plasma and melt radiation; stability and spatter event detection
Coaxial thermal camera	Melt pool map; fit-up variation detection
Post-weld camera / profiler	Crack, porosity, spatter, seam geometry
4-wire milliohm meter	Joint resistance — the functional verification
Clamp force load cell	Contact force before the beam fires
Cover slide monitor	Prevents slow, invisible power loss
Sampling pull test	The reference for correlating all inline signals

Consumables and wear parts

Item	Basis
Cover slides	Interval from measured power decay. Stock generously
Crossjet nozzles	Spatter erosion; counted hours
Clamp tooling	Wear affects contact force distribution; dimensional inspection at interval
Extraction filters	Differential pressure

Acceptance criteria

- **Penetration map:** power × speed with OCT depth measured, validated against metallographic cross-sections at selected points
- Process window **boundaries** documented explicitly — minimum energy for full bonding, maximum before through-penetration — with the operating point set at the centre, not at an edge
- OCT depth, photodiode signal, joint resistance and pull strength correlated on the same samples
- Clamp force requirement established by deliberately introducing gaps and measuring the resulting false-weld rate
- Cover-slide interval established from measured power decay
- Characterisation performed on coupons representing the *actual folded tab plane*, not flat foil stacks — compaction state matters

15.1 Remaining assembly stations — condensed datasheets

AS-340 Can Deep-Draw Press Line	
Function: form the cylindrical can from strip Environment: parallel line, normal environment Maturity: mature	
Aspect	Specification
Route, steel	Deep drawing, multi-stage. Typically 3–6 draws for an 80/46 aspect ratio
Route, aluminium	Impact extrusion
Material	Nickel-plated steel (Hilumin) — nickel gives corrosion resistance, steel gives strength for internal pressure and mechanical stress. Or aluminium. Some 46 mm cells use different materials to meet specific requirements
Wall thickness	~0.5 mm, tolerance ±20 µm
Roundness	< 50 µm — out-of-round cans jam at AS-350
Lubrication	Must be completely washable. Residue contaminates electrolyte and defeats welding
Instrumentation	Press load cell recording the full force–displacement curve for every part (a wrinkled or thinned draw shows in the curve before it shows in a measurement); ultrasonic wall thickness gauge; vision for diameter and roundness; leak detection for cracked cans
Acceptance	Wall thickness and roundness across a production batch; washability verified by residue analysis; force–displacement reference envelope established

AS-345 Cap Assembly and Vent Scoring Unit	
Function: produce the cap with its safety vent score Maturity: mature	

Aspect	Specification
Vent score	The explosion-proof line is etched on the cover plate; the cover plate connects to a notch in the housing. Laser-etched or pressed
Score depth tolerance	Tens of microns. Too shallow: no vent at design pressure. Too deep: premature leak. Both failure modes are safety-relevant
Instrumentation	Laser profilometer for score geometry; vision for crack detection; sampled burst pressure test — the only true functional verification
Acceptance	Score depth mapped across the score length; burst pressure distribution measured against specification with a stated capability index

AS-350 Jelly Roll Insertion Station

Function: insert the jelly roll and insulators into the can | **Environment:** dry room | **Maturity:** mature

Aspect	Specification
Sequence	Insulator ring placed on the anode-tab side; can pushed over the jelly roll so the insulator rests on the can bottom; connections welded; second insulator placed. The stack is often additionally wrapped in insulation foil
Critical quality parameter	Low mechanical stress on the jelly roll during handling — specifically avoidance of shifting of the layers due to friction during insertion . Layer shift destroys the alignment achieved at winding and raises internal short risk
Instrumentation	Insertion force load cell on every part — a sudden rise means binding, and the part should be rejected rather than forced. Coaxiality control; insulator position camera; sampling X-ray to confirm no layer shift
Acceptance	Force curve envelope established; X-ray confirmation of internal position on a sample; reject-on-force-rise logic demonstrated

AS-360 Can and Terminal Welding Station

Function: join collectors to can and terminal; pre-weld the cap | **Environment:** dry room | **Maturity:** moderate

Aspect	Specification
Typical sequence	(1) anode collector to can wall or base, laser welded from outside the can ; (2) cathode collector to positive terminal, ultrasonic or laser; (3) cap pre-welding; (4) final sealing weld after filling
Welding through the wall	Penetration must equal wall thickness plus a small controlled excess — not less (false weld), not much more (perforation or jelly-roll damage). The window is a few tenths of a millimetre wide. OCT depth monitoring converts this from unmanageable to controllable
Ultrasonic parameters	20 or 40 kHz; amplitude 10–60 µm; pressure; energy or displacement control rather than fixed time , which compensates stack-height variation
Quality parameters	Material compatibility at joint areas; ultrasonic parameter set (frequency, amplitude, pressure, time); flexible beam guidance and shaping for laser welding; tightness of the weld seam between housing and cap assembly
Acceptance	Penetration map versus wall thickness; joint resistance distribution; pull test correlation; ultrasonic energy control repeatability across deliberate stack-height variation

AS-370 Beading and Swaging Machine

Function: form the circumferential bead and reduce the opening angle | **Maturity:** mature

Aspect	Specification
Operation	A bead is impressed into the can body as a seat for gasket and cap. If the can opening is conical, the angle can be reduced to 90 degrees by swaging
Tolerances	Bead depth to tens of microns; bead distance from rim sets the crimp or weld geometry; forming force monitored per part
Zero-tolerance defect	Cracking. A crack in the bead is a guaranteed leak path
Instrumentation	Laser profilometer for bead geometry; vision for crack detection; sampling X-ray

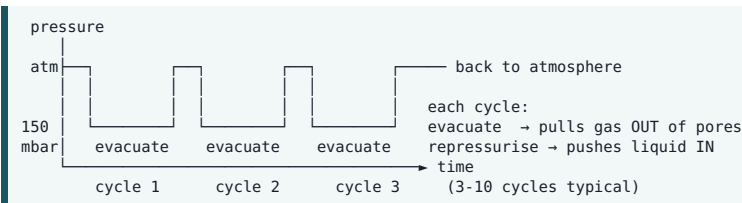
AS-380 Electrolyte Filling Machine

Function: dose electrolyte and drive it into the electrode pores | **Environment:** particularly dry / inert | **Maturity:** mature
Utilities: vacuum, N₂, extraction, containment, specialised fire suppression

Process physics

Filling is the easy half. **Wetting** is the hard half. Electrolyte must penetrate pores of order 100 nm to 1 µm throughout a jelly roll 44.5 mm in diameter and 71 mm tall. Capillary action drives it, but capillary flow through a long, tortuous, low-porosity path is slow. The vacuum-pressure cycle works by first evacuating gas out of the pores, then using the pressure differential on repressurisation to drive liquid in.

Process and pressure profile



Aspect	Specification
Working pressure	approximately 150 mbar
Dosing accuracy	±0.5 % or better; gravimetric measurement is the stated method
Vacuum-pressure cycles	3–10 typical; partial filling and wetting can be repeated several times
Pressure instrument	Capacitance manometer — gas-species independent, essential at 150 mbar with solvent vapour present. A Pirani gauge reads incorrectly here
Lance positioning	Precision of positioning significantly determines electrolyte contamination on the packaging. A drop on a surface that must later be welded ruins the seal weld. Instrument and verify per cycle , not once at commissioning
46-series-specific challenge	The compacted tab plane restricts penetration paths; wetting time is significantly longer than for a 21700. This is the direct cost of the AS-320 method choice
Emerging instrumentation	Inline wetting degree measurement is a named innovation direction — ultrasonic or impedance based
Safety	Tightened safety measures required specifically because of electrolyte use: vapour detection, spill containment, appropriate fire suppression. Calcium gluconate gel immediately available for HF exposure
Acceptance	Dosed mass distribution over a sustained run; lance position repeatability; wetting verified indirectly by EoL capacity on a build; no residue on sealing surfaces

AS-390 Sealing and Helium Leak Test Station

Function: close the cell and verify tightness | **Maturity:** mature

Aspect	Specification
Sealing methods	Crimping (Gen 1 — in the Tesla 4680 the crimped closure is on the negative, bottom side); laser seal welding (Gen 2); sealing nail welding of the fill port
Leak detector	Magnetic-sector mass spectrometer tuned to mass 4
Sensitivity	10 ^{−8} to 10 ^{−10} mbar·L/s
Typical accept limit	< 10 ^{−6} mbar·L/s
Cycle time	Seconds to tens of seconds; chamber pump-down dominates. Multi-chamber carousels hide it
Test placement	Two test points. After cap pre-welding <i>before filling</i> — catches a leaking cell before electrolyte is committed, avoiding a scrapped filled cell and an electrolyte spill. And after final sealing

REFERENCE ASSEMBLY LINE CONFIGURATION

A representative 46-series assembly line comprises: hot press, X-ray testing machine, tab pre-welding machine, soft-connection laser welding machine, Mylar wrapping machine, shell pre-welding machine, cap seal welding machine, and first helium testing machine. Reported single-line capacity for a high-speed intelligent 46-series line is on the order of **150 PPM**. Processes covered include welding of positive and negative current collectors to the core, collector-to-shell-inner-wall welding, collector-to-positive-pole welding, cap pre-welding, sealing welding and sealing nail welding.

16. Cell Finishing — FN-400 to FN-450

The cell is sealed, so the environment requirement relaxes: cell finishing takes place in a normal environment because the cell is already sealed and degassing occurs in a vacuum chamber, giving lower requirements for particle environment and humidity. But this is where a third of the capital sits.

FN-400 High-Temperature Soaking Chamber	
Function: accelerate electrolyte penetration into the electrode pores Maturity: mature	
Aspect	Specification
Principle	Raising temperature reduces electrolyte viscosity, improving contact and access to the mesopores of the electrode, so penetration into electrode and separator pores is faster
Temperature	30–50 °C, homogeneously applied
Uniformity	±2 °C across the rack — specify and verify by mapping
Alternatives	Roll pressing (pouch); vibrating table (prismatic and cylindrical); pre-charging (all formats)
Acceptance	Temperature map across a fully loaded chamber; soaking effectiveness verified indirectly by EoL capacity on a controlled build

FN-410 Formation System — LARGEST CAPITAL ITEM	
Function: initial charge/discharge cycles that build the SEI Duration: up to 15 h per cell	
Footprint: large — sized by channel count, see §4.3.1 Utilities: power (very high), HVAC, gas detection	
CAPEX: EUR 65–85 m for a 10 GWh/y line — the most expensive station in the plant	

Process physics

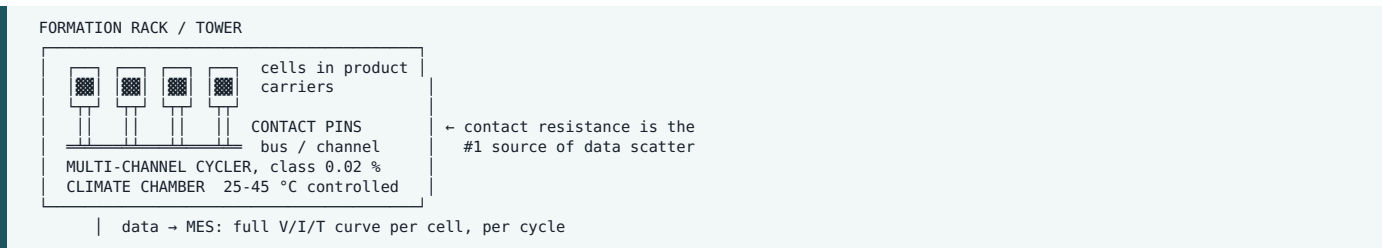
During formation, lithium ions are deposited into the crystal structure of the graphite on the anode side. Simultaneously the electrolyte decomposes at the anode surface, and its decomposition products form the **solid electrolyte interphase (SEI)**.

The SEI is a compromise made physical. It must be **electronically insulating**, so decomposition stops rather than continuing forever; **ionically conducting**, so the cell works; **mechanically flexible**, so it survives the graphite's volume change on cycling; and **thin**, because every molecule of SEI consumes lithium that is then permanently unavailable.

WHY THE FIRST CYCLE MUST BE SLOW
SEI quality is governed by nucleation and growth kinetics. At low current, decomposition proceeds slowly and the layer grows uniformly and densely from a large number of nucleation sites. At high current, local overpotential drives rapid heterogeneous growth: the layer becomes thick, porous and non-uniform, consuming more lithium (higher first-cycle loss) and providing worse long-term protection. In the worst case local overpotential drives lithium <i>plating</i> rather than intercalation — metallic lithium that will nucleate dendrites over subsequent cycles.

THE MOST EXPENSIVE TRADE ON THE LINE
Every hour of formation is rack capital, electricity, floor space and working capital. Every increase in rate risks a worse SEI and shorter cycle life — consequences that appear months or years later, far too late to correct. At EUR 65–85 m for this station, halving formation time is worth tens of millions. That is exactly why every manufacturer works on it, and exactly why none publish the answer.

Machine architecture



Specification	Value / note
First charging cycle	approximately 0.1–0.5 C
Subsequent cycles	Successive increase of C-rates with each charge/discharge cycle
Process duration	up to 15 h
Cycler accuracy class	0.02 % current and voltage. Channel accuracy sets grading resolution — a 0.1 % cycler cannot grade into 0.5 % capacity bins
Contact pins	Low contact resistance is a stated process requirement. Pin wear and oxidation is the leading source of data scatter. Treat pins as a consumable with a counted life, not as machine structure
Chamber temperature	Controlled, typically 25–45 °C
Channel density	Channels per square metre drives the capital cost per GWh — a key commercial comparison between suppliers
Throughput reference	A 355 PPM formation and grading line has been delivered, covering 18650/21700 and 4680 cylindrical cells
Instrumentation	DCIR at defined points; EIS per cell after formation , recommended for high-end EV applications specifically to detect micro-shorts; NTC per cell; thermal camera scanning the rack
Safety	Gas detection (H ₂ , CO); exhaust after-treatment (RTO) where required; lockout/tagout for high voltage

WHAT THE SUPPLIER CANNOT GIVE YOU

Formation parameters vary by manufacturer, have a great impact on cell performance, and **represent the core knowledge of a cell manufacturer**. The supplier sells you channels, accuracy, throughput and data. The *protocol* is yours to develop by designed experiment on your own chemistry — budget months, not weeks. Do not accept a supplier's "standard recipe" as a substitute.

Acceptance criteria

- Channel-to-channel accuracy verified against a calibrated reference load across the current and voltage range
- Contact resistance measured across a full rack; distribution recorded as the baseline for pin-life monitoring
- Chamber temperature uniformity mapped under full electrical load, not empty
- Data integrity: full V/I/T curve captured per cell per cycle and delivered to MES without loss at full channel count

FN-420 Degassing Unit

Function: remove gas generated during formation | **Applicability:** format-dependent | **Maturity:** mature

Aspect	Specification
Cylindrical applicability	Cylindrical cells typically retain gas internally, relying on the safety vent. In smaller cells with lower gas generation, and depending on manufacturer, gas is not separated after degassing
Plan for it anyway	The stated innovation direction is integration of degassing steps for ever larger cylindrical formats . A 4680 generates roughly five times the gas of a 21700; at 46120 the quantity is harder to ignore. Reserve the footprint even if the current product does not need it
Requirements when performed	Particularly dry or inert environment; sealing against moisture and oxygen; residual gas is the key quality parameter; damage-free handling
Exhaust	Extracted gases must be after-treated (e.g. by RTO) before entering the exhaust system, depending on occupational health and environmental regulations

FN-430 Aging Racks and Towers

Function: detect latent internal defects through self-discharge monitoring | **Duration:** up to 3 weeks
Footprint: warehouse-scale — ~6.9 million positions for a 10 GWh/y 4680 line (§4.3.1) | **CAPEX:** EUR 15-30 m

Process physics

A cell with a latent internal short discharges itself faster than a good cell. Over days to weeks this difference becomes measurable as an abnormal decline in open-circuit voltage. Aging is, in essence, a very slow and very sensitive short-circuit test that simultaneously allows the SEI to stabilise.

Specification	Value / note
Method	High-temperature (HT) and room-temperature (RT) aging. Cell properties monitored by regular OCV measurements. If no significant change occurs over the period, the cell is OK and can be shipped
SOC at start of aging	30-80 %
Aging time	up to 3 weeks
Room temperature	approximately 20-25 °C
High temperature	approximately 30-50 °C
Decisive quality criterion	Self-discharge below 5 mV/week. An increased loss rate above 5 mV per week may indicate cell-internal short circuits
Voltmeter	6.5-digit DMM or dedicated multi-channel voltmeter, microvolt resolution
Rack temperature uniformity	Part of the measurement system, not a comfort setting. A 1 °C difference across a rack produces an OCV difference of the same order as the 5 mV/week criterion itself. Specify and map it
Safety	Gas detection (H ₂ , CO); thermal camera tower scans; fire strategy designed for a store holding megawatt-hours of charged cells, not for a general warehouse

THE HIGHEST-LEVERAGE COST REDUCTION IN THE PLANT

Aging is three weeks of finished-goods inventory, a warehouse of racks and a delay between spending money and receiving it. The named innovation direction is **measurement technology for early quality detection** — identifying the defective cell in the first hours rather than after three weeks. Any supplier offering credible early-detection capability deserves serious evaluation, because the prize is enormous.

FN-440 End-of-Line Tester and Grader

Function: verify specification and classify cells | **CAPEX:** EUR 5-15 m | **Maturity:** mature

Test	Purpose
Pulse test	Dynamic response, power capability
Internal resistance (DC)	Joint quality, electrode contact, wetting completeness
Optical inspection	External damage, contamination, marking legibility
OCV test	Final self-discharge confirmation
Capacity test	Primary grading parameter
Leak test	Seal integrity
Hi-Pot	Latent internal shorts, at 500-1000 V

After testing, cells are discharged to the shipping state of charge — typically around 30 % SOC per transport regulations. Many manufacturers

grade cells according to performance data so that matched cells go into the same pack. The grading algorithm is proprietary and is yours to develop.

FN-450 Marking and Packing Line	
Function: apply unique identity and prepare cells for shipment Maturity: mature	
Aspect	Specification
Marking	Laser-etched DataMatrix on the can. Applied before assembly so the identity exists for the whole genealogy chain, not just at the end
Verification	Read-after-write on 100 % of cells; an unreadable code breaks traceability permanently
Packing	Cells covered with a plastic cover and stacked in a cardboard box for transport
Documentation	UN 38.3 test summary must accompany shipments

Part V — Measurement

17. The Dry Electrode Inspection Problem

17.0 Measurement fundamentals

Term	Definition	Practical implication
Resolution	Smallest change the instrument can display	Sales literature quotes this. It says nothing about correctness
Repeatability	Spread of repeated measurements of the same thing	Determines whether you can detect a real process shift
Reproducibility	Spread across operators, instruments or days	The one that kills multi-shift comparisons
Accuracy / bias	Difference from the true value	Only knowable against a traceable standard
Stability / drift	Change in bias over time	Sets the calibration interval
Spot size	The area actually averaged by the measurement	A gauge with a 10 mm spot cannot see a 2 mm edge defect regardless of its resolution

GAUGE R&R RULE OF THUMB

Before trusting any inline gauge to drive a control loop, run a gauge repeatability and reproducibility study. Target total measurement variation below 10 % of the process tolerance band; 10-30 % is conditionally acceptable; above 30 % the gauge is measuring itself, not the process. Lines frequently run closed-loop control on gauges that have never had an R&R study, then blame the process for the resulting oscillation.

CONVENTIONAL MACHINE VISION DOES NOT WORK ON A DRY ELECTRODE

Without solvent there is **no contrast under standard RGB illumination**. A dry-coated electrode is a matte, uniformly dark surface with essentially no optical signature distinguishing a good region from a defective one. A line-scan camera and lighting package specified from wet-route experience will detect almost nothing on dry material.

The industrial answer is **computational imaging**: multi-spectral and polarisation-based imaging fused via neural networks, which unlocks dry-electrode inspection. Anyone budgeting a dry electrode line must budget this class of system, not a conventional installation.

17.1 Three measurement gaps unique to the dry route

Gap	Why it exists	Equipment answer
Optical contrast	No solvent, no gloss differential, no colour variation	Multi-spectral and polarisation imaging with neural-network fusion
Porosity	In the wet route, porosity is created by solvent evaporation and can be inferred from mass/thickness with reasonable confidence. In the dry route it must be <i>engineered</i> by roll pressure and the inference is less reliable	Ultrasonic pulse-echo, which responds to thickness, density and porosity together. Calibrated against pycnometry
Adhesion	The bond is purely mechanical. A perfect-looking electrode can have almost no bond to its foil, and no inline gauge sees it	Automated sampling 90° peel tester adjacent to the line, feeding the MES with a control limit

FIBRILLATION STATE HAS NO INLINE SENSOR AT ALL

There is currently no production sensor that measures fibrillation directly. The practical qualification set is offline: SEM imaging of the powder, green-film tensile strength, film formability trial, electrode resistivity, and powder flow / bulk density as a calibrated proxy. Plan the laboratory accordingly — a dry electrode plant needs more analytical capacity than a wet one, not less.

18. Metrology Equipment Catalogue and Priority

18.1 MT-1xx — Mass and thickness gauges

Tag	Instrument	Performance	Notes for a dry line
MT-101	Beta transmission gauge	±0.2 % of coat weight	Industry standard for areal mass. Measures the quantity that matters, independent of surface finish. Requires radioactive source licensing, dose monitoring and trained personnel
MT-102	X-ray / XRF gauge	±0.3 %	No radioactive source, so no licence burden. XRF adds elemental information. Sensitive to formulation change
MT-103	Confocal chromatic laser	Sub-micron	Excellent at a point. Struggles on strongly absorbing diffuse black coatings — verify on <i>your</i> dry material before ordering
MT-104	Paired laser triangulation	±0.5 µm	Robust, works on dark surfaces. Needs a thermally stable C-frame; frame growth appears directly as thickness error. Specify frame temperature compensation
MT-105	Ultrasonic pulse-echo	Method-dependent	High priority on a dry line. Sensitive simultaneously to thickness, density and porosity. Frequency-domain analysis with machine-learning inversion is an active development area; open datasets exist for benchmarking
MT-106	Eddy current	Micron-level	Measures the metal foil only — separates foil thickness from film thickness when paired with another sensor

MEASURE TWO, COMPUTE THE THIRD

The parameter that determines cell performance is **density**, and no instrument measures it directly inline. Measure *areal mass* (MT-101/102) and *thickness* (MT-104) at the same web position and compute density. A single-sensor installation cannot distinguish “more material” from “same material, less compacted” — two conditions with opposite corrective actions.

Scan speed and coverage — a specification most people omit

Thickness gauges scan perpendicular to the processing direction, tracing a zig-zag path. At slower scan speeds, gaps in the pattern can leave defects undetected. At faster speeds, sensor precision falls because averaging time per point drops. Sensor performance depends on resolution, precision and beam size. **Record the scan pattern and coverage fraction as part of the recipe** — changing line speed without changing scan speed silently changes detection capability.

18.2 MT-2xx — Optical inspection

Tag	Instrument	Application and notes
MT-201	Line-scan camera	High-speed high-resolution web imaging, resolutions to 16k pixels. On dry electrode, useful only with the illumination below
MT-202	Contact image sensor (CIS)	Telecentric imaging in space-limited positions; no perspective distortion, so well suited to metrology including margin inspection where edges are measured dimensionally — directly relevant to the DE-190 margin specification
MT-203	Area-scan camera	Discrete parts: caps, cans, jelly-roll ends, tab folding
MT-204	3D laser profiler	Height maps: tab-plane flatness (AS-320), bead geometry (AS-370), weld seam profile
MT-205	Multispectral / polarisation imaging	The enabling technology for dry-electrode inspection. Multiple wavelength bands or polarisation states fused by a neural network
MT-206	Thermal imaging	Roll temperature uniformity, formation rack hot-cell scanning, weld monitoring

Illumination — the half of vision that gets neglected

Illumination	Reveals	Use for
Bright field (coaxial)	Reflectivity change	Contamination, stains. Weak on dry electrode
Dark field (grazing)	Surface relief and edges	Burrs , scratches, particles, cracks — the single best method for slitting edge inspection at DE-210
Backlight (transmission)	Through-holes	Foil pinholes, separator defects
Structured light	3D shape	Tab folding, bead geometry
Multi-angle / photometric stereo	Surface normals	Distinguishes a stain (flat) from a bump (raised) — different causes, different actions

TWO NUMBERS TO DEMAND FROM ANY VISION SUPPLIER

Detection rate alone is meaningless. Insist on both: **detection rate** at your smallest defect of interest, and **false-call rate** at that setting — both measured on *your* material under *your* lighting, on a seeded-defect coil that you supply. Current-generation systems achieve sub-50 ms inference at the edge with detection above 99.5 % on defects below 0.05 mm, but a system with a 2 % false-call rate will scrap so much good product that operators will raise the threshold until it detects nothing.

18.3 MT-3xx — Non-destructive internal inspection

Tag	Instrument	Throughput	Notes
MT-301	2D X-ray radiography	100 % capable on many lines	Electrode overhang and alignment, jelly-roll position, foreign objects, gross weld defects. The practical default
MT-302	Inline CT	~10-15 cells/min	Full 3D internal structure of a closed cell. System pricing pushed below approximately USD 400k, which is what changes CT from a sampling tool to a potentially inline one

MT-303	Laboratory micro-CT	Offline	Highest resolution; failure analysis
MT-304	Ultrasonic sealed-cell inspection	Emerging	Electrolyte wetting state and gas pocket presence — both otherwise invisible

THE BLIND SPOT THIS CLOSES

Inline inspection is already used for the optically exposed early steps. The problem is quality control **after assembly and before finalisation** — once the electrodes and electrolyte are inside a closed housing, no optical method sees anything. Development work targets X-ray CT systems enabling non-destructive testing directly within line production at the end of assembly, precisely to close this gap.

18.4 MT-4xx — Process monitoring

Tag	Instrument	Role
MT-401	Inline OCT weld monitor	Keyhole depth in real time at the process point. The only method measuring penetration <i>directly</i> . Enables closed-loop penetration control. The highest-value instrument on the assembly line (AS-330, AS-360)
MT-402	Multispectral photodiodes	Plasma and melt radiation; instability and spatter detection. Cheap, fast, indirect
MT-403	Coaxial thermal camera	Melt-pool geometry; fit-up variation
MT-404	4-wire milliohm meter	Joint resistance — the functional test
MT-405	Torque / net power monitoring	DE-130 and DE-140 batch fingerprinting; the basis of specific shear energy calculation
MT-406	Load cells (bearing housing)	Actual line pressure on DE-170, DE-190, DE-200
MT-407	Electrostatic field meter	Verifies ioniser effectiveness. Static is invisible until it has already attracted contamination or torn a film

18.5 MT-5xx — Electrical test

Tag	Measurement	Instrument	Detects
MT-501	OCV	6.5-digit DMM or multi-channel voltmeter	Self-discharge during aging. Allowable rate below 5 mV/week ; above that may indicate cell-internal short circuits
MT-502	DCIR	Cycler with pulse capability	Joint quality, electrode contact, wetting completeness
MT-503	ACIR (1 kHz)	Milliohm meter	Fast production screen; less informative than DCIR
MT-504	EIS	Potentiostat / FRA	Separates ohmic, charge-transfer and diffusion contributions. Recommended per-cell after formation for high-end EV applications to detect micro-shorts
MT-505	Hi-Pot	High-voltage insulation tester	Latent internal shorts, at 500–1000 V
MT-506	Capacity	Formation-grade cycler	Primary grading parameter

18.6 MT-6xx — Environmental and safety instruments

Tag	Instrument	Application
MT-601	Chilled-mirror hygrometer	The reference dew-point method. Slow but accurate; used to verify the fast sensors
MT-602	Capacitive dew-point sensor	Fast, cheap, drifts. Suitable for control if verified regularly against MT-601. Required range to -60°C
MT-603	Airborne particle counter	Cleanroom classification and continuous monitoring at critical points
MT-604	Differential pressure transmitter	Room pressurisation and filter loading. The first indicator of a dry-room boundary failure — alarms before dew point moves
MT-605	Oxygen analyser	Nitrogen inerting verification on DE-130, DE-140 and conveying
MT-606	Combustible dust / spark detector	Duct monitoring downstream of laser and mechanical cutting
MT-607	VOC / electrolyte vapour detector	Leak detection around AS-380
MT-608	H ₂ and CO detector	Formation and aging areas; early thermal-event warning
MT-609	Helium mass spectrometer	10 ⁻⁸ to 10 ⁻¹⁰ mbar-L/s. AS-390 leak testing

18.7 MT-7xx — Laboratory instruments

Tag	Instrument	Use in a dry electrode plant
MT-701	Karl Fischer coulometric titrator with oven sampler	Incoming materials, electrode after DE-220, electrolyte. The most-used lab instrument on any cell line
MT-702	Laser diffraction particle sizer	Incoming active material; and before/after DE-140 to confirm the mixer is not over-atritting brittle cathode powder
MT-703	SEM with EDX	The reference method for fibrillation assessment. Direct visual evidence of fibril formation, length and distribution. A dry electrode plant cannot operate without SEM access
MT-704	BET surface area analyser	Incoming material qualification
MT-705	Mercury or gas pycnometer	True density and porosity — calibrates the MT-105 ultrasonic porosity measurement and validates calendaring
MT-706	90° peel / adhesion tester	The direct test for the dry route's characteristic failure mode. Specify an automated sampling version adjacent to DE-190, not only a bench instrument
MT-707	Tensile tester (film)	Green-film tensile strength — the functional consequence of fibrillation quality
MT-708	XRD	Incoming material phase purity
MT-709	ICP-MS / ICP-OES	Magnetic and metallic impurity verification to ppb. Verifies DE-120 effectiveness and incoming certificates
MT-710	Powder rheometer / shear cell	Flow properties for hopper design; a proxy for fibrillation state once calibrated against SEM

MT-711	Laboratory battery cycler	Capacity vs C-rate and cycle life. Published characterisation uses C/20 through 10C at 25 °C
MT-712	Three-electrode cell rig	Separates anode and cathode potentials; definitive degradation attribution
MT-713	Glovebox (Ar, <1 ppm O ₂ /H ₂ O)	Mandatory for teardown and post-mortem work

17.8 Instrumentation priority on a finite budget

In the order that returns the most quality per unit of spend, for a **dry electrode** line specifically.

Rank	Instrument	Why at this position
1	Aging OCV monitoring (MT-501) with rack temperature mapping	Cheapest instrument for the most dangerous failure mode. The 5 mV/week criterion works immediately and needs no development
2	Beta or XRF areal mass gauge (MT-101/102) + paired laser thickness (MT-104)	Enables density control — the master variable of electrode quality. Neither alone is sufficient
3	Automated sampling peel tester (MT-706)	Dry-specific. Adhesion is this route's characteristic failure mode and is invisible to every other inline gauge
4	Inline OCT weld monitor (MT-401)	Converts AS-330, the hardest and lowest-yielding station, into a controllable one
5	Multispectral / polarisation imaging (MT-205)	Dry-specific. Without it you have no working surface inspection at all on dry material
6	SEM access (MT-703)	The only reference method for fibrillation. Buy, share or contract — but have it
7	Dark-field line-scan on the slit (MT-201 + grazing illumination)	Directly detects the burr — one of the three internal-short causes
8	ICP-MS for incoming magnetic impurity (MT-709)	Directly detects another of the three internal-short causes
9	Karl Fischer + multi-point dew point (MT-701, MT-601/602)	Guards the HF pathway that limits cell lifetime
10	Ultrasonic pulse-echo (MT-105)	Dry-specific. The only credible inline porosity signal, and porosity must be engineered rather than inherited on this route
11	CCD dimensional inspection after notching (AS-300 integrated)	Catches the error that gets magnified at tab folding and welding
12	Hi-Pot at EoL (MT-505)	Last-chance detection of latent shorts before shipment
13	EIS after formation (MT-504)	Micro-short detection; recommended per-cell for automotive
14	Inline X-ray / CT after assembly (MT-301/302)	Closes the post-assembly blind spot. Highest cost, so last — but genuinely valuable

BEFORE CLOSING ANY CONTROL LOOP

Run a gauge repeatability and reproducibility study. Target total measurement variation below 10 % of the process tolerance band; 10–30 % is conditionally acceptable; above 30 % the gauge is measuring itself, not the process. A loop driven by a gauge with 40 % measurement variation will oscillate the process, and the oscillation will be blamed on the material.

Part VI — Quality and Control

19. Defect Engineering and FMEA

19.1 Internal short circuit — the failure mode that matters most

The three primary causes, as stated in industrial practice:

Cause	Mechanism	Mitigation
Metallic particle contamination (iron, nickel, copper)	The particle penetrates the separator	Magnetic separation of slurries; incoming ICP-MS verification; cleanroom discipline; segregated equipment
Electrode edge burrs from poor notching or slitting	A sharp edge cuts through the separator under winding and assembly pressure	High-resolution vision inspection after notching; dark-field slitting inspection; knife change on count
Separator wrinkles or pinholes	Direct electrode-to-electrode path	Separator ceramic coating to improve puncture resistance; correct separator tension; incoming separator inspection

Detection: Hi-Pot testing at 500–1000 V at end of line identifies cells with latent shorts; OCV decline above 5 mV/week during aging; EIS after formation.

19.2 Defect catalogue by station

Station	Defect	Root cause	Detection
Lamination (DE-190)	Low adhesion	Foil Ra too low; nip temperature or pressure low	90° peel test only — invisible to gauges and cameras
	Margin position drift	Registration open-loop; foil edge guide error	Margin vision measurement
	Blocking in the coil	Laminate wound warm	Visual on unwind
Film forming (DE-170)	Film tears at take-off	Under-fibrillation; doctor blade wear; speed ratio wrong	Green-film tensile; SEM on powder
	CD loading variation	Powder curtain non-uniform (DE-160)	Scanning gauge CD profile
	Periodic MD variation	Roll runout or feeder metering element	Gauge trace; match period to circumference
Fibrillation (DE-140)	Under-fibrillation	Insufficient specific shear energy; temperature too low	SEM; green-film tensile — invisible to mass and thickness gauges
	Over-fibrillation	Excess shear energy; long residence tail	Electrode resistivity; cell impedance
	Cathode over-attrition	Shear too aggressive on brittle powder	Particle size before vs after
Calendering (DE-200)	Wavy edge	CD tension gradient; crown mismatch	Vision; thickness CD profile
	Coating cracking	Excess line pressure	Vision; microscopy
	Porosity off target	Line pressure; roll deflection	Ultrasonic; mass/thickness ratio
Slitting	Burr	Knife wear; incorrect overlap	Dark-field line scan; sampling microscopy
	Width out of tolerance	Knife position drift; tracking error	Laser width gauge
Notching	Flag position error accumulating	Cam synced to motor not web; web stretch	CCD dimensional inspection
	Gradual cut-quality loss	Cover slide contamination	Power measurement at workpiece
Winding	Layer misalignment	EPC saturated; coil camber	CCD edge; finished roll dimensions
	Separator necking	Separator tension set too high	Width measurement
	Compaction gradient core-to-edge	Tension not tapered with radius	Teardown thickness profile
Tab forming	Untidy folding, gaps	Flag position error from notching, magnified	Folding inspection camera; 3D profilometer
	Torn flags	Excess compaction force	Vision
Welding	False weld	Focus off; power low; clamp gap; oxidised surface	OCT depth; joint resistance
	Over weld	Power high; speed low; plane thinner than assumed	OCT depth; X-ray
	Spatter	Keyhole instability; no beam shaping	Photodiode; post-weld vision
Insertion	Layer shifting	Friction during insertion	Insertion force curve; X-ray
	Insulator missing or misplaced	Feeder fault	Vision
Filling	Incomplete wetting	Compacted tab plane; low porosity; too few cycles	Low capacity at EoL; EIS
	Residue on sealing surface	Lance misalignment; splashing	Vision; failed seal weld
Formation	Non-uniform SEI	Rate too high; temperature wrong	First-cycle efficiency; DCIR; EIS
	Data scatter across channels	Contact pin resistance variation	Compare scatter against pin service history

19.3 FMEA summary

Failure mode	Effect	Root cause	Detection	Prevention
Internal short circuit	Thermal runaway, fire	Metal particle, burr, separator defect	Hi-Pot, OCV > 5 mV/wk, EIS, CT	Magnetic separation, edge inspection, ceramic-coated separator
False collector weld	High resistance, hot spot, power loss	Focus deviation, clamp gap	OCT, DCIR	Closed-loop OCT control, guaranteed clamp force
Over weld / perforation	Jelly-roll damage, delayed failure	Excess power, low speed	OCT, X-ray	Narrow validated window, beam shaping
Incomplete wetting	Low capacity, local lithium plating	Over-compacted tab plane, low porosity	EoL capacity, EIS	More vacuum cycles, HT soaking, tab-forming method choice
Residual moisture	HF generation, cathode corrosion, short life	Vacuum drying insufficient, dry-room excursion	Karl Fischer, dew-point log	Longer/hotter drying, dry-room integrity, queue-time limits
Non-uniform SEI	Capacity loss, short cycle life	Formation rate too high	First-cycle efficiency, DCIR, EIS	Slower first cycle, tighter temperature control
Porosity below target	Poor fast charge, lithium plating	Excess calender line pressure	Ultrasonic, mass/thickness	Closed-loop line pressure, crown compensation
Over-fibrillation	High impedance, poor rate capability	Excess shear energy	Resistivity, cell EIS	Residence time distribution control
Low adhesion	Delamination; capacity loss	Foil Ra; nip conditions	90° peel test	Foil spec + inline peel monitoring
Layer shift in jelly roll	Alignment loss, short risk	Insertion friction	X-ray, insertion force	Force monitoring with reject, coaxiality control
Untidy tab folding	Joint gaps, high resistance	Notching position error	Folding camera	Cam sync to actual web length, registration correction
Cell leak	Dry-out, safety hazard	Bead crack, incomplete seal weld	Helium test x2	Bead profilometry, seal weld monitoring
Thermal runaway	Fire, propagation	Combination of the above	Cell-level abuse testing	Vent, CID, shutdown separator, cell design margin

19.4 4680-specific challenges

Challenge	Why it is specific to this format
Low initial yield (~80 % reported)	Three novel technologies introduced simultaneously: full-tab forming, collector plate welding, laser sealing
Long wetting time	The compacted tab plane closes capillary paths
Internal thermal gradient	Lower surface-to-volume ratio at 46 mm diameter
Variable flag geometry	Flag length must grow with winding radius — a continuously varying notch pattern
High weld volume	Transition from spot welding to surface welding; more joints, more heat, narrower window
Can mass penalty	0.5 mm wall thickness is inactive mass, directly reducing gravimetric energy density
Coating precision on the arc edge	The arc-shaped full-tab coating edge demands higher equipment precision than a straight edge

20. Troubleshooting Decision Trees

Tree A — Low capacity at end of line

Low capacity at EoL

- Is DCIR also high?
 - YES → suspect JOINT or WETTING
 - Check weld OCT depth & joint resistance data for that lot
 - abnormal → welding station (\$9.4)
 - normal → wetting (AS-380): check fill cycles, soaking, tab-plane compaction data (AS-320)
 - NO → suspect ACTIVE MATERIAL or LOADING
 - Check areal loading gauge data for the coil
 - low → film forming (DE-170) or feeder (DE-160)
 - normal → check incoming material CoA and retain sample
 - Check first-cycle efficiency from formation
 - low → SEI / formation protocol (FN-410)
- Is the effect coil-correlated?
 - YES → trace to coil position; electrode manufacturing
 - NO → trace to assembly machine / carrier / formation channel

Tree B — Rising internal short rate

Internal short rate rising (Hi-Pot fails or OCV > 5 mV/week)

- Correlate with SLITTER KNIFE service life
 - shorts cluster near end of knife life → BURR. Shorten knife interval
- Correlate with MATERIAL LOT
 - shorts cluster on one lot → METALLIC PARTICLE
 - ICP-MS the retain sample
 - inspect magnetic separators; check saturation
- Correlate with SEPARATOR LOT
 - shorts cluster on one lot → separator defect; check incoming data
- Correlate with WELDING station / parameter change
 - over-weld perforation. Review OCT depth distribution for that period
- No correlation found
 - Widen the window: check cleanroom particle counts, recent maintenance events, and any equipment change in the period.
 - Do NOT conclude "random" – every short has a cause.

Tree C — Weld penetration drifting

OCT depth trending away from target

- Direction: DECREASING over days/weeks
 - Almost always COVER SLIDE contamination or optics degradation. Measure delivered power at the workpiece. Replace slide.
- Direction: SCATTER increasing, mean unchanged
 - Upstream variation. Check:
 - tab-plane height data from folding profilometer (AS-320)
 - clamp force records
 - can wall thickness (if welding through the wall)
- Direction: STEP CHANGE
 - Something was changed. Check the MES recipe version log, maintenance log, and consumable change records for that timestamp.

Tree D — Film will not form, or tears at take-off

Film tears at DE-170 take-off, or never becomes self-supporting

- Check GREEN-FILM TENSILE on the batch
 - LOW → fibrillation problem (DE-140)
 - specific shear energy below setpoint? → check net power log
 - product temperature below band? → jacket / chiller
 - residence time short? → throughput too high
 - confirm by SEM: are fibrils present and extended?
 - NORMAL → the powder is fine; the problem is at the roll pair
 - speed ratio too low → film has no preferred roll
 - line pressure too low → insufficient consolidation
 - roll temperature wrong → PTFE not deforming
 - doctor blade worn → tearing at take-off, not in the nip
 - web speed above the limit → residence time in nip too short
- Is the failure CD-localised (one side, one band)?
 - YES → powder curtain uniformity (DE-160). Check CD feed profile. Do NOT adjust the roll pair for a feeder problem.
- Did it start suddenly after a change?
 - Check MES: recipe version log, wear-part changes, incoming lot change. A step change is always caused by something; find the timestamp.

Tree E — Adhesion below specification

90° peel result below control limit

- └ Check INCOMING FOIL surface roughness Ra for that coil
 - └ below spec → foil problem. Quarantine the coil, not the machine.
- └ Check LAMINATION nip temperature and line pressure (measured, not commanded)
 - └ temperature low → film not deforming into the foil topography
 - └ pressure low → insufficient mechanical interlocking
- └ Check SURFACE PREPARATION
 - └ corona/plasma electrode worn? → dyne test the treated foil
 - └ foil contaminated? → rolling oil residue kills adhesion
- └ Check the FILM itself (green-film tensile)
 - └ if cohesion is low, peel will fail cohesively rather than adhesively – this is a DE-140 problem presenting at DE-190.
Examine the peel FRACTURE SURFACE: film-side residue on the foil means cohesive failure; clean foil means adhesive failure.
- └ Did adhesion pass at DE-190 and fail after DE-200?
 - └ Calender line pressure is over-working an already marginal bond.
Peel test AFTER calendering, not only after lamination.

21. SPC, Traceability and Manufacturing Data

21.1 The six critical parameters

Advanced battery manufacturing focuses on controlling six critical parameters. Variations in these cause capacity fade, increased internal resistance, and lithium plating.

#	Parameter	Target
1	Electrode loading uniformity	±1.5 % or better
2	Coating thickness profile	Uniform MD and CD
3	Calendered density	Per porosity target
4	Moisture content	< 50 ppm cathode, < 20 ppm anode
5	Electrolyte wetting consistency	Homogeneous distribution
6	SEI formation quality	Stable, uniform, thin

21.2 Capability targets

Metric	Industrial target
Process capability index Cpk	Above 1.33
Defect rate (mature line)	Below 10 ppm
Vision detection rate, defects < 0.05 mm	> 99.5 %
Vision inference latency	< 50 ms at the edge
Ramp-up period to mature yield	Typically several years for a novel format

21.3 Closed-loop control

Gigafactories deploy inline metrology (X-ray, laser, optical) with closed-loop feedback to maintain Cpk above 1.33. The practical loops:

Loop	Measured	Actuator
Areal loading	Beta / XRF gauge	Powder feeder metering rate (DE-160)
Post-calender thickness	Paired laser triangulation	Calender line pressure
Web lateral position	Edge sensor (coating edge preferred)	EPC steering guide
Weld penetration	Inline OCT	Laser power (AS-330)
Film adhesion	Sampling peel tester	Lamination nip temperature / pressure (DE-190). Dry-route-specific
Fibrillation state	Net power → specific shear energy	Mixer throughput or shear setting (DE-140). Dry-route-specific
Web tension	Load cell	Drive torque
Winding tension taper	Roll diameter laser	Unwind torque profile
Dry-room dew point	Hygrometer + differential pressure	Desiccant rotor speed / regen temperature

DO NOT CLOSE A LOOP ON AN UNQUALIFIED GAUGE

Every loop above amplifies measurement error into process error. Complete a gauge R&R study (§5.1.1) before enabling closed-loop control. A loop driven by a gauge with 40 % measurement variation will oscillate the process and the oscillation will be blamed on the material.

21.4 Traceability

Every cell needs a unique identifier — typically a laser-etched DataMatrix on the can before assembly — linked to a complete genealogy.

Data element	Why it must be captured
Raw material lot numbers	Cathode, anode, electrolyte, separator, foil, can. The first thing any investigation asks for
Mother coil ID	Links to coating and calendering process data
Daughter coil ID and slit position	A CD-position-dependent defect affects only some daughter coils
Longitudinal position on the coil (metres from start)	The single most valuable field, and the one most often omitted
Machine, tool and shift ID per station	Correlates defects to a specific knife, sonotrode or operator
Actual measured parameters, not setpoints	Setpoints tell you what was asked for; measurements tell you what happened
Recipe version and effective date	Distinguishes a process change from a material change
All measurement results	Every gauge, camera, weld monitor and electrical test
Alarms and maintenance events	The correlation between a chiller alarm and a coating drift is frequently the answer
Retain sample reference	Links back to physical evidence

WHY COIL POSITION IS WORTH BUILDING FOR

If a field failure traces to a coil and you have *no* position data, you must recall every cell made from that coil — potentially tens of thousands. With metres-from-start recorded and propagated, you can often narrow it to the 200 m of web that was actually affected, and recall hundreds. The engineering cost of capturing position is small. The cost of not having it is measured in orders of magnitude.

21.5 Sampling strategy

Inspection type	Coverage	Applied to
100 % inline	Every unit	Coat weight, thickness, surface vision, notch dimensions, weld OCT, leak test, EoL electrical
Continuous scanning	Partial, zig-zag	CD thickness profile — understand the coverage fraction (§5.1.2)
Skip-lot	Reduced frequency	Incoming material Tier 3 tests once a supplier is qualified
Periodic destructive	Fixed count or interval	Weld cross-sections, pull tests, peel tests, burr microscopy
Event-triggered	On demand	After any parameter excursion, maintenance event or material change
Retain	Every lot	Physical samples for future investigation

21.6 Yield accounting

Yield must be tracked by station and by defect code, not as a single plant number. A plant reporting “92 % yield” without station attribution has no basis for improvement priorities.

Metric	Definition and use
First-pass yield per station	Units passing without rework, per station. Identifies the actual constraint
Rolled throughput yield	Product of all station first-pass yields. The honest number
Scrap by defect code	Pareto by cause, not by station. The same defect code may originate upstream
Scrap cost weighting	A cell scrapped after formation costs vastly more than a metre of web scrapped at coating. Weight the Pareto by accumulated value, not unit count
Latent defect escape rate	Field failures traced back to a manufacturing cause. The metric that matters most and is measured least

22. Environment: Cleanroom and Dry Room

22.1 Requirements by process step

Process	Cleanroom class	Dry room (dew point)	Temperature	Rationale
Blending (DE-130)	ISO 7-8	—	22 ±2 °C	Electrode production takes place under cleanroom conditions because foreign particles in the coating cannot be removed by cleaning methods such as suction later in the process
Fibrillation (DE-140)	ISO 7-8	N ₂ inerted; controlled process conditions	22 ±2 °C	
Film forming (DE-170)	ISO 7-8	N ₂ inerted; controlled	22 ±2 °C	
Lamination (DE-190)	ISO 7-8	—	22 ±2 °C	
Calendering	ISO 7-8	—	22 ±2 °C	
Slitting	ISO 7-8	—	22 ±2 °C	
Vacuum drying	Encapsulated process (vacuum or inert)		60-150 °C	Also functions as air lock into the dry room
Cutting / notching	min. ISO 7	Very dry (–40 to –60 °C)	20-50 °C	Cell assembly must be carried out under dry conditions, as water inside the cell leads to severe quality losses (service life) and a safety risk (formation of hydrofluoric acid)
Stacking / winding	min. ISO 7	Very dry (–30 to –60 °C)	22 ±2 °C	
Packaging	min. ISO 7	Very dry	22 ±2 °C	
Electrolyte filling	min. ISO 7	Very dry (–40 to –60 °C)	20-50 °C	
Soaking	—	—	20-50 °C	Cell finishing takes place in a normal environment. Since the cell is already sealed and degassing takes place in a vacuum chamber, there are lower requirements for the particle environment and humidity
Formation	—	—	22 ±2 °C	
Degassing	Encapsulated process		—	
Aging	—	—	20-50 °C	
EoL testing	—	—	22 ±2 °C	

22.2 Dry room economics

THE DEW POINT SPECIFICATION IS A LARGE RECURRING COST DECISION

Achieving –60 °C dew point is substantially more expensive than –40 °C, because desiccant regeneration energy rises steeply as the target falls. Regeneration heating is the single largest energy line item in a dry room.

Two levers reduce it: **size the pre-cooling coil properly**, so bulk moisture is removed by condensation (cheap) rather than by desiccant (expensive); and **specify the dew point your chemistry actually requires**, rather than the tightest number available. If the chemistry tolerates –40 °C, specifying –60 °C is a permanent operating cost with no return.

22.3 Practical monitoring

Monitor	Alarm rationale
Dew point at multiple room points	A single sensor near the supply diffuser will read good while a corner is out of specification
Room differential pressure	The earliest indicator of a boundary failure. Alarm before dew point moves
Filter differential pressure	Change filters on loading, not on a calendar
Airborne particle count at critical points	Continuous at notching, winding and filling
Air-lock cycle logging	A propped-open lock is the most common human cause of an excursion
Compressed dry air dew point	CDA is piped directly onto the product; a CDA excursion bypasses the room entirely

23. Facility Equipment and Utilities

23.1 What the dry route demands of the building

Facility system	Wet route	Dry route
Solvent recovery plant	Required, >99 % NMP	Not required
Solvent storage and handling	Bulk NMP storage, bunding, permits	Not required
Dryer exhaust and abatement	Large volume, solvent-laden, RTO or equivalent	Not required for the electrode block (still required for formation off-gas)
Thermal energy for drying	Very large — the dominant electrode-step load	Largely eliminated
Floor area, electrode block	Baseline	~60 % of baseline
Dust extraction	Moderate — slitting and notching only	Substantially higher — the whole electrode block handles dry powder
Combustible dust protection	Localised	Plant-wide in the electrode block; a different hazard assessment
Nitrogen supply	Moderate	Higher — inerting on blending, fibrillation, conveying, storage
Humidity-controlled powder storage	Moderate	Higher — no downstream drying oven to correct moisture pickup
Dry room	Required from cell assembly onward	Unchanged — still required, same specification
Chilled water	Moderate	Higher on the mixer — all fibrillation shear energy becomes heat that must be removed

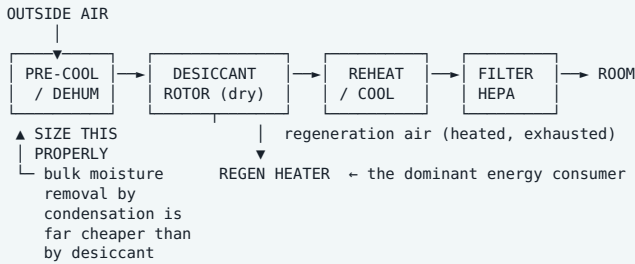
WHERE THE SAVING IS REAL AND WHERE IT IS TRADED AWAY

The dry route genuinely deletes a chemical plant (solvent recovery), a thermal plant (the dryer) and a large abatement train. It genuinely reduces floor area by roughly 40 %. In exchange, it upgrades dust handling from a peripheral utility to a plant-wide safety system, and it raises nitrogen and chilled-water demand. **Net, the facility is much cheaper — but the savings are not free, and the dust safety case is not optional.**

23.2 Facility equipment catalogue

FC-100 Dry Room Dehumidification Plant

Function: hold the assembly area at -40 to -60 °C dew point | **Serves:** AS-300 through AS-390
Utilities: power (very high), chilled water, steam or electric regeneration heat | **Maturity:** mature



Component	Specification points
Desiccant rotor	Rotor speed, regeneration temperature and purge sector setting determine achievable dew point. Regeneration heating is the largest single energy line item
Pre-cooling coil	Where dry-room operating cost is won or lost. Doing bulk moisture removal by condensation is far cheaper than by desiccant. Size it properly
HEPA / ULPA filtration	Particle control to ISO 7-8. Monitor differential pressure, not elapsed time
FFU (fan filter unit)	Local clean airflow at notching and winding to protect open product
Room pressurisation control	Loss of positive pressure is the most common cause of a slow dew-point excursion. Alarm on differential pressure
Air lock / pass-through	DE-220 vacuum dryers are often used as a lock into the dry room — design this deliberately

SPECIFY THE DEW POINT YOUR CHEMISTRY REQUIRES, NOT THE TIGHTEST AVAILABLE

Achieving -60 °C is substantially more expensive than -40 °C because regeneration energy rises steeply as the target falls. If the chemistry tolerates -40 °C, specifying -60 °C is a permanent operating cost with no return. This is one of the largest recurring cost decisions in the building.

FC-200 Cleanroom Air Handling

Function: ISO 7-8 particle control in the electrode block | **Serves:** DE-100 through DE-220

Electrode production takes place under cleanroom conditions because **foreign particles in the coating cannot be removed by cleaning methods such as suction later in the process**. Temperature 22 ±2 °C. Continuous particle monitoring at critical points; filter differential pressure trended.

FC-300 Dust Extraction and Explosion Protection System — ELEVATED DUTY

Function: capture, convey and safely collect combustible electrode dust | Serves: the entire DE block plus AS-300
Maturity: mature hardware, but the hazard assessment is plant-specific and must precede layout freeze

Component	Specification points
LEV hoods	Capture velocity specified <i>at the source</i> , not at the fan. Measured at commissioning and periodically thereafter
Machine-integrated de-dusting	Brush, negative-pressure de-dusting, FFU and dust collector — standard on notching and slitting machines
Dust collector	Must be rated for combustible dust. Explosion venting, suppression or isolation per the hazard assessment
Explosion isolation valve	On the duct between process and collector, to prevent flame propagation back. Frequently omitted in first designs
Explosion relief or suppression	Relief must vent to a safe place — which constrains collector siting, which constrains duct routing, which constrains machine positions
Spark detection and extinguishing	Particularly downstream of laser processes and mechanical cutting
Earthing and bonding	Every metallic component bonded and earthed; continuity tested and recorded , not assumed
HEPA final filtration	Extraction discharging into the room defeats the cleanroom
Certified cleaning equipment	Vacuum rated for combustible dust. No compressed-air blow-down — it creates the dust cloud that is the precondition for an explosion

FOUR DESIGN RULES FOR THIS SYSTEM

1. **Commission the hazard assessment before the layout is frozen.** Relief venting requirements determine collector siting, which determines everything else.

2. **Anode and cathode extraction must be separate systems.** A shared duct is both a cross-contamination path and a mixing of two different dust hazards in one vessel.

3. **Design for cleaning.** Specify sloped or eliminated horizontal ledges on machine frames in the RFQ.

4. **Record capture velocity at commissioning.** A slowly blocking duct looks exactly like a process drift.

FC-400 Central Utilities

Function: chilled water, compressed dry air, inert gas, process vacuum, exhaust treatment

Utility	Served	Failure signature
Chilled water	DE-130, DE-140 (high duty) , DE-170/190/200 roll tempering, laser cooling, chambers	Temperature swing appears as fibrillation drift, film density drift or laser power drift — all of which look like process problems
Compressed dry air	Pneumatics, air knives, air support	A CDA dew-point excursion introduces moisture directly onto the product inside the dry room, bypassing the room entirely. Monitor CDA dew point separately
Nitrogen	Blending, fibrillation, conveying, storage, laser shielding, vacuum purge, welding	Purity and flow stability. Higher demand than a wet-route plant
Process vacuum	DE-220, AS-380, AS-390, FN-420	Shared headers couple stations — one station's demand affects another's setpoint. Prefer dedicated pumps for control-critical processes
Exhaust after-treatment (RTO)	Formation and degassing off-gas	Still required despite the absence of solvent. Extracted gases must be after-treated before entering the exhaust system per occupational health and environmental regulations
Electrical	Formation dominates; drying no longer does	Power quality issues surface first as cycler channel faults

FC-500 Fire Protection and Emergency Systems

Function: protect against the three distinct fire scenarios in a cell plant

Scenario	Protection
Combustible dust deflagration (DE block)	Explosion venting, isolation, suppression; ignition source control; housekeeping regime
Electrolyte fire (AS-380)	Vapour detection, containment, appropriate suppression medium; calcium gluconate gel immediately available for HF exposure, with trained personnel — HF burns are deceptive and pain can be delayed while damage progresses
Charged cell store (FN-430)	An aging warehouse holds megawatt-hours of charged cells. Fire strategy, compartmentation and detection must be designed for that specific hazard, not for a general warehouse

Part VIII — Procurement and Execution

24. CAPEX Structure and Capacity Model

24.1 Published station-group figures

Reference basis: a **10 GWh/year** line. For cylindrical cells the study basis is approximately 550 million cells/year at 5 Ah; other stations reference pouch cells. Read these as *relative magnitudes*, not quotations.

Station group	Wet route (EUR m)	Dry route	Comment
Mixing	32 – 40	Replaced by DE-130 + DE-140	Two machines instead of one; fibrillation mixer is a specialised, low-volume product
Coating and drying	30 – 57	Largely eliminated	Replaced by DE-160 + DE-170 + DE-190. This is where the saving is
Calendering and slitting	6 – 12	6 – 12	Unchanged in cost; different duty
Vacuum drying	3 – 8	Lower — reduced duty	Shorter cycles, smaller chambers, or shifted to cell baking
Separation / notching	20 – 30	20 – 30	Unchanged
Winding	20 – 30	20 – 30	Unchanged
Packaging (cylindrical)	35 – 45	35 – 45	Unchanged
Electrolyte filling	35 – 45	35 – 45	Unchanged
HT soaking	10 – 20	10 – 20	Unchanged
Formation	65 – 85	65 – 85	Unchanged — and still the largest single item
Degassing	10 – 15	10 – 15	Unchanged
Aging	15 – 30	15 – 30	Unchanged
EoL testing	5 – 15	5 – 15	Unchanged
Total equipment	286 – 432	Reduced by the coating/drying delta	Tesla reports ~90 % reduction in equipment CAPEX for the electrode step; another source reports ~50 %

TWO CAUTIONS ON THESE NUMBERS

1. **The published figures exclude** building shell, cleanroom and dry-room construction, HVAC and dehumidification plant, chillers, electrical infrastructure, dust collection and explosion protection, internal logistics, MES and IT, spare parts, installation and commissioning labour, and working capital. Depending on site and region these typically add a comparable amount again. A EUR 300–430 million equipment figure is not a project budget.
2. **The dry-route saving applies to one block only.** Formation, aging, filling and assembly are unchanged and together dominate the total. Deleting the dryer is a large saving on a moderate fraction — not a transformation of the whole plant budget.

24.2 Two observations that should shape the budget

OBSERVATION 1 — CELL FINISHING IS A THIRD OF THE CAPITAL AND TRANSFORMS NOTHING

FN-400 through FN-440 together run to roughly EUR 105–165 m — comparable to the entire electrode line, and this is the phase that produces *no physical transformation* of the product. It only conditions and verifies it. That is why early quality detection and formation acceleration are the industry's most-funded process research topics, and why a supplier offering credible capability there deserves disproportionate attention.

OBSERVATION 2 — COST AND IMPORTANCE ARE WEAKLY CORRELATED

Calendering and slitting cost EUR 6–12 m, the cheapest major group on the line, and they set porosity (which governs fast charge and lithium plating) and edge quality (which governs internal short rate). On a dry line, DE-140 and DE-160 are similarly modest purchases carrying most of the block's technical risk. **Do not allocate engineering attention in proportion to CAPEX.**

24.3 Capacity model and machine count

24.3.1 Reference basis

The published CAPEX and capacity references used throughout this handbook assume a **10 GWh per year** line. For cylindrical cells the reference basis is approximately **550 million cells per year at 5 Ah cell capacity**. A 4680 at roughly 26 Ah and 98 Wh is a very different cell, so scale by energy, not by cell count.

24.3.2 Working the numbers for 4680

Quantity	Calculation
Cell energy	~98 Wh (from ~26 Ah at ~3.7–3.8 V)
Cells per GWh	1,000,000,000 Wh ÷ 98 Wh ≈ 10.2 million cells
Cells per year for 10 GWh	≈ 102 million cells
Operating hours per year	8,760 × availability. At 85 % OEE → ~7,450 h
Required rate	102 M ÷ 7,450 h ≈ 13,700 cells/h ≈ 228 cells/min

24.3.3 Machine count from published throughput

Machine	Published throughput	Units for 228 cpm	Note
AS-310 Winder	up to 30 cells/min (round cell)	8–10	Add margin; winding is rarely the published

			maximum in practice on a tabless format
AS-3xx Assembly line (integrated)	Reported ~150 PPM single-line capacity for a high-speed 46-series line	2	An integrated line hides the individual station counts
FN-410 Formation	Reported 355 PPM formation and grading line delivered	1 nominal	But formation <i>duration</i> (up to 15 h) drives channel count, not PPM — see below
AS-300 Laser notcher	~0.2 s per sheet reference	Rate-matched to winders	Often integrated with the winder in a combined notch-and-wind machine

24.3.3.1 FORMATION AND AGING ARE SIZED BY RESIDENCE TIME, NOT RATE

THE CALCULATION THAT DETERMINES A THIRD OF YOUR CAPITAL

Formation channels required = cell rate × formation duration.
228 cells/min × 15 h × 60 = **~205,000 formation channels**.

Aging positions required = cell rate × aging duration.
228 cells/min × 21 days × 1,440 = **~6.9 million aging positions**.

This is why formation is the most expensive station on the line at EUR 65-85 m, why aging consumes a warehouse, and why every manufacturer funds research into shortening both. Halving formation time halves 205,000 channels of capital. Nothing else on the line offers that leverage.

24.3.4 Electrode block sizing

Parameter	Basis
Electrode area per cell	Derived from jelly roll geometry: ~44.5 mm diameter, ~71 mm height, no mandrel. Electrode length exceeds several metres per cell
Web speed required	Total electrode length per hour ÷ number of parallel lines ÷ web width utilisation
Dry route speed constraint	Uniformity degrades above roughly 50 m/min. This is the binding constraint on the dry route and it is the number to challenge every supplier on
Wet route comparison	Coating and drying run at 35-80 m/min; calendering at 60-100 m/min; slitting at 80-150 m/min

THE NUMBER THAT DECIDES WHETHER A DRY LINE IS VIABLE AT YOUR VOLUME

If the dry film-forming line runs at 50 m/min where the wet line runs at 80 m/min, you need roughly 60 % more parallel electrode lines to reach the same output. That partially offsets the capital saving. **Make achievable line speed at specified uniformity a contractual performance guarantee, not a datasheet claim**, and define exactly how uniformity will be measured to verify it (see §26).

5. How to Read Any Machine: The Six-Subsystem Model

Every machine in this catalogue decomposes into the same six subsystems. Learning to identify them turns an unfamiliar machine into a familiar one within an hour, and it structures both the RFQ and the fault-finding.

Subsystem	What it does	Where it fails
1. Material transport	Moves product through the machine: web path, powder conveying, conveyor, indexing table, robot, carrier	Tension excursions, tracking drift, powder bridging, jams, carrier wear
2. Process actuator	Does the physical work: roll pair, mixer tool, laser head, sonotrode, press, pump, heater	Wear, thermal drift, contamination, alignment loss
3. Metrology	Measures the result: gauges, cameras, load cells, thermocouples	Calibration drift, fouling, cabling noise, wrong installation position
4. Control	Closes the loop: PLC, motion controller, PID, recipe manager	Loop tuning, sample rate too slow, deadband too wide
5. Environment and utility	Supplies and conditions: vacuum, dry air, inert gas, chilled water, extraction, filtration	Usually the actual root cause when a “process problem” is intermittent
6. Safety and interlock	Guards, light curtains, laser enclosure, e-stop chain, dust explosion protection, gas detection	Nuisance trips that operators then defeat — the classic path to an incident

USE THIS AS A DIAGNOSTIC ORDER, NOT JUST A DESCRIPTION

When a station starts producing defects, walk the six subsystems in order before touching a recipe parameter. Subsystem 5 — utilities and environment — is the root cause of a disproportionate share of intermittent problems and is the one nobody checks first. A chilled-water temperature swing, a compressed-air dew-point excursion or a partially blocked extraction duct looks exactly like a process drift.

5.1 Use it to structure the RFQ too

A specification organised by these six subsystems is harder for a supplier to answer vaguely. For each subsystem, ask: what is supplied, what is the performance, what is the interface, what is excluded. Most disputes at acceptance trace to subsystem 3 (metrology — whose gauge, calibrated how) and subsystem 5 (utilities — who supplies what to which specification at which boundary).

25. RFQ Specification and Supplier Evaluation

Structure the specification by the six subsystems (§5). A supplier finds it much harder to answer vaguely when the question is subsystem-specific.

25.3.1 Universal sections — every machine RFQ

Section	Content to require
1. Scope and battery limits	Exactly what is supplied and what is excluded. Define the utility connection points physically — most acceptance disputes trace to an ambiguous boundary
2. Product specification	Materials, dimensions, tolerances, formats to be run, changeover requirements
3. Performance guarantees	Throughput <i>at a stated quality</i> , availability, yield, changeover time. Each must have a defined verification method — a guarantee without a test method is not a guarantee
4. Subsystem 1 — material transport	Web path, tension architecture, EPC, handling, carriers
5. Subsystem 2 — process actuator	The core process hardware and its adjustable range
6. Subsystem 3 — metrology	Which sensors, whose brand, what accuracy, calibrated how, R&R evidence. State whether gauges are in supplier or buyer scope
7. Subsystem 4 — control	PLC platform, motion platform, HMI, recipe management, data interface to MES with a tag list
8. Subsystem 5 — utilities	Demand figures for every utility, with quality requirements (CDA dew point, chilled water temperature and stability, N ₂ purity)
9. Subsystem 6 — safety	Machine directive compliance, laser class, dust explosion protection, interlock architecture, risk assessment deliverable
10. Data and traceability	Every measured value tagged and timestamped; actual values, not setpoints ; OPC UA or equivalent; sample rate
11. Documentation	Layout drawings, P&ID, electrical schematics, parts lists, manuals, risk assessment, calibration certificates
12. FAT and SAT protocol	Agreed in the contract, not negotiated after delivery (see §26)
13. Spares and consumables	Recommended spares list with prices and lead times; consumable life figures <i>for your material</i>
14. Training and support	Operator and maintenance training scope; response times; remote support
15. Warranty and liability	Duration, coverage, performance shortfall remedies

25.3.2 Dry-electrode-specific RFQ questions

ADD THESE TO THE RFQ FOR DE-140, DE-160, DE-170 AND DE-190	
Machine	Question
DE-140 Fibrillation	What is the residence time distribution of <i>my</i> material in your machine at guaranteed throughput, and how did you measure it?
	How is specific shear energy measured and controlled, and may I see a per-batch record?
	Is product temperature measured in-product or only at the wall?
	What is your scale-up rule from pilot to production, and what evidence supports it?
	What is the wear-part life on an abrasive high-nickel cathode powder specifically, not generically?
	How many production installations exist on this duty, and may I speak to one?
DE-160 Feeder	Is the CD feed profile adjustable, by what mechanism, and in how many independent zones?
	What areal loading CD uniformity do you guarantee on the resulting film?
	What is the maximum permissible hold time in the feeder before the fibrillated powder consolidates?
DE-170 Roll pair	What line speed do you guarantee at ±1.5 % areal loading uniformity, and how will that be verified?
	Is crown compensation adjustable, and over what load range does it hold the CD profile?
	Are the rolls independently servo-driven, and what is the speed-ratio resolution?
	Is line pressure measured on the bearing housings or inferred from hydraulic pressure?
	What is the roll regrind interval and lead time, and what does a spare roll set cost?
DE-190 Lamination	What 90° peel adhesion do you guarantee on my film and my foil, and how is it measured?
	What foil surface preparation do you recommend, and what incoming Ra do you require?
	Is margin registration closed-loop from a vision sensor, or set mechanically?
	Is an automated sampling peel tester included, or is it my scope?
All DE	How is inspection performed on a low-contrast dry surface? If the answer is a conventional RGB line-scan camera, the supplier has not run this material.
	What is the cleaning and changeover procedure for tacky fibrillated material, and may I see it performed and timed at FAT?

25.3.3 Supplier evaluation matrix

25.3.1 Weighted scoring

Suggested weightings for a **dry electrode** machine purchase. Adjust, but keep process guarantee and reference installations dominant — on a TRL 6-7 technology, demonstrated capability outweighs specification promises.

Criterion	Weight	What to score
Demonstrated process capability	25 %	Trial results on <i>your</i> material. Not a datasheet, not a similar material, not a customer's material. This is the single most important criterion on the DE block
Reference installations	15 %	Number on this duty; willingness to arrange a visit or a call. A supplier who cannot produce a reference has not solved the problem

Technical specification compliance	15 %	Line-by-line against the RFQ, with deviations listed rather than silently omitted
Metrology and data package	10 %	Which gauges, whose brand, accuracy, R&R evidence, MES tag list
Service and spares	10 %	Local presence, response time, spares availability and lead time, consumable pricing
Price	10 %	Total cost of ownership over five years, not purchase price. Include consumables and expected downtime
Delivery and commissioning	5 %	Lead time, installation resource, commissioning duration
Safety and compliance	5 %	Directive compliance, dust explosion protection, risk assessment quality
Financial stability	5 %	You are buying a twenty-year asset from this company

25.3.2 Disqualifying answers

If the supplier says...	Conclusion
"We haven't run your material but it will be similar"	On DE-140 or DE-170, this is disqualifying. Insist on a trial
"Line pressure is derived from hydraulic pressure"	Commanded, not measured. Push for bearing-housing load cells
"The rolls are geared at a fixed ratio"	Removes your most important process variable. Reject the design
"We use a standard line-scan camera for inspection"	They have not inspected a dry electrode. See §18
"Mixing time is the control parameter"	They are controlling a proxy, not the variable. Specific shear energy is the parameter
"Formation recipe is included as standard"	Formation protocol is the core knowledge of a cell manufacturer and varies by chemistry. A standard recipe is a starting point at best — do not treat it as a deliverable

26. FAT and SAT Acceptance Protocol

AGREE THE PROTOCOL IN THE CONTRACT, BEFORE THE ORDER

Acceptance testing negotiated after delivery is negotiated from a position of weakness — the machine is on your floor and the money is largely paid. Put the full protocol, the pass criteria and the consequences of failure in the purchase contract.

26.1 Factory acceptance test (FAT)

Element	Content
Material	Your material. Ship it to the supplier in advance. A FAT on a surrogate proves very little on a dry electrode line
Duration	Long enough to see drift, not just a demonstration. Insist on a sustained run, not a ten-minute showcase
Performance verification	Every contractual guarantee tested with its agreed method. Throughput measured <i>at the guaranteed quality</i> , not in isolation
Metrology verification	Every supplied gauge calibrated against a traceable standard, multi-point. Gauge R&R on any gauge intended for closed-loop control
Failure mode demonstration	Deliberately induce a fault — a film break, a gap at the weld clamp, a feeder blockage — and verify detection, safe stop and clearing. Time the clearing
Cleaning and changeover	Performed and timed, especially on machines handling tacky fibrillated material
Data verification	Confirm every required tag is present, correctly named, correctly timestamped and delivered at the specified rate without loss
Wear part baseline	Inspect and dimension wear parts after the FAT run; record as the wear baseline
Documentation	Complete before shipment. Documentation promised after delivery arrives late or never

26.2 Site acceptance test (SAT)

Element	Content
Utility verification first	Confirm every utility meets specification <i>at the machine connection</i> before starting. A machine underperforming on out-of-specification chilled water is not the machine's fault, and you need that established
Repeat of key FAT tests	In the real environment, with real material, at real ambient conditions
Integration test	With upstream and downstream machines running, not standalone. Interaction effects only appear here
Extended run	Multi-shift, to expose thermal drift and shift-to-shift variation
Environment verification	Dry-room dew point and cleanroom particle counts held while the machine runs
Operator competence	Trained operators demonstrate normal operation, changeover and fault recovery unassisted

26.3 Dry-electrode-specific acceptance tests

Machine	Test that must be in the protocol
DE-140	Residence time distribution study on your material; SEM fibrillation assessment on samples from start, middle and end of three consecutive batches; specific energy reproducibility; particle size before and after to confirm no over-attribution
DE-160	Areal loading CD profile across the full width at three line speeds — the single most important test for this machine; deliberate CD error introduced and corrected using the machine's own adjustment
DE-170	Line speed ramp with uniformity measured at each step to establish and document the actual speed limit; crown compensation demonstrated at two very different loads; green-film tensile; repeating-defect check against every roll circumference
DE-180	Break count over a sustained run; deliberate break induced and clearing timed
DE-190	90° peel at multiple CD and MD positions with a stated capability index ; bend test at winding curvature; margin width and position over a sustained run; adhesion after a deliberate thermal excursion to establish sensitivity
DE-200	Porosity verified against pycnometry; no cracking or delamination at target density — peel test after calendering, not only after lamination
DE-210	Burr height by microscopy at multiple CD positions; dust generation baseline recorded; knife life established in metres
DE-220	Moisture target on a fully loaded chamber verified at centre and extremities; temperature map; exhaust dew-point curve recorded
AS-330	Penetration map with OCT validated against cross-sections; window boundaries documented; OCT / photodiode / resistance / pull strength correlated on the same samples
FN-410	Channel-to-channel accuracy against a calibrated reference load; contact resistance distribution across a full rack; chamber uniformity under full electrical load

27. Commissioning and Ramp-Up Playbook

This chapter is the applied summary: the order in which to do things, and the traps at each stage.

27.1 Phase sequence

Phase	Objective	Key activities and exit criteria
0	Definition	Freeze the cell design and chemistry. Decide the novelty budget (see §18.2). Complete a freedom-to-operate review. <i>Exit: design frozen, IP reviewed, one novelty maximum.</i>
1	Utilities and environment	Commission dry room, cleanroom, chilled water, CDA, vacuum, extraction. Verify dew point at multiple points, room pressurisation, CDA dew point. <i>Exit: environment holds specification for 7 consecutive days with no process running.</i>
2	Mechanical and web handling	Roller alignment to 0.05 mm/m. Load-cell calibration at correct wrap angle. Tension loop tuning zone by zone. Speed ramp with bare web. Splice and accumulator test. <i>Exit: stable tracking and tension at full line speed with no product.</i>
3	Metrology qualification	Calibrate every gauge against traceable standards. Run gauge R&R on every instrument intended for closed-loop control. Establish scan coverage fractions. <i>Exit: R&R below 10 % of tolerance on all control gauges. No loop closed before this.</i>
4	Station process windows	Establish each station's window independently on coupons or short runs. Map boundaries explicitly — minimum and maximum, not just the setpoint. <i>Exit: documented window with boundaries for every station.</i>
5	First article	Build cells end to end at low rate. Full teardown of a sample. Correlate every inline signal against destructive measurement. <i>Exit: inline signals demonstrably predict destructive results.</i>
6	Formation DOE	The longest single activity. Design the formation protocol experimentally on your own chemistry. Months, not weeks. <i>Exit: protocol meeting first-cycle efficiency, DCIR and cycle-life targets.</i>
7	Rate ramp	Increase speed in steps. At each step, re-verify quality before proceeding. Expect new failure modes to appear at each speed threshold. <i>Exit: target rate with stable yield.</i>
8	Yield ramp	Pareto by accumulated value, not unit count. Attack the constraint station. Close loops progressively. <i>Exit: Cpk > 1.33 on the six critical parameters, defect rate trending toward the ppm range.</i>

27.2 The novelty budget

THE NOVELTY BUDGET — AND HOW IT APPLIES WHEN DRY IS MANDATORY

Introduce one novelty per programme. Four of the world's most capable cell manufacturers each took four to six years to reach limited 46-series production, and Tesla's leadership has publicly acknowledged that pursuing the dry electrode process for the 4680 from the outset was a mistake.

Since the wet route is not available to you, the dry electrode process is a fixed constraint rather than a choice — that is already one novelty spent. Do not add a second. Keep the chemistry conventional (graphite anode, established cathode) and the format proven until the dry line is producing to specification. A dry-electrode 4680 with a silicon anode is two coupled research programmes running simultaneously, in which every failure has two plausible causes and neither can be isolated.

27.3 The ten traps

#	Trap	Avoidance
1	Commissioning process before web handling	No process parameter can compensate for unstable tension or tracking. Finish §4.2.6 first, every time
2	Closing a control loop on an unqualified gauge	Gauge R&R before any closed loop. A bad gauge in a loop makes the process worse than open loop
3	Controlling fibrillation with a mixing <i>time</i>	Time is a proxy. Control specific shear energy, measured from net power over mass, and qualify the residence time distribution
4	Fixing a calender problem that started at lamination or fibrillation	Run a peel test and a green-film tensile before touching line pressure. Examine the peel fracture surface to tell cohesive failure (DE-140) from adhesive failure (DE-190)
5	Using one tension setpoint for all four webs at winding	Separator runs at roughly one fifth the electrode tension, with its own loop
6	Synchronising the notching cam to motor position	Sync to a low-slip measuring wheel on the actual web, with registration correction
7	Chasing a laser “material problem” that is a dirty cover slide	Measure delivered power at the workpiece periodically. Monitor cover-slide contamination
7b	Adjusting the roll pair for a powder feeder problem	A CD-localised film defect is almost always DE-160. Check the CD feed profile before touching DE-170
8	Interpreting aging OCV without rack temperature	1 °C across a rack produces an OCV difference comparable to the 5 mV/week criterion itself
9	Omitting coil longitudinal position from traceability	Without it, a coil-level defect condemns the whole coil instead of 200 metres of it
10	Allowing untracked HMI recipe edits	Recipes download from MES, versioned and approved. Untracked edits are the leading cause of unexplained lot-to-lot variation

27.4 Instrumentation priority — if the budget is finite

In the order that returns the most quality per unit of spend:

Rank	Instrument	Why first
1	Aging OCV monitoring (6.5-digit DMM + rack temperature)	Cheapest instrument for the most dangerous failure mode. The 5 mV/week criterion works immediately
2	Beta or XRF areal mass gauge + paired laser thickness	Enables density control — the master variable of electrode quality
3	Inline OCT on collector welding	Converts the hardest, lowest-yielding station into a controllable one
3b	Automated sampling peel tester	Dry-specific. Adhesion is this route's characteristic failure mode and is invisible to every other inline gauge
4	Dark-field line-scan on the slitter	Directly detects the burr — the second of the three internal-short causes
5	ICP-MS for incoming magnetic impurity	Directly detects the first of the three internal-short causes
6	Karl Fischer + multi-point dew-point monitoring	Guards the HF pathway that limits cell lifetime
7	CCD dimensional inspection after notching	Catches the error that gets magnified at tab folding and welding
8	Hi-Pot at EoL	Last-chance detection of latent shorts before shipment
9	EIS after formation	Micro-short detection; recommended per-cell for automotive
10	Inline X-ray / CT after assembly	Closes the post-assembly blind spot. Highest cost, so last — but genuinely valuable

28. Spares, Consumables and Maintenance

28.1 Consumables master list

Machine	Consumable / wear part	Replacement basis	Criticality
DE-110	Feeder screws	Counted hours	Medium
DE-120	Sieve meshes; magnetic grid seals	Inspection interval	High — a torn mesh passes everything silently
DE-130	Scraper blades; seals	Counted hours	Medium
DE-140	Screw elements / rotor blades; barrel liners; seals	Counted hours or tonnage	High — wear debris is contamination; wear changes shear delivery
DE-160	Metering element; edge dam seals	Counted hours	High — wear degrades CD uniformity
DE-170	Doctor blades; roll cleaner media; edge dam seals; rotary union seals	Counted hours	High
DE-170/200	Roll regrind / spare roll set	Long interval, long lead time	Critical — hold a spare set; a damaged roll otherwise stops the line for months
DE-190	Nip roll covers; corona electrodes; web cleaner media	Counted hours; dyne test for corona	High — affects adhesion directly
DE-210	Slitter knives	Counted metres — change on count, not on defect	Critical — burr is a safety defect
AS-300	Laser cover slides; extraction filters; de-dusting brushes	Power decay measurement; DP; hours	High
AS-310	End tape; winding needles; separator guides	Continuous; counted cycles	Medium
AS-320	Folding tools; press tooling	Dimensional inspection	High
AS-330	Laser cover slides ; crossjet nozzles; clamp tooling	Power decay; counted hours	Critical — stock generously; slide contamination is the leading cause of weld drift
AS-340	Draw tooling; lubricant	Counted parts	High
AS-360	Sonotrodes and anvils	Counted cycles — knurl wear changes coupling efficiency	High
AS-380	Dosing lance tips; seals; pump wet ends	Counted cycles	High
FN-410	Contact pins	Counted cycles — treat as a consumable, not machine structure	Critical — pin wear and oxidation is the leading source of formation data scatter
FC-100	HEPA filters; desiccant rotor	Differential pressure; performance	High
FC-300	Dust collector cartridges	Differential pressure	Critical — a blocked collector is both a process and a safety failure

28.2 Spares strategy

Category	Stocking rule
Long-lead critical	Roll sets, laser sources, specialised drives. Hold on site. Lead time, not price, decides this
High-turnover consumables	Cover slides, knives, contact pins, filters. Hold to a defined reorder point based on measured life
Instrumentation spares	At least one spare of every gauge in a closed loop. A failed gauge otherwise stops production while a replacement is sourced
Common mechanical	Bearings, seals, belts — standard stores
Supplier-held	Negotiate guaranteed availability with a response time for items too expensive to stock

28.3 Maintenance principles

- Change wear parts on count, not on defect.** Waiting for a burr to appear means the preceding coil is already suspect. This applies to slitter knives, sonotrodes, cover slides and contact pins alike.
- Record consumable serials against the product they made.** When a defect is traced to a coil, you need to know which knife cut it.
- Measure delivered power at every laser workpiece periodically.** This one habit prevents the most common and most confusing drift on the line.
- Trend the leading indicators, not the failures.** Rising dust at the slitter precedes burrs. Rising OCT scatter precedes weld failures. Rising motor torque precedes mixer wear.
- Calibrate on observed drift.** An instrument that never drifts is being over-calibrated; one always out of tolerance is being under-calibrated.
- Write the out-of-tolerance procedure before you need it.** Quarantine scope, product impact assessment and notification path, decided in advance.

28.4.4 Supplier landscape and procurement risk

28.4.1 Companies referenced in public sources

Descriptive, not an endorsement. The market moves; verify current capability directly.

Category	Companies named in public sources
Turnkey line integration	Wuxi Lead Intelligent Equipment, Manz AG, Schuler Group, Applied Materials (battery manufacturing technology division)

Production-scale dry coating lines	Wuxi Lead, Manz — named in public sources as offering production-scale dry coating lines
Cylindrical assembly and laser lines	Wuxi Lead (formation and grading lines covering 18650/21700/4680, including a delivered 355 PPM formation and grading line), SHINHOP Laser (46-series line at a reported 150 PPM single-line capacity), CENJE, TOB
Machine vision (non-China)	Keyence, Cognex, ISRA Vision, Basler, LMI Technologies, Teledyne (line-scan and CIS), Allied Vision, Photoneo, Sensofar
Machine vision (China)	OPT Machine Vision, Mech-Mind, Luster LightTech, Hikrobot
Vision software	MVTec (HALCON, MERLIC)
Hyperspectral imaging	Headwall Photonics, Specim Spectral Imaging
X-ray and CT	Nikon Industrial Metrology, Yxlon (Comet Group), Viscom, VisiConsult (VCbattery)
Laser sources	Trumpf, Coherent
Weld process monitoring	Precitec
Roll cleaning	Kadant (continuous calender roll cleaner systems)
Coating (R&D and pilot)	FOM Technologies (slot-die), Coating Edge Solutions (consulting)
Can material	Speira (aluminium 4680 cell can)
Cell test equipment	Neware, Maccor, Arbin, Basytec
Research partners for process development	PEM RWTH Aachen, Fraunhofer ISI / IWS, Argonne CAMP, WMG Warwick

28.4.2 Procurement risk register

Risk	Severity	Mitigation
Dry film-forming equipment is TRL 6-7	High	Pilot or demonstration line before the production order. Process guarantees written into the contract with defined verification. Extended qualification budget
Few suppliers on DE-140 and DE-170	High	Qualify two if at all possible. Accept a higher unit price for a second source on the critical machines
Line speed guarantee not met	High	Make speed-at-uniformity contractual with a defined verification method and a stated remedy. Model the parallel-line count at both the guaranteed and the fallback speed before committing to a building
Adhesion below specification	High	Peel adhesion as a contractual guarantee; foil surface preparation as a joint scope item with a defined incoming Ra
Scale-up from pilot does not hold	High	Require the supplier's scale-up rule and its evidence in writing. Stage the order: pilot → one production module → the rest
Dust explosion protection retrofitted late	High	Hazard assessment before layout freeze. Retrofitting relief venting after the building is built is very expensive
Inspection package inadequate on dry material	Medium	Require a demonstration on your material. A conventional RGB camera answer is a red flag
Formation protocol assumed to come with the machine	Medium	Budget an in-house DOE campaign of several months. Engage a research institute early
Long-lead spares not ordered with the machine	Medium	Order roll sets and laser sources with the original purchase order, when commercial leverage is highest
Compound novelty	High	See the box below

THE NOVELTY BUDGET — THE MOST IMPORTANT PROCUREMENT RULE IN THIS HANDBOOK

Introduce one novelty per programme. Four of the world's most capable cell manufacturers each took four to six years to reach limited 46-series production, and Tesla's leadership has publicly acknowledged that pursuing the dry electrode process for the 4680 from the outset was a mistake that delayed the programme.

A dry-electrode line on a proven format is a hard but tractable project. A dry-electrode 4680 with a silicon anode is three coupled research programmes running at once, in which every failure has three plausible causes and none can be isolated. **Sequence them.** If the business case requires all three simultaneously, the business case is the thing that needs revisiting — not the engineering plan.

29. Public Patent Landscape

READ THIS BEFORE USING ANY NUMBER IN THIS CHAPTER

The numbers below are collected from public sources. **Verify every one directly in Google Patents, Espacenet or the USPTO register before any legal or commercial use.** Legal status (granted, pending, expired, abandoned) and claim scope can only be established from the official document. Several publication dates are reported inconsistently across secondary sources and are flagged. This chapter is for technical study. It is not legal advice and not a freedom-to-operate opinion.

29.1 Tesla — tabless architecture

Number	Title	Notes
US11749842B2	Cell with a tabless electrode	Granted. Current assignee Tesla Inc. Inventors: Kunio Tsuruta, Mikel Ehrlich Dermer, Rajeev Dhiman. Application US16/673,464. Prior-art keywords: substrate, conductive, cell, cap, conductive portions
US20200144676A1	Same family	Original application publication
US12463256B2	Continuation	Application US18/154,748
US20260038882A1	Continuation	Application US19/354,692
—	Tabless energy storage devices and methods of manufacturing thereof	Filed 17 Sept 2021, published 24 March 2022. Describes a method of making a cell with anode or cathode having a series of flags formed from foil portions at the upper and lower ends of each electrode, the flags folded over at each end to form an interleaved flower shape
—	Energy storage cell	Covers the cylindrical 4680 cell itself

Core method claim of this family: providing an electrode layer with active material disposed over a foil → forming a series of flags in the foil to produce a flagged electrode → winding the flagged electrode → forming an electrode roll comprising a series of rolled flags.

29.2 Tesla / Maxwell — dry electrode

Number	Title	Notes
US20150303481A1	Dry energy storage device electrode and methods of making the same	Originally Maxwell Technologies; assigned to Tesla, Inc. on 22 Oct 2021. The foundational document of the family. Application US14/690,153
US11876230B2	Dry energy storage device electrode and methods of making the same	Granted. Inventors: Hieu Minh Duong, Haim Feigenbaum, Jian Hong. Application US16/939,616. Prior-art keywords: PTFE, binder, anode, PVDF, electrode film
US20200358100A1	Same family	Application publication
US12406995B2	Continuation	Application US18/528,614
US20250364562A1	Electrode Manufacturing Process	Application US19/289,989, filed 4 Aug 2025. Publication date reported inconsistently (27 Nov 2025 vs 29 Jan 2026 in different secondary sources) — verify
US20260031317A1	Dry electrode manufacturing method	Reported published 29 Jan 2026

Technical claims reported in this family: non-destructive mixing of active material (particles larger than 10 μm) with porous carbon *before* adding a single PTFE binder; PTFE below 2 wt%; conductive carbon limited to 8 wt% or less; self-supporting films in three or fewer calender passes; active material content up to 98 %; binder as low as 1.25 wt%; approximately 90 % capacity retention after 2,000 cycles.

29.3 Tesla — chemistry and materials

Number	Subject
WO 2024/147993	Doped Manganese-Rich Cathode Active Materials — specific dopants lock the manganese structure in place to prevent manganese dissolution
Granted Jan 2023	Dry electrode film with a micro-particle non-fibrous binder to improve battery life
Granted Jan 2023	Electrode binder able to withstand higher operating voltages without rapid degradation
Filed June 2022	Mixture of metallic elements and carbon for energy storage devices — dry electrode film containing lithium metal, carbon particles, porous carbon particles and an SEI layer
Filed Aug 2022	Method for preparation of 3-substituted-1,4,2-dioxazol-5-one compounds — electrolyte additive that significantly extends life during long cycling
US20250357544 (weak source)	Heat-resistant electrolyte environment. Reported from a low-reliability secondary source — verify before citing

29.4 LG Energy Solution

LG was the first South Korean company to successfully mass-produce cylindrical lithium-ion batteries, in 1998. Its portfolio spans 18650 and 21700 through to next-generation 46-series. The company's IP Center has shifted focus to **cylindrical 46-series batteries, dry electrodes and solid-state batteries**, and has established an advanced IP R&D system with regional IP offices.

Number	Subject
US20240006722	Electrode Assembly, Battery Cell, Battery Pack, and Vehicle — electrode terminal riveted through a through-hole in the bottom of the battery can , gasket between terminal and through-hole outer diameter, fixing part on the active material layer bonded to the separator. This riveted-terminal architecture is characteristic of the LG 46-series design
US20240250307A1	Electrode assembly and electrochemical device including same — addresses stress accumulation from electrode expansion and contraction in jelly-roll assemblies, which causes deformation, non-uniform electrode spacing, performance deterioration and internal short-circuit risk

29.4.1 Active litigation — July 2026

On **21 July 2026**, LG Energy Solution filed a patent infringement suit in the **US District Court for the Eastern District of Texas** against **EVE Energy**, and in parallel submitted a complaint to the **ITC** seeking an investigation under **Section 337 of the US Tariff Act**.

Element	Detail
Patents asserted	Five US patents: four covering cylindrical cell construction including tabless structures, one covering separator technology
Licensing vehicle	Tulip Innovation — a lithium-ion licensing programme combining IP from LG Energy Solution and Panasonic Energy
Portfolio scale	More than 5,000 patents across upwards of 1,500 patent families; also covers cell-safety features
Scope note	Available filings do not suggest that EVE's large-format energy storage cells are directly covered by the dispute

THE POINT THAT MATTERS FOR ANYONE ENTERING THIS SPACE

Tesla's own tabless patents are **separate** from the ones LG and Panasonic are asserting. Multiple independent and potentially overlapping portfolios cover the same underlying concept — connecting a large part of the electrode edge to the current collector rather than routing current through discrete tabs. A tabless 46-series programme therefore faces a layered IP landscape, not a single owner. Budget for a proper freedom-to-operate analysis by qualified patent counsel before committing capital.

29.5 Panasonic

Panasonic Energy licenses its IP jointly with LG through the Tulip Innovation programme. Its older Panasonic/Sanyo filings on cylindrical current-collection structure are technically foundational and worth reading for the engineering rationale:

Number	Subject
US6653017	Nonaqueous electrolyte secondary cells — laser welding of the collector plate to the current collector foil edge with the junction positioned at 90 degrees (or close) to the direction of laser beam projection, so the beam acts effectively on the junction and gives high weld strength through increased joint area
US7273678	Secondary cell — collector plate welded to the foil edge over a large contact area; the plate is then laser-welded to the can bottom from outside the cylinder , with the beam moved along the flat region of the plate

These two documents describe, twenty years earlier, precisely the welding geometry and the outside-in welding strategy discussed in §9.4 and §9.7. The engineering logic of the 4680 collector architecture did not appear from nothing.

29.6 Samsung SDI, CATL, EVE, BYD

Public 46-series patent information for these companies is thinner. Suggested search routes:

Company	Route
Samsung SDI	Focus on 4695. Search Espacenet by applicant name plus the relevant IPC classes below; check KIPRIS for Korean filings without US equivalents
CATL / EVE	Predominantly Chinese (CN) filings with limited PCT equivalents. Search CNIPA
BYD	LFP-based large cylindrical development

29.7 IPC classes for searching

Class	Subject
H01M 10/0525	Rocking-chair / lithium-ion batteries with lithium insertion or intercalation in both electrodes
H01M 4/04	Electrode manufacturing processes
H01M 4/13	Secondary cell electrodes
H01M 50/40	Separators, membranes, diaphragms, spacing elements inside cells
H01M 50/46	Separators characterised by their combination with electrodes
H01M 50/461	Separators with adhesive layers between electrodes and separators
H01M 50/531	Current collectors / tabs
H01M 50/107	Cylindrical casings
H01G 9/00	Electrolytic capacitors — relevant for the Maxwell legacy documents
US Class 429/94	Plural concentric or single coiled electrode — legacy US classification, useful for historical searching

29.8 Search methodology

- Google Patents** — best starting point. Always read the “Similar Documents”, “Cited By” and “Also Published As” sections; they surface the family and the competitive landscape faster than keyword searching.
- Espacenet** (EPO) — better international coverage and stronger family tools.
- USPTO Patent Public Search** — the official document and legal status.
- KIPRIS** — Korean filings (LG, Samsung SDI) that have no US equivalent yet.
- CNIPA** — Chinese filings (CATL, EVE, BYD).
- Justia Patents** — convenient browsing by company and classification.

METHODOLOGICAL POINT THAT SAVES WEEKS

Do **not** search for “4680”. The format designation almost never appears in patent text — it is a marketing term, not a patent term. Search technical concepts instead: *tabless electrode*, *current collector plate cylindrical*, *dry electrode film PTFE*, *electrode assembly winding notched foil*, *flagged electrode*. Then use IPC classes to bound the result set.

30. Safety, Standards and Testing

30.1 Cell-level standards

Standard	Subject
UN 38.3	Transport testing — mandatory for every shipment. Covers altitude simulation, thermal cycling, vibration, shock, external short circuit, impact/crush, overcharge and forced discharge
IEC 62660-1	Performance testing of lithium-ion cells for electric vehicle propulsion
IEC 62660-2	Reliability and abuse testing
IEC 62660-3	Safety requirements
ISO 6469-1	Electrical safety of electric vehicles — rechargeable energy storage system
UL 1642 / UL 2580	Cell safety / batteries for use in electric vehicles
SAE J2464 / J2929	Abuse testing and safety standard
GB 38031	Mandatory Chinese standard for EV traction batteries
GB/T 33824-2017	Aluminium and aluminium alloy plates, sheets and strips for cans and caps of new-energy power batteries
GB/T 3190-2020	Chemical composition of wrought aluminium and aluminium alloys
DIN EN 573-3:2024	Aluminium and aluminium alloys — chemical composition and form of wrought products

30.2 Abuse testing

Test	Simulates
Nail penetration	Internal short circuit
Crush	Collision
Overcharge	BMS failure
External short circuit	Wiring fault
Thermal abuse (external heating)	Adjacent fire
Thermal propagation	Whether one failing cell ignites the pack — the test that matters most at system level

30.3 Occupational safety

Hazard	Control
Electrolyte	Flammable and toxic. Tightened safety measures are required specifically because of electrolyte use: vapour detection, spill containment, appropriate fire suppression
Hydrofluoric acid	Formed from LiPF ₆ reacting with moisture. Calcium gluconate gel must be immediately available at every point where electrolyte is handled , with trained personnel. HF burns are deceptive — pain can be delayed while damage progresses
Metallic and carbon dust	Respiratory hazard and combustible dust explosion hazard . On the dry route this applies across the <i>entire</i> electrode block, not just at cutting stations. Local extraction, rated collectors, explosion venting/isolation/suppression, spark detection, full bonding and earthing, certified vacuum cleaning — and no compressed-air blow-down
Lasers	Class 1 enclosure with interlocks. Laser protection during laser welding is a stated requirement
Beta radiation (gauges)	Licensing, dose monitoring, trained radiation protection officer
Formation and aging gases	After-treatment (e.g. RTO) before exhaust. H ₂ /CO detection in dense stores of charged cells
High voltage	Lockout/tagout at formation and EoL stations
Stored energy	An aging warehouse holds megawatt-hours of charged cells. Fire strategy, compartmentation and detection must be designed for that, not for a general warehouse

Appendix A — Equipment Register Template

Tag	Machine	Phase	Route	Maturity	Governing specification
DE-100	Powder receiving	Electrode	Shared	Mature	Containment; <20 % RH; segregation
DE-110	LIW dosing	Electrode	Shared	Mature	±0.1 % of setpoint
DE-120	Sieve + magnetic sep.	Electrode	Shared	Mature	2 stages >10,000 G
DE-130	Pre-blender	Electrode	Dry	Mature hw	Low shear; no attrition
DE-140	Fibrillation mixer	Electrode	Dry	Low	RTD; specific shear energy
DE-150	De-agglomeration	Electrode	Dry	Moderate	Break lumps, no further fibrillation
DE-160	Powder feeder	Electrode	Dry	Moderate	CD uniformity → ±1.5 %
DE-170	Film-forming roll pair	Electrode	Dry	Mod-low	Speed at uniformity; crown
DE-180	Free-film handling	Electrode	Dry	Moderate	Minimum span; clearing time
DE-190	Lamination	Electrode	Dry	Moderate	Peel adhesion; margin
DE-200	Final calender	Electrode	Shared	Mature	Porosity; ≤3 total passes
DE-210	Slitter	Electrode	Shared	Mature	Burr-free; ±150-250 µm
DE-220	Vacuum conditioning	Electrode	Shared	Mature	<20/50 ppm; air lock
AS-300	Laser notcher	Assembly	Shared	Moderate	Variable pitch; web-length sync
AS-310	Winder	Assembly	Shared	Mature	<200-300 µm alignment; tension taper
AS-320	Tab-forming press	Assembly	Shared	Moderate	Plane flatness and height
AS-330	Collector welder	Assembly	Shared	Moderate	OCT penetration control
AS-340	Can draw press	Assembly	Shared	Mature	0.5±0.02 mm; roundness <50 µm
AS-345	Cap / vent scoring	Assembly	Shared	Mature	Score depth; burst pressure
AS-350	Insertion	Assembly	Shared	Mature	No layer shift; force monitoring
AS-360	Can/terminal welder	Assembly	Shared	Moderate	Penetration = wall + margin
AS-370	Beading	Assembly	Shared	Mature	Bead depth; no cracking
AS-380	Electrolyte filling	Assembly	Shared	Mature	±0.5 % gravimetric; lance position
AS-390	Sealing / leak test	Assembly	Shared	Mature	<10-6 mbar·L/s; two test points
FN-400	Soaking	Finishing	Shared	Mature	30-50 °C, ±2 °C uniformity
FN-410	Formation	Finishing	Shared	Mature	0.02 % class; pin resistance
FN-420	Degassing	Finishing	Shared	Mature	Reserve footprint for larger formats
FN-430	Aging	Finishing	Shared	Mature	<5 mV/week; rack temp uniformity
FN-440	EoL test / grader	Finishing	Shared	Mature	Hi-Pot 500-1000 V
FN-450	Marking / packing	Finishing	Shared	Mature	DataMatrix, read-after-write

Appendix B — Utilities Matrix

Machine	Power	Chilled water	CDA	N ₂	Vacuum	Extraction	Environment
DE-100	Low	—	Yes	Yes	—	High	ISO 7-8, <20 % RH
DE-110	Low	—	Yes	Yes	—	Medium	ISO 7-8
DE-120	Low	—	—	Yes	—	Medium	ISO 7-8
DE-130	Medium	Yes	Yes	Yes	—	Medium	ISO 7-8, N ₂
DE-140	High	High	Yes	Yes	—	Medium	ISO 7-8, N ₂
DE-150	Low	—	—	Yes	—	Medium	ISO 7-8, N ₂
DE-160	Low	—	Yes	Yes	—	Medium	ISO 7-8, N ₂
DE-170	High	Hot + chilled	Yes	Yes	—	Medium	ISO 7-8, N ₂
DE-180	Low	—	Yes	—	Optional	Medium	ISO 7-8
DE-190	Medium	Hot + chilled	Yes	—	—	Low	ISO 7-8
DE-200	High	Hot + chilled	Yes	—	—	Low	ISO 7-8
DE-210	Medium	—	Yes	—	—	High	ISO 7-8
DE-220	High	Yes	—	Yes	Yes	—	Encapsulated
AS-300	Medium	Yes	Yes	Yes	—	High	Dry room –40/–60 °C
AS-310	Medium	—	Yes	—	—	Medium	Dry room
AS-320	Low	—	Yes	—	—	Low	Dry room
AS-330	High	Yes	Yes	Ar/N₂	—	Medium	Dry room, laser Class 1
AS-340	High	Yes	Yes	—	—	Low	Normal
AS-350	Low	—	Yes	—	—	Low	Dry room
AS-360	High	Yes	Yes	Ar/N ₂	—	Medium	Dry room
AS-370	Medium	—	Yes	—	—	Low	Dry room
AS-380	Medium	Yes	Yes	Yes	Yes	VOC	Dry room, containment
AS-390	Medium	Yes	Yes	He	Yes	Low	Dry room
FN-400	Medium	—	—	—	—	Low	Normal
FN-410	Very high	Yes	Yes	—	—	RTO	Normal, gas detection
FN-420	Medium	—	Yes	Yes	Yes	RTO	Encapsulated
FN-430	Medium	—	—	—	—	Low	Normal, gas detection
FN-440	Medium	—	Yes	He	Yes	Low	Normal

Appendix C — Layout Planning

C.1 Sequencing rules

1. **Commission the dust hazard assessment before freezing the layout.** Explosion relief venting must discharge to a safe place, which constrains collector siting, which constrains duct routing, which constrains machine positions.
2. **Anode and cathode trains physically separated** from powder receiving through to the dry room. Separate equipment, separate extraction, separate cleaning tools. Cross-contamination is unrecoverable.
3. **Minimise the free-film path** (DE-170 → DE-180 → DE-190). If the supplier offers an integrated unit, take it.
4. **Minimise transfer count for fibrillated powder** (DE-140 → DE-150 → DE-160). Fibrillated material consolidates on standing and segregates on conveying.
5. **Place DE-220 at the dry-room boundary** so it can serve as the air lock.
6. **Reserve footprint for FN-420** even if the current cell format does not require degassing.
7. **Formation and aging dominate area.** Size them from residence time (§4.3.1) before laying out anything else — they will determine the building envelope more than the process line does.

C.2 Footprint blocking factors

Item	Allowance
Machine envelope from datasheet	Baseline
Maintenance access and roll change	+40–60 % over the envelope
Coil staging and material buffers	Sized from batch size and queue-time limits
Operator space and walkways	Per local regulation
Utility risers and duct routing	Coordinate with FC-300 relief venting constraints early
Laser enclosures	Class 1 enclosure adds envelope around AS-300, AS-330, AS-360
Electrode block versus wet route	~60 % of the wet-route baseline

Appendix D — What Is Not Public

Listed explicitly so that no one spends weeks searching for it. These are trade secrets, and the industry says so:

Item	Status
Formation protocol — exact current/voltage/temperature profile and step sequencing	Explicitly described in industrial literature as <i>the core knowledge of a cell manufacturer</i> . Varies by manufacturer and chemistry. Not published by anyone
Exact electrolyte additive formulations and ratios	Proprietary. Patents disclose compound classes, not working formulations
PTFE fibrillation parameters — specific shear energy, temperature, residence time	The critical variable of the dry process. Patents disclose composition and sequence, not the mixing energy
Laser welding process windows per configuration	Developed empirically per machine, per material stack. Not transferable between installations
Grading algorithms	Bin definitions and parameter weighting are proprietary
Predictive quality ML models	The primary competitive asset of a mature plant
Actual yield figures by station	Only aggregate ramp figures (~80 %) appear publicly

LEGITIMATE ROUTES TO THIS KNOWLEDGE

Collaboration with research institutes (Fraunhofer ISI / IWS, Argonne CAMP, WMG Warwick, PEM RWTH Aachen); purchase of specialist market and technology reports; recruitment of engineers with production line experience; or your own designed-experiment campaign. Budget months, not weeks, for the formation protocol in particular.

Appendix E — Glossary

Term	Definition
Areal loading	Dry active-material mass per unit electrode area, mg/cm ² . Industry uniformity target ± 1.5 % or better
Beading	Circumferential groove formed in the can body as a seat for gasket and cap
CD / MD	Cross direction (across web) / machine direction (along web travel)
Crown	Deliberate barrel profile ground into a roll to compensate bending under load
Direct calendering	Dry route variant where powder is compacted directly onto the foil, with no free-standing film
Dry coating	Family of methods based on dry processing of materials to manufacture an active material layer, without solvent
False weld	Insufficient metallurgical bonding despite apparently normal process signals
Fibrillation	Drawing of PTFE particles into nanoscale fibrils under shear, forming the binding network that holds the dry electrode together
Flag	Tongue cut into the uncoated foil margin, later folded into the current-collecting plane
Free-standing film	Dry route variant where a coherent film is formed first, then laminated onto foil
Green-film tensile strength	Tensile strength of the unsupported dry film — the functional consequence of fibrillation quality
Hilumin	Nickel-plated steel strip used for cell cans
Jelly roll	The wound assembly of anode, cathode and two separators
Keyhole	The vapour cavity in deep-penetration laser welding
LIW	Loss-in-weight feeder — gravimetric dosing by continuous hopper weighing
Notching	Pre-forming the electrode contour in the web state
OCT	Optical coherence tomography — direct real-time measurement of laser weld penetration depth
Over weld	Excessive penetration, potentially reaching into the jelly roll and perforating the separator
PPM (throughput)	Parts per minute
Residence time distribution	The spread of time that different material elements spend in a mixer. The governing specification for DE-140
SEI	Solid electrolyte interphase — the passivating layer formed on the anode during formation
Specific shear energy	Energy input per unit mass, in kWh/kg or J/kg. The primary process variable for fibrillation
Tabless / full-tab	Current collection using the electrode edge itself rather than discrete welded tabs
Taper tension	Reducing web tension as rewind diameter grows, to avoid crushing inner wraps
TRL	Technology readiness level. Dry coating is assessed at TRL 6–7

Appendix F — Cited Sources Register

Every source drawn on by this handbook, numbered, with its public access route and the chapters that rely on it. All entries are publicly obtainable. Nothing in this register requires privileged access.

F.1 Tier A — Peer-reviewed and DOI-indexed

Ref	Source	Public access route	Used in
S1	<i>Lithium-Ion Cells in Automotive Applications: Tesla 4680 Cylindrical Cell Teardown and Characterization</i> — Journal of The Electrochemical Society, 2023	DOI 10.1149/1945-7111/ad14d0	Ch. 1, 2, 4, 15
S2	<i>Thermal management of 4680 battery cells: cell design and cooling concept</i> — Journal of Energy Storage, 2025	Journal platform; DOI-indexed	Ch. 2, 30
S3	Baazouzi et al. — comparative study of 19 cylindrical cells across four manufacturers and four formats (18650, 20700, 21700, 4680)	Journal platform; DOI-indexed	Ch. 1, 2
S4	Guk, Faraji Niri, Farhadi Tolie, Marco (WMG Warwick / Faraday Institution) — open-access ultrasonic frequency-domain dataset for electrode coating and calendering parameterisation, incl. FFT data, mass, thickness, density and full DOE factors	DOI 10.1016/j.dib.2025.112433 — open access	Ch. 14, 17, 18
S5	<i>A review of lithium-ion battery electrode drying: mechanisms and metrology</i> — Queen's University Belfast	Journal platform; DOI-indexed	Ch. 14, 17
S6	Open pilot-scale convective drying dataset — coating and drying process variables with matching half-coin electrochemical data (C/20 to 10C)	Open data repository; DOI-indexed	Ch. 17, 21

F.2 Tier B — Institutional engineering references (ISBN)

THE SINGLE MOST IMPORTANT SOURCE IN THIS HANDBOOK

S7 supplies the large majority of the process parameter windows in Parts III and IV. It is a publicly distributed reference work with an ISBN, produced by a university chair jointly with a national industry association. If a reader wishes to check one source only, it should be this one.

Ref	Source	Public access route	Used in
S7	PEM of RWTH Aachen University & VDMA — <i>Production Process of a Lithium-Ion Battery Cell</i> , 4th ed., January 2023. Authors incl. Prof. Achim Kampker, Dr.-Ing. Heiner Hans Heimes, Sarah Wennemar, Gerrit Bockey, Lorenz Plocher; VDMA contributors Dr. Sarah Michaelis, Jörg Schüttrumpf	ISBN 978-3-947920-27-3	Parts III-IV throughout ; Ch. 19, 21, 22
S8	PEM / VDMA — <i>Component Manufacturing of a Lithium-Ion Battery Cell</i>	ISBN 978-3-947920-06-8	Ch. 4, 10
S9	PEM / VDMA — <i>Battery Module and Pack Assembly Process</i> , 3rd ed.	ISBN 978-3-947920-05-1	Ch. 1, 30
S10	PEM / VDMA — <i>Production Process of an All-Solid-State Battery Cell</i> , 2nd ed.	ISBN 978-3-947920-29-7	Ch. 3 (context)
S11	PEM / VDMA — <i>Recycling of Lithium-Ion Batteries</i>	ISBN 978-3-947920-11-2	Ch. 30

F.3 Tier C — Published patent documents

Publication by a patent office is the act of placing a disclosure in the public domain for reading and study. All numbers below are searchable at Espacenet, USPTO Patent Public Search or Google Patents. Full detail and prior-art keyword strategy are in Chapter 29.

Ref	Subject and documents	Public access route	Used in
S12	Tabless / current-collection architecture — US11749842B2 (granted, Tesla Inc.); US20200144676A1; US12463256B2; US20260038882A1	Espacenet, USPTO PPS	Ch. 2, 15, 29
S13	Dry electrode, foundational family — US20150303481A1 (Maxwell Technologies; assigned to Tesla, Inc. 22 Oct 2021)	Espacenet, USPTO PPS	Ch. 3, 11, 29
S14	Dry electrode, process and sequence claims — US11876230B2; US20200358100A1; US12406995B2; US20250364562A1; US20260031317A1	Espacenet, USPTO PPS	Ch. 11, 12, 14, 29
S15	Chemistry and materials — WO 2024/147993 (doped manganese-rich cathode); US20250357544 (<i>weak source — flagged in place</i>)	Espacenet, WIPO Patentscope	Ch. 4, 29
S16	Competing portfolios — US20240006722; US20240250307A1 (LG); US6653017; US7273678 (Panasonic, foundational cylindrical current collection)	Espacenet, KIPRIS, J-PlatPat	Ch. 29
S17	Litigation of public record — LG v. EVE, complaint filed 21 July 2026, E.D. Texas and ITC Section 337, five patents asserted via the Tulip Innovation joint licensing programme	Court and ITC public dockets	Ch. 29
S18	Search infrastructure — Espacenet (EPO), USPTO Patent Public Search, Google Patents, KIPRIS, CNIPA, J-PlatPat, Justia	All free to use	Ch. 29

STANDING CAVEAT ON EVERY PATENT REFERENCE

Numbers, status and assignee are reproduced for technical study only and must be verified against the official register before any legal or commercial reliance. One publication date in this set is reported inconsistently between databases; both are shown in Chapter 29 rather than one being chosen silently. This is not a freedom-to-operate opinion.

F.4 Tier D — Public corporate disclosure

Ref	Source	Public access route	Used in
S19	Tesla public statements on dry-electrode process economics (~90 % electrode-step CAPEX and energy reduction, ~70 % emissions reduction, ~40 % footprint reduction) and on large-scale deployment status	Investor relations material, earnings calls, shareholder presentations, press releases	Ch. 3, 24
S20	Public programme-risk acknowledgements by Tesla leadership regarding dry-electrode timing	Recorded public remarks	Ch. 3, 27

F.5 Tier E — Supplier and standards literature

Ref	Source	Public access route	Used in
S21	Equipment and instrument suppliers — Teledyne Vision Solutions, Precitec, VisiConsult, Keyence, Kadant and others	Public websites, downloadable brochures, trade-fair material	Parts II, V
S22	Aluminium can and enclosure technical material — Speira whitepapers; BatteryDesign.net cell benchmarking	Public whitepapers and website	Ch. 2, 15
S23	Published standards — UN 38.3 (transport), IEC 62660 series, ISO 6469, SAE J2464, UL 1642 / 2580, GB 38031	Purchasable from issuing body; scope and abstract free	Ch. 30

F.6 Tier F — Analyst and independent teardown reporting

Weakest tier. Used for market and programme context, and for a small number of secondary figures each flagged as “reported” at the point of use.

Ref	Source	Public access route	Used in
S24	SNE Research — <i>4680 Battery Technology Development Trend and Market Outlook</i> (2025)	Commercial market report	Ch. 1, 24
S25	About:Energy — laboratory teardown and differential capacity analysis of production-vehicle 4680 cells	Published laboratory report	Ch. 1, 4
S26	Independent teardown analysis — Munro & Associates; Neware disassembly and characterisation; The Limiting Factor; Matthew J. Lacey cell-model commentary	Public reports and published video analysis	Ch. 2, 15
S27	Trade press — Automotive World, InsideEVs, electrive, KED Global	Public publication	Ch. 1, 29

SUMMARY OF PROVENANCE

27 source entries across six tiers. The technical backbone — every process window in Parts III and IV — rests on **S7**, an ISBN-published institutional reference. The architectural description rests on **S1**, a peer-reviewed teardown with a DOI. The dry-process mechanism rests on **S13-S14**, published patent documents. No statement in this handbook depends on any source that a reader cannot independently obtain.

F.7 Suggested next steps by objective

Objective	Recommended path
Deep process understanding	Read the PEM/VDMA document in full, then the JES teardown paper, then Parts II-III of this handbook as the bridge between them
Line design	Pilot scale first — mandatory on a TRL 6-7 process. Engage turnkey suppliers early but specify metrology independently; do not let the line supplier also define how the line is measured
Patent research	Begin from the IPC classes in Ch. 29 in Espacenet. Never search “4680”
Investment evaluation	Chapter 24 for CAPEX and capacity, §1.4 for the honest performance picture, SNE Research for market
Academic research	JES and Journal of Energy Storage; the open WMG ultrasonic and drying datasets are directly usable
Formation development	Accept that this must be done in-house. Engage a research institute; budget a multi-month DOE campaign

Dry Electrode 4680 — Complete Manufacturing Handbook

Process and Equipment · Revision 1.0 · August 2026
Compiled and edited by **Maziar Abedzadeh**

Scope: dry route only. The wet slurry route is out of scope and is referenced only for comparison.

Compiled from public engineering literature, peer-reviewed teardown studies,
published patent applications and equipment supplier documentation.
Process windows are industrial reference ranges requiring DOE validation for any specific line and chemistry.
Footprints are indicative envelopes excluding maintenance access, staging and operator space.
CAPEX figures are published station-group costs for a 10 GWh/year reference line.
Patent numbers require verification against the official register before any legal or commercial use.
Dry electrode film-forming equipment is assessed at TRL 6-7; procurement should reflect that maturity.
This document is not legal advice and not a freedom-to-operate opinion.